

California Lutheran University

Modeling and Forecasting a Size, Value, and Profitability-Sorted
Equity Portfolio

ECON 513: Econometrics IV

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Introduction

This project will investigate if and how a portfolio of “growth” companies might be modeled by firm characteristics, market risk, macroeconomic factors, and market volatility. More specifically, the “excess return” of an asset will be under investigation. Returns are typically measured in unitless percentage-based terms. An “excess return” can be thought of as the difference in return someone earns above a comparison investment. Most commonly, the excess return people study is the return they can earn above the investment that is riskless. The risk-free return can be a difficult concept to grasp because it is common knowledge that moving one’s money from a cash position into a non-cash investment vehicle introduces the opportunity for that investor to lose their principal. But there does exist one such asset (or investment vehicle) that is thought of to be so safe that it involves no risk. That asset is the Treasury Bond issued by the US Treasury department. This asset is termed “riskless” because in the history of the US, this government agency has rarely failed to repay those who have invested in them. This is true for US government debt of all the different maturities (short term and long term). This portfolio of small-sized, profitable companies was chosen because it can be argued these stocks exhibit desirable characteristics to investors with a long-time horizon. In an ideal world, these small companies make good use of their capital by executing their business plans, effectively manage themselves through changes in the macroeconomy, and their competitive position to grow their operations, and return cash to shareholders through dividend payments.

The existence of riskless assets makes excess return an interesting concept to study because while an investor in US Treasury Bills will not lose their principal investment, they are still exposed to the risk posed by currency inflation. Meaning that, after the investor trusted the US government to re-pay them their principle plus interest at the maturity date of their bond, the purchasing power within that payment is reduced due to seigniorage, that is the persistent inflation that the bond-issuing agency and the central bank allow.

Motivation

Many investors (and individuals) will not accept the reduction to their purchasing power and will assume risk to protect themselves from inflation. The US capital markets provide a good platform for studying excess return because of their size and availability of data. To put their size into perspective, as of 2025 the total market capitalization of US equities was 2.7 times greater than the stock of Zero Maturity dollars (Siblis Research, 2025). To put this in perspective, the total market capitalization of US equities is approximately \$60 Trillion. And the stock of Zero Maturity Dollars is a measure inclusive of not only all physical cash, but also financial assets that are

redeemable on demand for their face value (i.e. checking & savings accounts and money market funds).

The risk-free return may not be enough for an individual to live a comfortable life after they spent their human capital. The Life-Cycle-Income hypothesis developed by Franco Modigliani in 1957 helps explain why this might be. It posits that individuals derive utility from consumption, and that they aim to keep this level of consumption at the constant or higher over their lifetime; While their income takes a “hump shape” over their lifespan (Modigliani, 1976). The individual will spend (on average) 40 years in the labor force according to the BLS (2024). This working career starts around the age of 20 and ends at 60. Before an individual starts working, they do not have income. They often borrow money to consume. During their working career they have great amounts of energy and robust health. They’re able to do amazing things: pay off debts, increase their consumption, provide welfare to others, and even save some of these earnings to fund consumption later on. Sadly, the individual at age 60 is no longer the same person they were at age 20. They may have lost significant energy, or their physical health now requires more care, for various reasons... they step away from laboring. This is ok, and a natural part of life, but it leaves the individuals’ utility vulnerable. They have shifted into a position where they are without a labor income and they are not saving at the same rate. The individual is now solely relying on that savings to fund their consumption. Investing some of their savings in a portfolio of companies instead of the riskless asset can help the individual keep their consumption “smooth” over their lifetime as the life-cycle theory posits.

The question of how much an individual chooses to allocate towards receiving the excess return is a difficult one. Seeking the excess return comes with risk and quantifying that risk has massive implications for the individuals retirement planning (Markowitz, 1952).

Background

Investment portfolios and stock returns are the primary topic in this paper, but they are merely a byproduct of a much greater phenomenon. That being the US equity market. Having an equity market present in a nation can facilitate fundraising for companies, wealth creation for individuals, and opportunities for risk-hedging for all. They are still used for this purpose today, but they have evolved substantially through the proliferation for broker-dealer agents, electronic trading, and sophisticated financial instruments.

An innovation to the tracking and performance measurement of financial markets has been the advent of market “indices” and “market-capitalization weighting.” Market indices function to measure the value of a group of assets, much like a portfolio. Today, the primary market indices that investors look at are the Standard & Poor’s 500 (S&P500 ticker ^SPX), the NASDAQ (ticker ^IXIC), and the Dow (Dow Jones Industrial average ^DJI). Among these, the Dow was the first index established. In its original form, it included 30 stocks. The overall price level of the Dow was based on the price of each

stock... however, they were not all treated equally. For example, if US Steel was \$50 per share, and US Timber was \$25 per share, the Dow Jones would weight (or rank) US Steel two times heavier than US Timber. This means the overall price of the Dow would be more sensitive to changes in US Steel than US Timber. This is known as a “price-weighted” index. The index value of a stock in a price-weighted index is given below:

$$\text{Price weighted Index Value} = \frac{\sum_{i=1}^n P_i}{D}$$

Where P_i = Price of Stock i

Where D = Index Divisor

The Index Divisor Is calculated by adding up all the stock prices then dividing by some number to reach a desired value in a base period. The divisor is necessary as it allows the market index to maintain its accuracy in the case events that could distort stock prices (stock splits, reverse-splits, dividends, company mergers).

A market-cap weighted index is calculated a little differently:

$$\text{Market Cap weighted Index Value} = \frac{\sum_{i=1}^n (P_i * S_i)}{D}$$

Where P_i = Price of Stock i

Where S_i = # of Shares Outstanding for stock i

Where D = Index Divisor

The divisor is calculated by adding up the market capitalizations of all the companies in the index and then dividing by the desired index value.

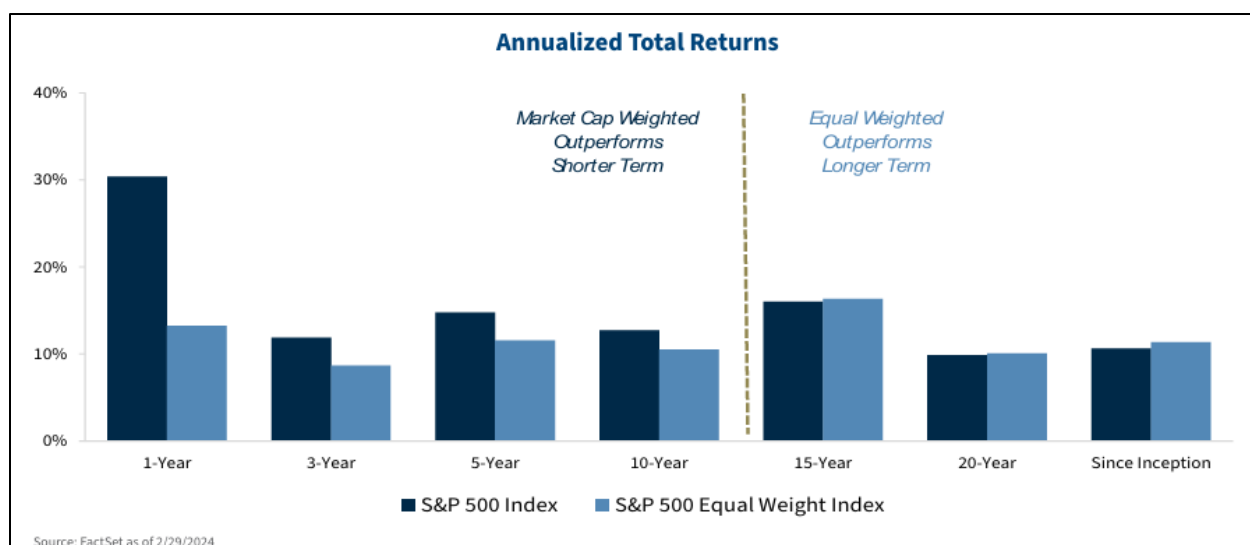
Table A: Descriptive Statistics for returns data of the SPX index and the individual stock portfolio

Excess Returns				
	MoM		YoY	
Statistics	SPX	Portfolio	SPX	Portfolio
Mean	0.32	1.02	4.1	13.5
Median	0.67	1.5	5.28	14.2
Standard Deviation	4.31	5.85	15.6	24.3
Skewness	-0.43	-0.48	-0.35	0.758
Kurtosis	4.62	6.18	3.32	5.77
Min	-22.36	-31.71	-46	-46.1
Max	15.79	27.93	54.1	161.2

Table A shows the Excess Returns of two Market Cap value-weighted assets. One is SPX and the other is the Portfolio of growth stocks used for analysis. Even in assets that are dubbed as “highly diversified” returns can be very volatile. Value-weighting the returns in an asset can also contribute to its volatility. When higher weighting move sharply in either direction the overall index/portfolio value will move sharply with it. Specific details on how the portfolio was constructed will be shared in the data section.

The value of Market-Cap weighted indices are agnostic to the individual share price of a stock, they consider a company's overall market value when weighing their contribution to the index. While this is a slightly more sophisticated method of indexing, it does make it immune to distorting the overall value of the stock market. Today's stock market has seen a small handful of companies become extraordinarily valuable (Apple, Microsoft, Tesla, Nvidia, Meta, Google, and Amazon). They even have a nickname (The Magnificent Seven). These companies' enormous value have an outsized impact on the overall SPX index. The visual below shows how market-cap weighted index of stocks can perform differently than the same group of stocks with equal weights (Adam et al, 2024).

Chart 1: Annualized Returns Comparison of Market Cap Weighted vs Equal Weighted



Economic activity is an important underpinning of equity markets. There have been periods throughout history where economic activity has been challenged. Naturally, these challenges work their way through companies and into financial markets

Data availability on financial products, financial performance, and general market indicators are a huge obstacle in research like this. However, data on economic indicators and aggregates are less of a challenge because of public, free, data aggregators such as FRED (Federal Reserve Bank of St. Louis). Financial Markets produce massive amounts of data. Even popular, once-free, data aggregators such as Yahoo! Finance now charge money for access to historical stock price data. Caroline Shaw, commissioner of the SEC in 2020, had this to say on the topic of "Market data infrastructure."

"Market participants need access to accurate, timely market data – the trades, prices, liquidity, and other information about what's happening in the securities markets. Our system is, in fact, premised on the notion that this data is broadly accessible to all participants. But the evidence suggests the current framework has created a two-tiered market: one for those

participants who can afford to pay ever-increasing prices for faster and better-quality market data, and another for those who cannot.” (Shaw, 2020)

Part of the reason why is because high-quality data allows people to answer more sophisticated questions. In the context of financial markets, high-quality data can be used to develop trading strategies that successfully identify security mispricing's and yield a profit. The high-quality data on stock prices, bids/asks, Implied volatility, yield, order-flow, shares-outstanding, etcetera are owned by the exchanges. The exchanges are responsible for administering the data to public. They do this through public data feeds. However, those data feeds are delayed. They also sell their real-time data to market participants through “proprietary data feeds.” (Shaw, 2020).

The intersection of numerous financial products, multiple exchanges, and inherent value of the data makes research in this area quite challenging. This is partially overcome by the help of data aggregators like FRED, and benevolent academics who believe in advancing research such as Kenneth French.

Theory

There are three theoretical concepts, and one excellent application that are most important to mention. In 1952, Harry Markowitz provided a framework that helps *anyone* who wants to build a portfolio answer the question, “how should I build my portfolio?” Markowitz simplified the portfolio selection process by offering an optimization. The optimization is concerned with the most fundamental aspects of investing: return and risk. Under Markowitz theory, an investor will seek to either maximize their expected return *given* a certain level of risk; or minimize their risk *given* a certain level of expected return (Markowitz, 1952). If the investor can this, then their portfolio is said to be “Mean-Variance” efficient (Markowitz, 1952). After Markowitz helped investors understand how to build a portfolio, Sharpe (1964) and Lintner (1965) helped investors understand the expected return on a risky asset. Their solution posited that any asset's return will be the risk-free return plus its sensitivity to the overall market. This became known as the “Capital Asset Pricing Model.”

$$E[R_i] = R_f + \beta_i(E[R_M] - R_f)$$

The CAPM is a single-factor model stating that an equity asset return can be explained by its sensitivity to the overall market. It does not explicitly model any firm characteristics or economic factors. An explicit assumption of this theory is that all market agents hold mean-variance efficient portfolios, and that the *market* under consideration is the overall equity market for a single country. Testing has been conducted and shows that CAPM can explain ~15% of the variation in stock returns in certain markets (Hasan, 2019). The CAPM assumptions are what other researchers have pointed towards when empirical testing of the this model failed (Fama & French, 2004). Nonetheless, the CAPM was the first asset pricing model for equities, and it was well received.

The single-factor model was a starting point for what came next. Stephen Ross shared a new equity asset pricing model in his 1976 paper titled the “The Arbitrage Theory of capital asset pricing.” Model given below:

$$\tilde{r}_i = E_{rf} + \beta_i \tilde{\delta}_1 + \cdots + \beta_k \tilde{\delta}_k + \tilde{\varepsilon}_i$$

$\tilde{r}_i$ = Return on Capital Asset i

E_i = Expected Return on Risk Free asset

$\tilde{\delta}_1 = E[R_{factor\ 1}] - R_f$ = Risk Premium for Risk Factor 1

R_f = Risk free return

β_i = Capital Asset i sensitivity to Risk Premium associated with Risk Factor i

$\tilde{\varepsilon}_i$ = unsystematic risk associated with the i th asset

Like the CAPM, Ross’s model of the equity asset return is a function of the risk-free rate... but also of k other risk factors. Notice that the risk factors are not specified. Ross’s pricing model posits that the risk components are numerous and unspecified. Roll & Ross found evidence that at least 3 factors are present in explaining asset returns (1980).

Unlike the CAPM that assumes that markets are always in equilibrium and securities are priced correctly, APT relaxes its assumptions. Instead, it believes that investors do not always hold efficient portfolios, and the market is not in a permanent equilibrium where prices reflect all information about the securities. APT grants that security mispricing’s do exist temporarily in markets, and arbitrageurs (a.k.a. investor who is looking for mispricing) takes advantage of these. A limitation of Arbitrage Pricing Theory is that it provides no answer as to what those risk factors impacting stock returns are (Ingersoll, 1984).

Researchers Eugene Fama and Kenneth French tried to explain what those factors are. Their research on asset returns is slightly different from Sharpe and Ross. In CAPM and APT, the dependent variable is the asset return on an individual stock. Fama and French examine the asset return on an *individual portfolio* (group of stocks). They did not venture into economic or policy indicators to look for an explanation behind asset returns. They focused their analysis on the financial characteristics that groups of publicly traded companies exhibit. They noticed that some companies are big and others are small, some companies trade at price close to their book value and others trade far away from that value. They used data on asset returns to create individual risk-factors from these characteristics. These measures are known as the “size” and “value” factors. They are specific, market-wide, measures of risk that an investor is exposed to if they choose to invest in an asset “ i ” or a portfolio of assets. The Fama-French 3 factor model that was developed in 1992 is given below:

$$R_i - R_f = \alpha_i + \beta_1(R_M - R_f) + \beta_2SMB_i + \beta_3HML_i + \varepsilon_i$$

The Market Risk premium that appeared in the CAPM is still relevant in this model, as it is the first explanatory variable on the Right-hand side. The *Size* (SMB) and *Value* (HML) factors were Fama and French's contribution. In 2015 they expanded on their model to include Profitability and Investment factors:

Fama French 5-factor Model:

$$R_i - R_f = \alpha_i + \beta_1(R_M - R_f) + \beta_2SMB_i + \beta_3HML_i + \beta_4RMW_i + \beta_5CMA_i + \varepsilon_i$$

Their work spurred similar research by others. Jegadeesh and Chan (1999) examined Momentum as a factor and found it explained differences in asset returns over a 12-month time horizon. Momentum was recognized by Fama and French as a factor in 2011.

The three-factor model was a big step forward in the research of asset pricing.

Literature

The literature on explaining stock returns has spanned many decades and has attracted researchers from the fields of statistics, mathematics, economics, and finance.

Academics who have written on this topic include Nobel Prize winners such as Eugene Fama and William Sharpe. When measuring excess return, most authors use the 1 month or 3-month Treasury Bill as a comparison. This research made a strong case that the single-factor model CAPM is not sufficient in explaining stock returns. Interestingly, empirical tests of the performance of the 3-factor vs 5-factor models offered mixed results depending on the markets tested and the portfolio sorting criteria. The 5-factor model was found to be superior in developed markets like the US and Australia but only offered marginal outperformance in Asian markets and other emerging markets (Cheong, 2023). Even finding that sometimes, the added factors were insignificant.

The Fama French 3 & 5 factor models are generally accepted as superior models in explain asset returns, but meaningful contributions have come from other researchers. Most notably, Jegadeesh & Titman developed the "momentum" factor using portfolio sorts on CRSP data (1999). Both Bergaoui & Trabelsi (2016) and . Ericsson & Karlsson (2004) used Fama & French factors in tandem with macroeconomic factors (inflation , industrial production, corporate bond spread, and growth rates of macroeconomic indicators). They found evidence that the Fama French factors are sensitive to macroeconomic conditions. Faust (2011) excluded the Fama French factors and alternative measures of the "value" factor (stock dividend to price ratio). Ammann et al (2010) found evidence that implied market volatility (VIX) is a stronger predictor than historical volatility (VOL) for stock returns and noted that this effect is stronger for smaller sized companies.

Post Great Financial Crisis in 2008, the Fama French factors are still the primary factors used to explaining asset returns, but researchers have not shied away from investigating the effects that monetary policy events have had on asset prices.

Laine (2023) uses “Overnight Index Swap” rates from the European Central bank as a proxy for a monetary policy shock, and Cepni et al (2021) uses 30 Day Fed Funds futures contracts to model in “monetary policy surprises.” Villanueva (2015) used the change in FFRT and constructed Quantitative Easing variables as change in MBS, and % change in Treasury securities held at the FED to explain forecast error in US Stock prices.

Data

Data on the dependent variable (ERP) was made available from the CMPT_ME_BEME_OP_INV_RETS file available on the Kenneth French online data library. The unique portfolio was titled “ME1 BM2 OP4.” Data was collected in April 2025 (it will not include any updates to accounting data that CRSP incorporates after this date), and is sourced from the Center for Research on Securities Prices (CRSP) database. The observations show the average, value-weighted, 1 month return on stocks sorted by market capitalization, valuation, and operating profitability. There are at least 10 different individual stocks in all portfolios from this file. The portfolios are constructed at the end of June. Portfolios were selected from all stocks on the NYSE, AMEX, and NASDAQ exchanges that satisfy the following criterion:

- Positive book equity in prior month
- Non-missing revenue in prior month
- Non-missing accounting in prior month for at least 1 category (Cost of goods sold, SG&A, interest expense).

While the dependent variable is a portfolio that is *only sorted on three factors* (Size, Value, and Profitability) that *does not* mean that it is necessary insensitive to the investment factor (CMA), systematic risk (ER), or Momentum (MOM). Each of the Fama-French factors is a calculation that measures the return spread between a portfolio of companies on opposite ends of the decile range for a specific factor. Depending on the Portfolio used on the left-hand side, the signs of these coefficients could vary. Each factor is described below.

The SMB factor is a measure of performance respective of Market Capitalization:

$$SMB_t = \frac{1}{3}(Small\ Value + Small\ Neutral + Small\ Growth) - \frac{1}{3}(Big\ Value + Big\ Neutral + Big\ Growth)$$

HML is a measure of performance respective of Valuation:

$$HML_t = \frac{1}{2}(Small\ Value + Big\ Value) - \frac{1}{2}(Small\ Growth + Big\ Growth)$$

RMW is a measure of performance respective of Size & Operating Profitability:

$$RMW_t = \frac{1}{2}(Small\ Robust + Big\ Robust) - \frac{1}{3}(Small\ Weak + Big\ Weak)$$

CMA is a measure of performance respective of Size & Asset Change (Investment):

$$CMA_t = \frac{1}{2} (Small\ Conservative + Big\ Conservative) - \frac{1}{3} (Small\ Aggressive + Big\ Aggressive)$$

MOM is a measure of market momentum over the prior 10 months. The calculation takes the return spread between 2 portfolios with high prior returns, less 2 portfolios with low prior returns:

$$MOM = \frac{1}{2} (Small\ High + Big\ High) - \frac{1}{2} (Small\ low + Big\ Low)$$

ERP is the *value-weighted* return spread on portfolio of stocks above the risk-free rate. Just as the SPX index weights bigger companies with higher values, so does ERP. The weighting values for each level of market capitalization are not publicly available. The portfolio sorting methodology was developed by Eugene Fama and Kenneth French. Each sort produced categories so that a group of stocks could be bunched together into a portfolio for analysis. These particular criterion divided stocks into 2 size groups (small and big) based on the NYSE median market cap value, 4 Value groups based on quartile values of Book Equity to Market Equity, and 4 Operating profit groups based on their quartile values. This factor-sort produced 32 unique portfolios for analysis. Each portfolio contains at least 10 stocks. ERP can be characterized as, “A group of small, modestly valued, profitable companies.” The table below shows the quartile values where these each stock (S_i) in the portfolio falls under:

Table 2: Quartile Sort for KDV Portfolio

Quartile	Book Eq / Mrkt Eq	Market Cap	Op Profit
Q1	$S_i < .35$	$S_i < \$1.2B$	$S_i < 3\%$
Q2	$.35 < S_i < .75$	$\$1.2B < S_i < \$2.7B$	$3\% < S_i < 8\%$
Q3	$.75 < S_i < 1$	$\$2.7B < S_i < \$10B$	$8\% < S_i < 15\%$
Q4	$S_i > 1$	$S_i > \$10B$	$S_i > 15\%$

Individual companies that would be included in this portfolio right now would be Kosmos Energy (KOS), Telecom Argentina ADR (TEO), and People's Bancorp (PEBO).

Data on ERP contained 0 missing observations. All data in my analysis is reported at the monthly or quarterly frequency. The data set consists of 30 variables. Not all of which will necessarily be used in regression analysis but were included for their appearance in the relevant literature regarding asset prices and returns. The most empirically relevant data was gathered from the Kenneth French online data library – Portfolio data on firm-

specific factors (ER, RF, ERP, SMB, HML, CMA, RMW, and MOM) is reported monthly starting in July 1963 through April 2025 (742 obs). Variable RF is the “Risk-free” rate of return, this variable was sourced by the Intercontinental Exchange (ICE) Bank of America 1-month treasury bill index. RF is also measured as the 1-month return as an Annual Percentage Rate on the 1-month US Treasury Bill.

17 Variables were collected from Data Aggregator FRED (Federal Reserve Bank of St Louis). These variables include macroeconomic indicators including (not exhaustive): Unemployment (M), personal consumption (M), Inflation expectations (M), Inflation rate (M), FED Total Assets (Q), % change in Corporate Debt (Q), Credit Spread on Corporate Bonds (M), Industrial Production (M), and FED Funds Effective Rate (M). Source include: The Federal Reserve Board of Governors, Moody’s, Bureau of Labor Statistics, Federal Deposit Insurance Corporation, National Bureau of Economic Research, and the Chicago Board of Options Exchange.

Data on the measure of implied Market volatility of the S&P500 (VIX) is sourced from CBOE (Chicago Board of Options Exchange) and aggregated by Yahoo Finance. Data on the measure of realized market volatility of the S&P500 (VOL) was sourced by CBOE, and is aggregated by FRED. This variable is included at the monthly frequency and includes 485 observations.

Table 3: Data Set

Capstone Data Table									
No.	Code	Description	Units	Source	SA or NSA	Frequency	Range	Nobs	
1	ER	Monthly Excess Return on the Overall Market	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
2	ERP	Monthly Excess Return -ME: BM2 OP4 Portfolio	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Sep 1981 - Apr 2025	742	
3	SMB	Small Minus Big	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
4	HML	High Minus Low	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
5	RMW	Robust Minus Weak	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
6	CMA	Conservative Minus Aggressive	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
7	RF	Risk Free	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	742	
8	MOM	Momentum	Percentage	Center for Research on Securities Prices (CRSP)	NSA	Monthly	Jul 1963 - Apr 2025	1180	
9	TP	Term Premium (10yr - 2yr)	Percentage	Federal Reserve Bank of St Louis	NSA	Monthly	Jun 1976 - Nov 2024	588	
10	TA	Fed Total Assets	\$ Millions	Federal Deposit Insurance Corporation (FDIC)	NSA	Quarterly	Jan 1984 - Jul 2024	165	
11	IORR	Interest Rate on Reserves	Percentage	Board of Governors Federal Reserve System	NSA	Monthly	Oct 2008 - Dec 2024	201	
12	FFRT	Fed Funds Effective Rate	Percentage	Board of Governors Federal Reserve System	NSA	Monthly	Jul 1954 - Dec 2024	852	
13	UE	US Unemployment Rate	Percentage	United States Bureau of Labor Statistics	SA	Monthly	Jan 1948 - Dec 2024	929	
14	IFD	Inflation Forecast Error	Percentage	Internally developed variable	NSA	Monthly	Jan 1982 - May 2025	521	
15	IFE	Inflation Expectations	Percentage	Federal Reserve Bank of Cleveland	NSA	Monthly	Jan 1982 - May 2025	522	
16	PCEI	Personal Consumption Expenditures Chain-Type Index	\$ Billions	United States Bureau of Economic Analysis	SA	Monthly	Jan 1959 - Nov 2024	791	
17	PPCE	PCE % change from Prior month	Percentage	United States Bureau of Economic Analysis	NSA	Monthly	Feb 1959 - Nov 2024	790	
18	PCED	Nominal Personal Consumption Expenditures	\$ Billions	United States Bureau of Economic Analysis	SA	Monthly	Jan 1959 - May 2025	791	
19	AAA	Yield to Maturity on Moody's AAA Corp Bond	Percentage	Moody's	NSA	Monthly	Jan 1919 - May 2025	1277	
20	BAA	Yield to Maturity on Moody's BAA Corp Bond	Percentage	Moody's	NSA	Monthly	Jan 1986 - Nov 2024	473	
21	DC	% Change in Corporate Debt	Percentage	Board of Governors Federal Reserve System	NSA	Quarterly	Jan 1952 - Jan 2025	293	
22	M2V	Velocity of M2 Money Supply	Ratio	Federal Reserve Bank of St. Louis	SA	Quarterly	Jan 1959 - Jul 2024	265	
23	EPI	Export Price Index: Gold	Percentage	United States Bureau of Labor Statistics	SA	Quarterly/Monthly	Dec 1984 - Nov 2024	408	
24	VIX	CBOE Volatility Index	Units	Chicago Board of Options Exchange (CBOE)	NSA	Monthly	Jan 1990 - May 2025	425	
25	IP	Industrial production related to capacity and utilization covering 5 sectors in US economy	Units	United States Bureau of Labor Statistics	SA	Monthly	Jan 1919 - May 2025	1277	
26	REC	Indicator Variable for US recessionary periods	Binary 1 or 0	National Bureau of Economic Research	NA	Monthly	Feb 1978 - May 2025	569	
27	QE1	Quantitative easing 1	Binary 1 or 0	Marquette Associates	NA	Monthly	Jul 1963 - Mar 2025	743	
28	QE2	Quantitative easing 2	Binary 1 or 0	Marquette Associates	NA	Monthly	Jul 1963 - Mar 2025	743	
29	QE3	Quantitative easing 3	Binary 1 or 0	Marquette Associates	NA	Monthly	Jul 1963 - Mar 2025	743	
30	VOL	Historical Market Volatility	Index	Stanford Institute of Economic Policy Research (Nick Bloom & Steven Davis)	NA	Monthly	Jan 1985 - May 2025	485	
31	X_TA	Change in Fed Total Assets	\$ Millions	United States Bureau of Labor Statistics	NA	Monthly	Apr 1984 - Jan 2025	164	
32	TBTY	YTM: 10 Year US Treasury Bond	Percentage	Board of Governors Federal Reserve System	NA	Monthly	Jan 1962 - June 2025	762	
33	MP	Unconventional Monetary Policy (Zero-lower Bound Interest rates)	Binary 1 or 0	Internally developed variable	NA	Monthly	Jul 1963 - June 2025	762	
34	M2	Supply of M2 Money	\$ Billions	Board of Governors Federal Reserve System	SA	Monthly	Jun 1963 - Apr 2025	748	
35	CPI	Consumer Price Index: All Items	Index	United States Bureau of Labor Statistics	NA	Monthly	Dec 1947 - May 2025	930	
36	BAA_FC	SCENARIO VARIABLE: FINANCIAL CRISIS PATH OF Yield to Maturity on Moody's BAA Corp Bond	Percentage	INTERNALLY DEVELOPED OOS PATH	NA	Monthly	May 2025 - May 2027	25	
37	VIX_FC	SCENARIO VARIABLE: FINANCIAL CRISIS PATH OF Supply of M2 Money	Index	INTERNALLY DEVELOPED OOS PATH	SA	Monthly	May 2025 - May 2027	25	
38	M2_FC	SCENARIO VARIABLE: FINANCIAL CRISIS PATH OF Consumer Price Index: All Items	\$ Billions	INTERNALLY DEVELOPED OOS PATH	NA	Monthly	May 2025 - May 2027	25	

Data gathered on QE comes from the Marquette Associates. This source provides dates (month & year) in which the Federal reserve began monetary policy measuring of quantitative easing, which allowed me to construct categorical variables for each QE phase.

Recession data was gathered from the National Bureau of Economic Research's (NBER) "Business Cycle Dating" report. Data is at monthly frequency ranging from January 1978 to December 2024.

A full data table listing Variable Codes, Units of Measurement, Description, Seasonal adjustments, range, and number of observations is included in the appendix section of the document.

Table 4: ERP Descriptive Statistics

Descriptive Statistics	ERP
Mean	1.05
Median	1.50
Std Deviation	5.83
Min	-31.71
Max	27.93
Skewness	-.50
Kurtosis	6.29
Jarque-Bera	341.04

This variable in “first-difference” form. The return on an asset is measured as: $\frac{P_t - P_{t-1}}{P_{t-1}}$

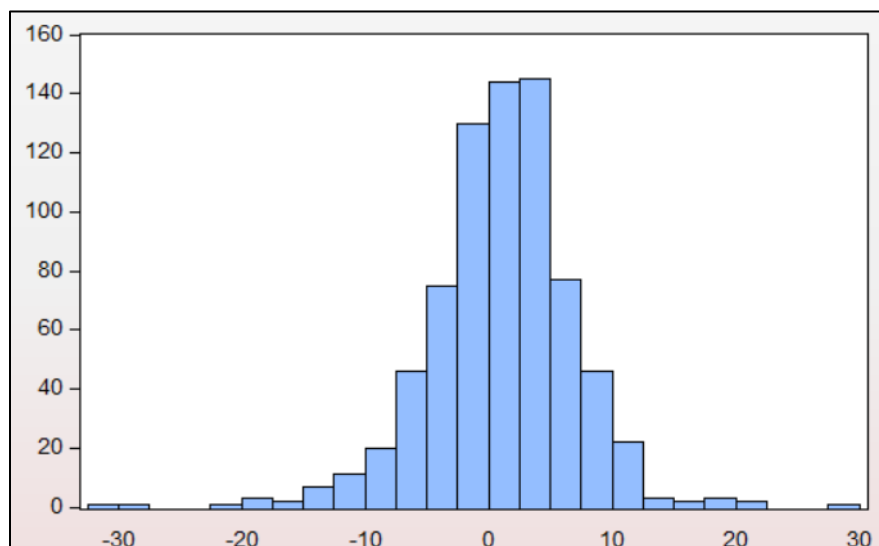
Much of the literature on the topic of capital assets uses returns because we observe them, whereas the true “intrinsic prices” of stocks are unobserved (Gray, 1997). Returns also have favorable statistical properties, and that they can be connected to economic theory. Returns are usually stationary over time, and, under the no-arbitrage conditions of APT their return is 0% in expectations (Ross, 1976).

Over the 64-year time series the mean 1 month return on this portfolio is 1.05%, and 1 standard deviation is almost 6%. Given this mean return, this portfolio is generating around a 12% annual rate of return. The minimum and maximum values are nearly the same magnitude in the positive and negative direction. Reading the skewness and kurtosis measures would suggest that this portfolio is more likely to experience large negative returns versus large positive returns. This is characteristic of equity return data.

Table 5: Stochastic Properties of ERP

Stochastic Properties	ACF(1) & ACF(2)	ADF Unit Root	DF-GLS Test	KPSS Unit Root Test
		Test Statistic	Statistic	Statistic
<u>Levels</u>	ACF(1) = .112 ACF(2) = -0.012	-0.22	-0.56	0.02
<u>Prewhitened (method)</u>	NONE	NONE	NONE	NONE
<u>Order of Integration:</u>	Evidence of I(0)	Evidence of I(0)	Evidence of I(0)	Evidence of I(0)
<u>Seasonal Lags (If relevant)</u>	N/A	N/A	N/A	N/A

Chart 2: Histogram (Excess Return of Portfolio)



The initial testing for evidence of autocorrelation and persistence in this time series is not found. The ACF decays faster than the geometric rate, and the test statistics across ADF, DFGLS, and KPSS provide evidence suggestive of this. Analysis of whether stationarity changes when considering shocks in the data provided confirming evidence that the dependent variable is stationary. We would not need to difference ERP for regression analysis, but it would be beneficial to apply a signed-log transformation to help the dependent variable have a more normal distribution. This would make the relationship less sensitive to outlier data points. If we were modeling the Price value of this portfolio we would need to apply differencing. This portfolio's price is not mean-reverting (stochastic properties shown below).

Table 6: Stochastic Properties of PRICE

Stochastic Properties	ACF(1) & ACF(2)	ADF Unit Root Test Statistic	DF-GLS Test Statistic	KPSS Unit Root Test Statistic
Levels	ACF(1) = .991 ACF(2) = .983	3.32	3.26	2.30
Prewhitened (method)	NONE	NONE	NONE	NONE
Order of Integration:	Evidence of I(1)	Evidence of I(1)	Evidence of I(1)	Evidence of I(1)
Seasonal Lags (If relevant)	N/A	N/A	N/A	N/A

Table 6: Unit Root Testing (ERP)

REGRESSION	ADF Test Statistic	NULL HYPOTHESIS
$ERP_t = \beta_0 + \varepsilon_t$	-24.166	REJECT
$ERP_t = \beta_0 + \beta_1 ERP_{t-1} + \beta_2 ERP_{t-2} + \beta_3 ERP_{t-3} + \varepsilon_t$	-27.011	REJECT
$ERP_t = \beta_0 + \beta_1 ERP_{t-1} + \beta_2 ERP_{t-2} + \beta_3 ERP_{t-3} + \beta_4' SPIKE_t + \varepsilon_t$	-26.44	REJECT

Given that volatility is often present in financial markets data, shock size measurements were performed to check whether the variable is non-stationary when accounting for shocks. Two separate regressions were run to measure how significantly shocks affect our lagged dependent variable. Statistical significance in the dependent variable, and unit root testing on the regression residuals did not provide evidence that ERP is non-stationary when accounting for shocks in the data (see appendix).

The McLeod-Li test was performed to check for Autoregressive Conditional Heteroskedasticity. In different test cases, evidence was found against the null hypothesis of No ARCH. The three regressions and their results are given below (full detail found in the appendix).

Table 7: ARCH Testing (ERP)

REGRESSION	Prob	NULL HYPOTHESIS
$ERP_t = \beta_0 + \varepsilon_t$	.048	REJECT
$ERP_t = \beta_0 + \beta_1 ERP_{t-1} + \beta_2 ERP_{t-2} + \beta_3 ERP_{t-3} + \varepsilon_t$	.016	REJECT
$ERP_t = \beta_0 + \beta_1 ERP_{t-1} + \beta_2 ERP_{t-2} + \beta_3 ERP_{t-3} + \beta_4' SPIKE_t + \varepsilon_t$	.031	REJECT

*The results rejecting the null hypothesis were consistent up to lag length 22.

*Spikes included 1975, 1987, 2008, and 2020

Table 8: Shock Size Measurements (ERP)

<i>ERP Largest Shocks</i>			
<i>Negative</i>		<i>Positive</i>	
2020M03	-5.6	1975M01	4.6
1987M10	-4.9	2020M11	3.4
1978M10	-3.7	1976M01	3.3
2008M10	-3.6	2020M04	3.2
1973M11	-3.3	2009M04	3.1

The Data contained shocks above 3 standard deviations from the mean in both directions. The most recent shock (Pandemic) was also the largest, nearly 6 standard deviations from the mean. Part of the reason that large shocks in financial markets data are commonplace is because markets contain widespread speculative behavior. Individual stock prices themselves are typically calculated as the present value of all future cash the company will earn over their lifetime. This is known as the “Discounted Cash Flow” method of valuation. Techniques such as this apply simple algebra to a firm’s accounting data to generate the stock price. This method is fine, but it relies on inherently builds in expectations of the future into the current price. Additionally, methods such as the DCF often are not capable of modeling in uncertainty related to macroeconomic or monetary/fiscal policy events. Analysts using a discounted cash model may try to model in uncertainty (crudely) by speculatively adjusting their discount rate (Bridges, 2025). When uncertainty is realized, the analyst has to update the assumptions they used in their DCF model, often resulting in a stock different stock price.

This macroeconomic uncertainty is reflected in the time series of ERP. In 1975, the US was exiting a period of geopolitical and economic uncertainty. In 1973 the US provided military support to Israel in a war against Egypt (Gramlich, 2004). This triggered OPEC to embargo oil exports to the US. Due to the outsized impact that oil had on the US economy, inflation surged *with* the increase in oil prices. In reaction to this, the US Central bank had increased interest rates from 7% to 12% to control inflation. By January 1975, interest rates had fallen back to 7% and the embargo had ended. October of 1987 is infamously known as “Black Monday” as US stock markets fell more than 20% in a single day. There were no major external market factors at work during the time. The Stock market had been in a “bull market” (positive trend) over the previous 4 years. The precise cause of this crash is not clear (Neumann, 2022). At the time, algorithmic trading programs and more sophisticated financial products (Portfolio Insurance and Index Arbitrage) were beginning to become popular at financial institutions (Neumann, 2022). It is believed that algorithms spiraled out of control and created overwhelming amounts of selling pressure on stocks. Human investors panicked, and sold as well, driving prices lower.

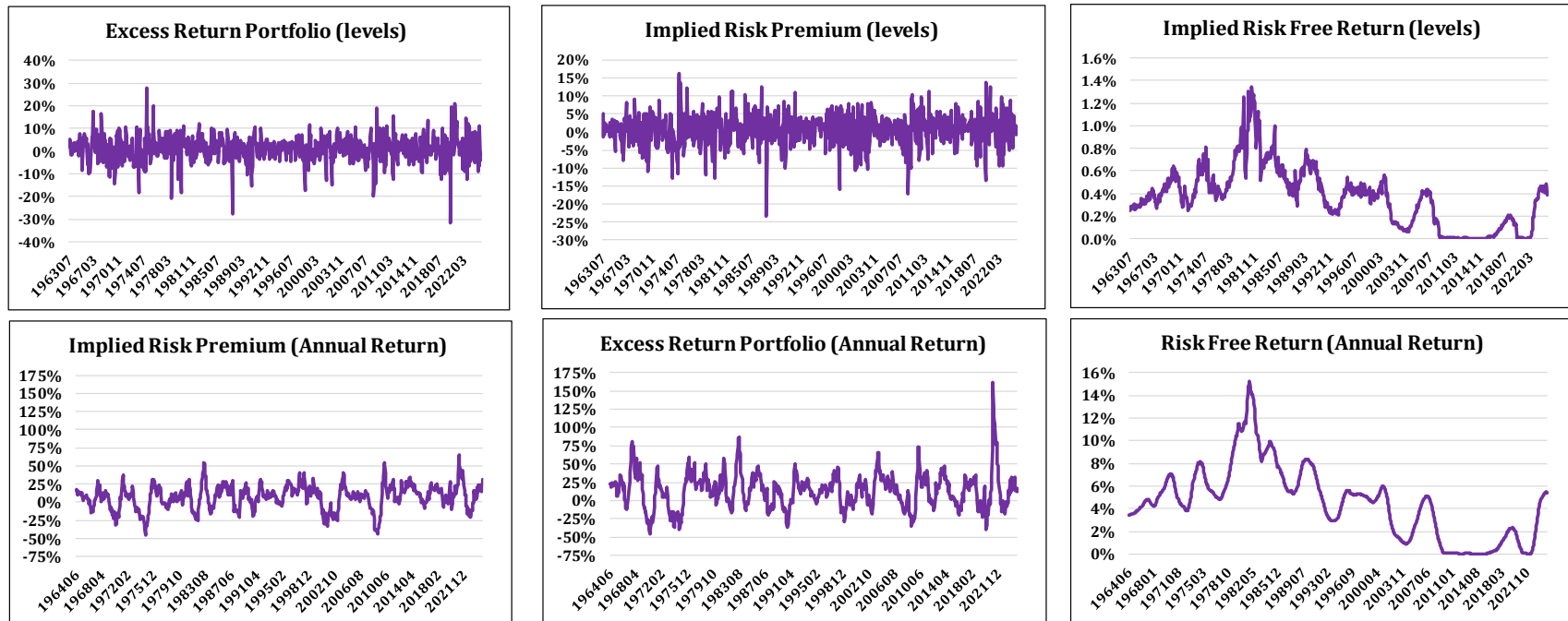
The largest drawdown in this time series (March 2020) was caused by an unprecedented world event in the form where the government official recognized the spread of a viral disease (corona virus) as a “global pandemic.” Governments across the globe issued “stay at home” orders to prevent the spread of disease. This temporarily hurt some businesses and eliminated others. Investors could not foresee how economic behavior would ultimately respond to this situation, so they broadly sold their equities, and the market entered a brief period of panic sending the market down 30% over a 2-month period. However, the panic was short lived. The legislative process initiated a fiscal policy maneuver through issuance of transfer payments (stimulus checks to citizens). The Central Bank took action by moving their overnight lending rate to near-zero and increasing the overall supply of money in the banking system (Ihrig, 2024). By the end

of 2020, the stock market recovered that entire loss and gained 30% over the remaining 8 months of the year.

The month over month volatility associated with ERP could make specification difficult in a regression with macroeconomic and credit market factors. I am going to proceed with the annualized, log price difference of the portfolio that was derived using the portfolio's excess return (ERP). This specification change may be a better fit for signal inside monetary policy and the credit market that may not have a meaningful relationship with the portfolio return month over month.

Charts

Charts 3-8: Chart Set for ERP, Equity Risk Premium and Risk-Free Rate of Return



The 1-month return (levels) for the Dependent variable and the Equity Risk Premium are volatile, but not easy to read. When the return is annualized, the month over month volatility becomes clearer. There doesn't appear to be a clear trend in mean for the Portfolio or the Risk premium.

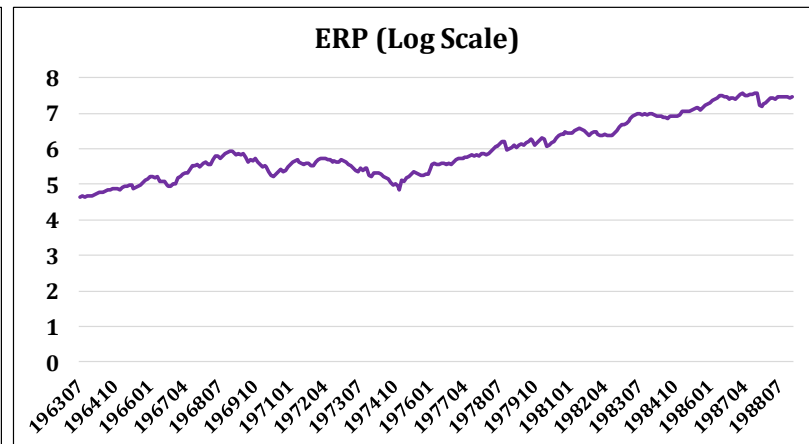
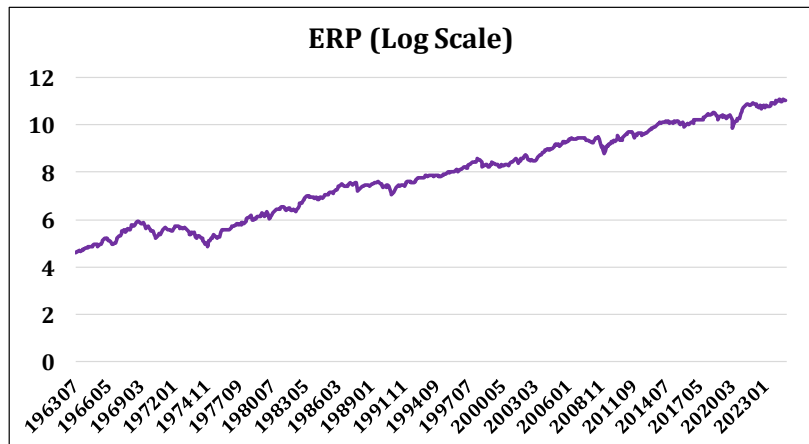
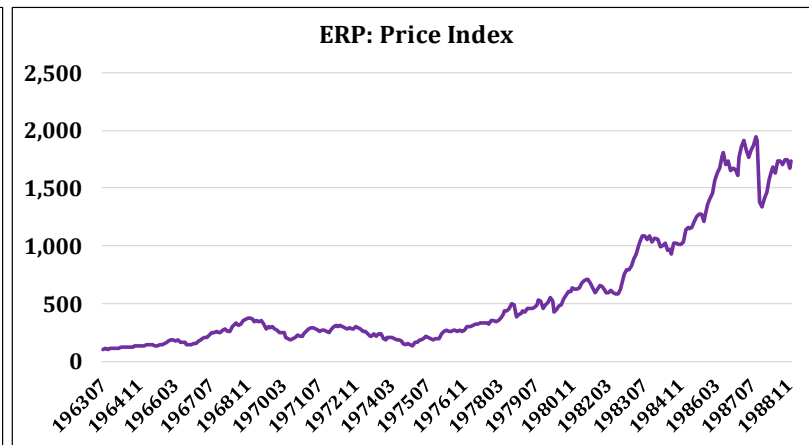
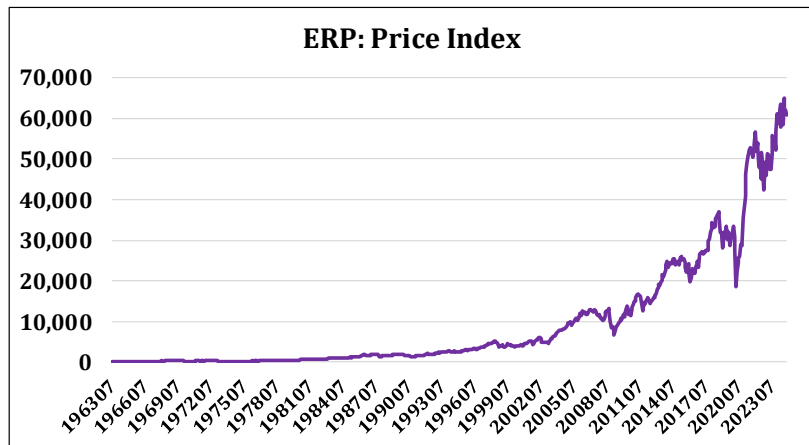
Charts 9-12: Chart Set for PRICE (Raw and Log Scale)

BASE PERIOD : 1963-06

BASE VALUE: 100

FULL SAMPLE (1963-

SUB-SAMPLE (1963-1988)



ERP data is only given in returns form. The structural model will contain risk factors macroeconomic and market structure based on their effect may not be found in month over month return differences in portfolio. Having the “price” of the portfolio can help address this issue. Using the monthly return data from ERP and setting a base price and period in the time frame 1 observation before historical data begins (Base period is June 1963), we can calculate the price of the portfolio in each period. The charts on the left show the portfolio’s price over the complete sample. It steadily trends until 2005 before the great recession, then increases at a faster rate post 2008. The two charts on the right-hand side are shown as a way to better visualize the price movement over the first 25 years of the sample. There is significant volatility throughout the period, but it is invisible over the entire sample.

Structural Regressions

The empirical approach used here examines multiple relationships that are central to asset returns. Each relationship incorporates fundamental economic theory. They are listed below:

$$Price_t = f(Investment\ Factor, Credit\ Spread, FED\ Policy, Volatility) + \varepsilon_t \quad (1)$$

$$VIX_t = f(Credit\ Spread, Macro\ Factor , Equity\ Returns , Exogenous\ Shock) + \varepsilon_t \quad (2)$$

$$AAA_t = f(YTM\ US\ Treasury\ Bond) + \varepsilon_t \quad (3)$$

Figure 3: Structural Regression 1 results

$\Delta_{12}Log(PRICE_t) = \alpha_0 + \beta_1\Delta_7CMA_{t-12} + \beta_2CS_t + \beta_3\% \Delta_{12}M2_{t-6} + \beta_4VIX_t + \varepsilon_t$					
Coefficients	-0.04	-0.004	-0.16	0.01	-0.01
t-statistics	6.11	-1.61	-2.81	2.69	-3.52
Estimation Technique	Least Squares				
Data Frequency	Monthly				
Estimation Range	Jan 1990 to Apr 2025 (424 observations)				
Net <u>d.f.</u>	419				
Resid Unit root test	-5.06				
Residuals ACF(1)	0.89				
Seasonal dummies	N/A				

Figure 4: Structural Regression 2 results

$$VIX_t = \alpha_0 + \beta_1 CS_t + \beta_2 \Delta IFD_t + \beta_3 TP_{t-9} + \beta_4 \Delta_{12} \log(PRICE_t) + \beta_5 SPIKE_{2020_t} + \varepsilon_t$$

Coefficients	8.08	13.37	1.41	-1.27	-13.11	28.52
t-statistics	5.03	9.54	2.56	-2.36	-2.87	14.42
Estimation Technique	Least Squares					
Data Frequency	Monthly					
Estimation Range	Jan 1990 to Apr 2025 (424 observations)					
Net d.f.	418					
Resid Unit root test	-6.70					
Residuals ACF(1)	0.80					
Seasonal dummies	N/A					

Figure 5: Structural Regression 3 results

$$AAA_t = \alpha_0 + \beta_1 TBTY_t + \varepsilon_t$$

Coefficients	1.63	0.90
t-statistics	8.11	29.23
Estimation Technique	Fully Modified Ordinary Least Squares (FMOLS)	
Data Frequency	Monthly	
Estimation Range	Feb 1962 to Apr 2025 (759 observations)	
Net d.f.	757	
Resid Unit root test	N/A	
Residuals ACF(1)	0.97	
Seasonal dummies	N/A	

Specification Challenges:

The functional form behind asset returns is not known with certainty. The literature tells us that it is a multi-factor model (Ross, 1964), but initial specifications of this relationship failed to capture the effects posited by theory and supported in the empirics.

Initially, the dependent variable was the 1-month excess return of this portfolio. The 1-month excess return showed in percent form showed sensitivity to the firm-factors contemporaneously (as expected). I was aiming for this structural regression to capture effects outside of the well-known Fama French factors. The excess return over 1 month failed to capture the effects of macroeconomic factors that affect asset prices over time.

To improve specification, the portfolio's "price" was derived from the 1-month excess returns, logged, then differenced on an annual basis. This transformation methodology

maintains the “excess return” measurement, removes some of the volatility and noise that is found in 1 month returns data. Using the 12-month Log Price difference of the portfolio instead of its 1-month return helped capture the longer-term effects found in other risk factors.

Firm characteristics were difficult to specify for this individual portfolio. The Momentum effect (MOM) for stocks in the literature, was not found for this portfolio along other risk factors. The investment effect was not found for this portfolio on a standalone basis. The Investment factor was present after a 12-period lag and examining its difference from 7 periods prior to that.

Credit Risk and Market Volatility yielded the expected results.

Other macroeconomic factors like the discrepancy between expected inflation and actual inflation failed to produce a statistically significant effect on this portfolio’s price change. This may have been because macroeconomic “surprise” news typically has a meaningful effect on asset prices “in the moment.” When the news is released, the markets update their expectations immediately. When studying this relationship with Year-on-Year data, those expectations may already be “priced in.”

How to best model volatility in the data was an important consideration as well. Volatility is apparent from looking at the time series on a chart or reading the summary statistics. However, volatility is not an idiosyncratic phenomenon to equity assets. Each asset has a level of volatility that is unique to it, but volatility is also a persistent, priced, market-wide phenomenon. Volatility’s place in the model should also be consistent with theory behind asset returns.

ARCH in-mean and GARCH models were estimated in an attempt to “find” best model of volatility for the forecasting.

Figure 6: Structural Regression 1 GARCH results

$\Delta_{12} \text{Log}(\text{PRICE}_t) = \alpha_0 + \beta_1 \Delta_7 \text{CMA}_{t-12} + \beta_2 \text{CS}_t + \beta_3 \% \Delta_{12} \text{M2}_{t-6} + \varepsilon_t$				
Coefficients	22.7	-0.003	-0.21	0.001
t-statistics	6.11	-1.75	-18.76	0.59
$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \varepsilon_t$				
Coefficients	0.004	0.93	-0.16	
z-statistics	4.63	6.08	-0.20	
Estimation Technique	Maximum Likelihood - ARCH			
Data Frequency	Monthly			
Estimation Range	Jan 1986 to Apr 2025 (472 observations)			
Net d.f.	464			
Resid Unit root test	-2.26			
Residuals ACF(1)	0.85			
Seasonal dummies	N/A			

The GARCH model allows us to examine the effects of inertial volatility in the data, and how the data behaves to recent shocks. Recent shocks given by ε_{t-1}^2 appear to be significant for this portfolio, suggestive that historical shocks raise volatility in the current period. Suggestive that a shock to portfolio increases the likelihood of more, but the inertial volatility σ_{t-1}^2 is not significant. Suggestive that this portfolio's historical volatility does provide a good measure into the future volatility.

The conditional variance equation from an ARCH In-Mean model provides results consistent with GARCH. However, the GARCH-M term does not show statistical significance, suggesting that this portfolio's conditional volatility doesn't imply a risk premium statistically different from zero.

Figure 7: Structural Regression 1 ARCH In-Mean results

$\Delta_{12} \text{Log}(\text{PRICE}_t) = \alpha_0 + \beta_1 \Delta_7 \text{CMA}_{t-12} + \beta_2 \text{CS}_t + \beta_3 \% \Delta_{12} \text{M2}_{t-6} + \beta_5 \sigma_t^2 + \varepsilon_t$					
Coefficients	0.30	-0.002	-0.22	0.0008	0.008
t-statistics	22.4	-1.75	-18.69	0.58	0.03
$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_2 \sigma_{t-1}^2 + \varepsilon_t$					
Coefficients	0.004	0.94	-0.01		
z-statistics	4.62	6.07	-0.20		
Estimation Technique	Maximum Likelihood - ARCH				
Data Frequency	Monthly				
Estimation Range	Jan 1986 to Apr 2025 (472 observations)				
Net d.f.	464				
Resid Unit root test	-27.95				
Residuals ACF(1)	-0.03				
Seasonal dummies	N/A				

Looking at the residuals from each model (SR #1, SR #1 ARCH In-Mean, and GARCH) we see that structurally modeling volatility with the VIX explains more variability than the dependent variable. This is evidence in favor of moving forward with VIX in my regression. An Augmented Dickey Fuller test of the residuals in GARCH fails to provide evidence suggesting that they are stationary at the 10% level.

Table 9: Descriptive Statistics of Structural Regression 1 residuals

Residual Diagnostics	Mean	SD
Structural Volatility Model	6.3497E-17	0.1632
ARCH In-Mean	-0.0059	0.1786
GARCH	-0.0059	0.1786

Causality & Functional Form:

$$Price_t = f(CMA, CS, M2, VIX) + \varepsilon_t \quad (1)$$

1. **(CMA)** When the spread between performance of companies who invest conservatively vs aggressively widens, the market perceives aggressive companies as lagging its competitors and reallocates towards higher yielding assets. If conservative and aggressive companies are generating the same profit and trade and the same price, the conservative companies will generate a higher return through its ability to execute share buybacks and issue dividends – generating a higher return over time and widening the spread. If the companies have the same profit – the aggressive company will have to issue stock and dilute shareholders to maintain its current profit.
2. **(CS)** Credit Spread: A widening Credit Spread serves as a signal to equity investors that the Cost of Capital for firms is increasing. Thus cash flows will be harder to maintain, and future profits will shrink. This negatively impacts smaller growth companies who typically have to invest heavily to maintain profitability or grow. When BAA rises investors reallocate towards higher yielding assets. When AAA falls reallocate towards safer assets.
3. **(M2)** M2 Money Stock: Growth in the Money supply originates from the FED and flows to its primary dealers. The primary dealers loan this new money to firms and investors. Their Cash-In-Advance constraint is relaxed, and they are now holding enough cash they require to purchase assets, raises stock prices in the market.
4. **(VIX)** Implied Volatility of SPX: An Increase in the implied volatility of the returns for S&P 500 companies decreases the returns for small growth stocks. Market volatility is a signal to investors that the likelihood of extreme movements up/down is high. Demand for downside protection INCREASES (Put Options), and investors buy protection against further downward movements.

$$VIX_t = f(CS, IFD, PRICE, SPIKE_ITV) + \varepsilon_t \quad (2)$$

1. **(CS)** Credit Spread: An increase in the Credit Spread between risky and safe corporate bonds will serve as a signal of increased risk in the equity market or higher return in the bond market. In either case, this motivates risk averse investors to reallocate their capital *away* from risky equity assets into safer assets leading to increased volatility in equity markets.
2. **(IFD)** Inflation Discrepancy: An increase in the difference between expected inflation and actual inflation serves as a “surprise” in the macroeconomy that equity investors were not anticipating and the volatility of future equity market returns increases.
3. **(PRICE)** Equity Returns: Increasing equity returns serve as a signal to investors that broad market sentiment is improving. Investors forego the cost of protection against

downside market performance (S&P 500 “Put” options). The lower demand in downside protection is priced into the VIX.

4. (SPIKE_2020) ITV: An intervention variable in the form of a Spike in time period 2020 M03 was included to control for an unanticipated, exogenous shock to equity markets in the form of a global pandemic.

$$AAA_t = f(TBTY) + \varepsilon_t \quad (3)$$

1. (TBTY) Yield to Maturity 10-Year US Treasury: An Increase in the 10 Year US Treasury will increase the Yield on a AAA rated corporate bond as the corporate bond is priced as a spread over a benchmark asset that is a key bellweather in financial markets.

Multi-Equation Model:

Asset returns are a complex phenomenon. A single regression model can provide insight into how they are related to a set of risk factors, but when those factors are interrelated, with causal factors also working in the opposite direction, the single regression does not have the capacity to capture this. Working under the multi-equation model framework allows us to strategically construct a system of equations where key financial indicators are treated as jointly determined endogenous variables. Each equation is designed to represent a key relationship underlying financial markets, and can expand the model's overall ability to capture various relationships among the various financial indicators.

Figure 8: Flowchart

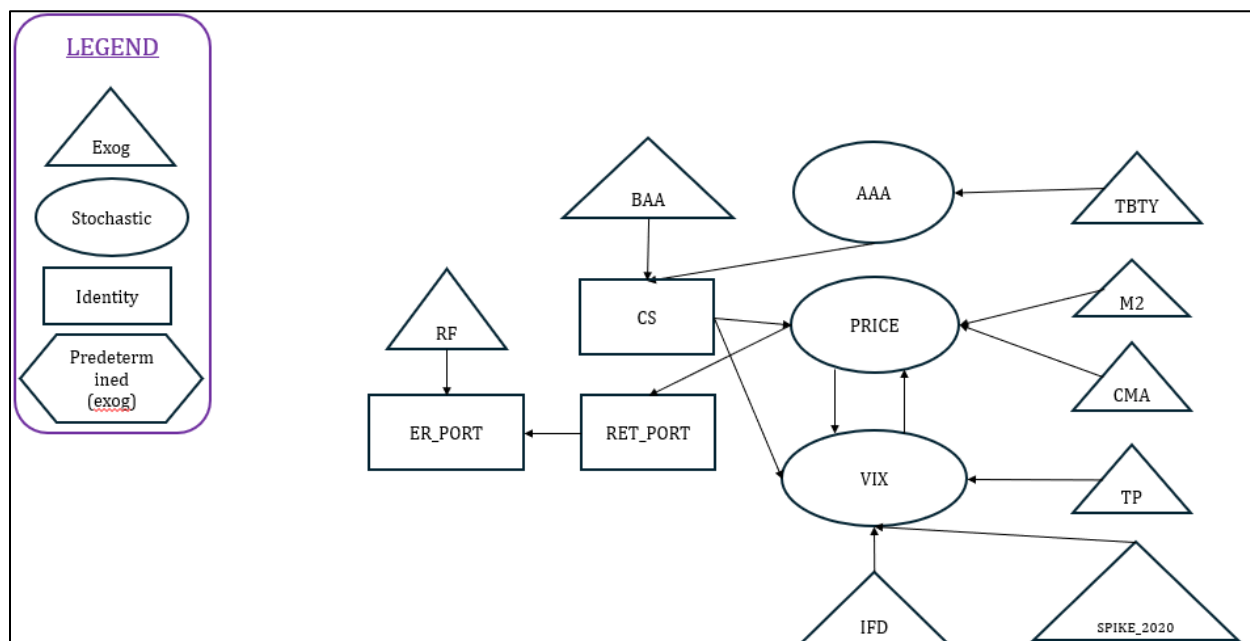


Figure 8b: Functional Form of Structural Multi-Equation Model

Structural:

$$Price_t = f(Investment\ Factor, Credit\ Spread, FED\ Policy, Volatility) + \varepsilon_t \quad (1)$$

$$VIX_t = f(Credit\ Spread, Macro\ Factor, Equity\ Returns, Exog\ Shock) + \varepsilon_t \quad (2)$$

$$AAA_t = f(YTM\ US\ Treasury\ Bond) + \varepsilon_t \quad (3)$$

Identity:

$$Return_t = \frac{\log(Price_t) - \log(Price_{t-12})}{\log(Price_{t-12})} \quad (4)$$

$$Excess\ Return_t = Return_t - RiskFree_t \quad (5)$$

$$Credit\ Spread_t = BAA_t - AAA_t \quad (6)$$

The MEM's construction captures endogenous relationships between a Portfolio of stocks (Price), Market Volatility (VIX), and Corporate Bond yields (AAA). Identity relationships in this model are given by the Credit Spread, Return of the portfolio, and Excess Return of the portfolio. Before discussion of the forecasting results produced by this model it is necessary to talk about the forecast results of the exogenous variables supporting MEM.

Exogenous Forecasts in Support of Baseline Forecast

Forecasting the model's exogenous variables provides the endogenous variable set a non-arbitrary "reality" to live in. Exogenous variables in this model were forecasted using univariate inertial techniques with long lag structures. This forecasting methodology ensures that the exogenous variables take on a path that is absent of unusual or unanticipated shocks while simultaneously capturing realized historical patterns in the data.

Charts 13-20: Chart Exogenous OOS Forecasts

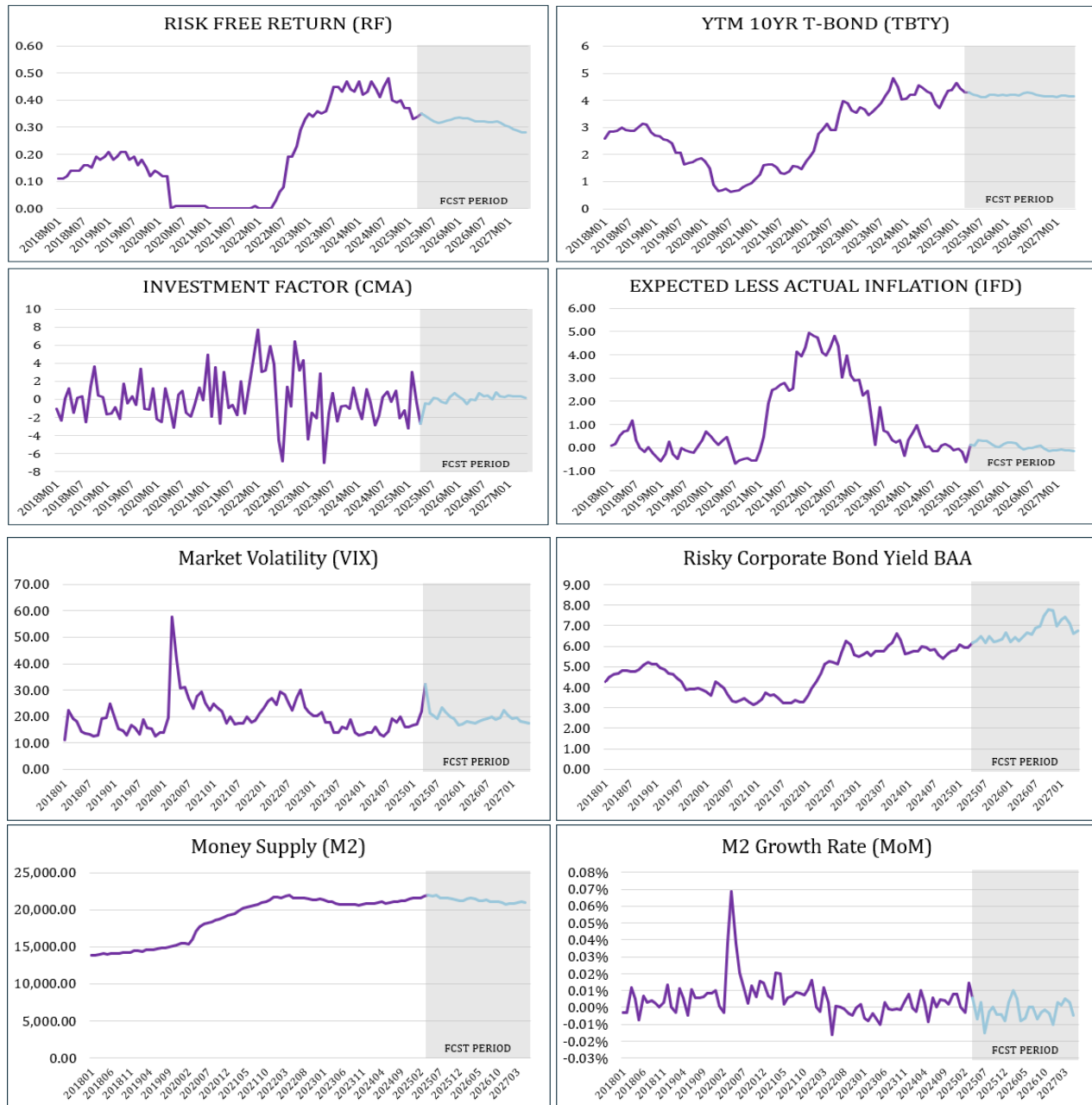


Table 10: ARIMA modeling support of Structural model

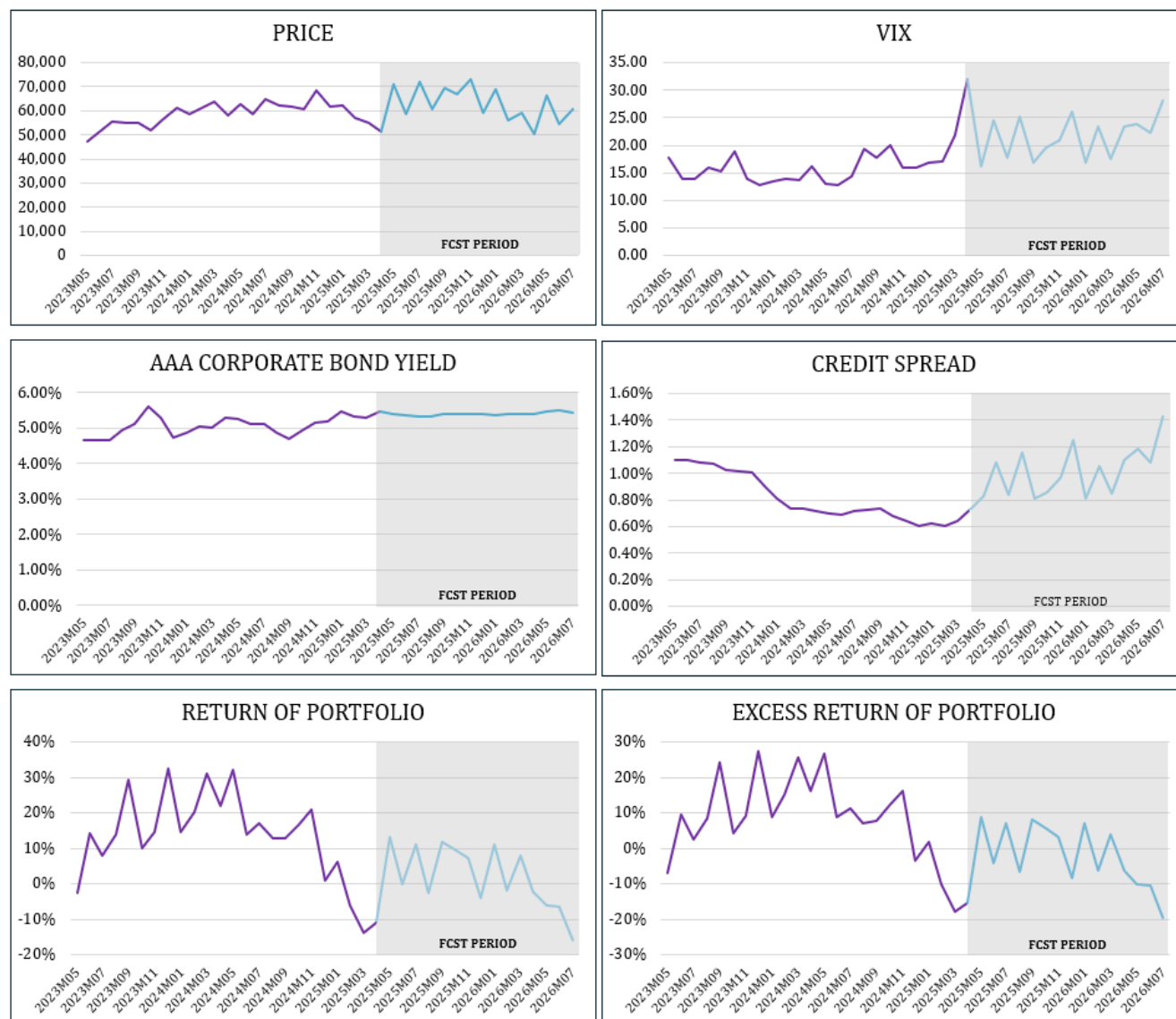
Variable Name	Variable Code	ARIMA Model	Q-Statistic (6 th Lag)
Term Premium	TP	ARIMA ((1,75),1,(1,70))	1.69
Inflation Discrepancy	IFD	ARIMA ((6,37),1,(6,50))	7.85
Yield to Maturity 10yr US Treasury	TBTY	ARIMA ((1,75),1,(1,70))	3.96
Risk Free Return	Risk Free	ARIMA ((1,25),1,(1,25))	8.65
Conservative Minus Aggressive	CMA	ARIMA ((1,10),0,(11,41))	8.72

Each exogenous variable's out of sample (OOS) trajectory is reflective of the historical dynamics in its time series and/or current events & trends existing in the macroeconomy right now. The OOS trajectories do not contain unusual shocks that could distort a baseline forecast.

Multi-Equation Model Baseline Forecast

The *baseline* should be thought of as “business as usual.” In the context of financial markets, as well as this forecasting model, there are several items that need careful consideration. In general, the US financial markets are not especially volatile, but they are not without volatility. Post the Great Financial Crisis of 2008, VIX has been more volatile on average. Even without the presence of a crisis, there have been times where market volatility has been especially high. Higher volatility, and bursts of volatility accurately characterize a large financial market of this size. The US Central Bank regularly reviews its policy stance, trying to guide the American economy towards prosperity. Pre-2008, the FED paid close attention to indicators like the “Taylor Rule” to set monetary. The baseline assumes the FED continues to guide the money supply in a way to keep the economy growing, but not in a way to over-stimulate inflation to avoid/mitigate a recession. Credit ratings agencies like Fitch & Moody's play the important role of evaluating the credit quality of firms who borrow money. Recent reporting from these agencies highlight that the quality of risky companies is being downgraded, and safer firms are not being upgraded quality. Not in startling or rapid fashion, but slowly over time. The baseline assumes the ratings agencies continue to report on this faithfully, and that this recent trend continues. As a result, the baseline assumes that investors become aware of this, and riskier corporate bonds require a higher rate of return over time. A “normal” US Financial market is not one with a lack of news, movement, price changes, policy decisions, and re-evaluation. These are the characteristics of strong, efficient, interconnected market.

Charts 21-26: Baseline Scenario Forecast results



Price: This asset experiences near-term appreciation over the first half of the forecast period. The month over month price change shows significant volatility-- consistent with its observed volatility observed over the 62-year historical period. The portfolio falls 19% in value from the start of the forecast to end under the baseline scenario.

VIX: The end of the historical period saw a spike in volatility. The baseline assumes that this is a "burst," and the VIX falls back towards the average value for nearly 12 periods out of sample. The final 10 periods of the forecast show an increased level of market volatility, 4 of which would be considered "highly volatile" for a monthly measure. This heightened level of volatility coincides with heightened bond yields for BAA corporates.

AAA: The AAA Yield trended upwards mildly during the historical period, and levelled off in the forecast. The baseline is projecting that AAA corporate yields will increase by 40 basis points when compared to the 2 years prior.

Credit Spread: This variable was historically, not volatile. The baseline forecast carries on the uptrend from the end of the historical period, but in more volatile fashion.

Return of the Portfolio: Both the historical and forecast periods show the volatility of this portfolio's annual return. The Historical period nicely starts in a month where the portfolio's value was nearly the same 12 periods earlier. Over a year's time the portfolio painfully climbs upwards, then falls down two times. The forecast shows painful stagnation for 12 months before a sharp slide downwards in the final half.

The excess return of the portfolio is very attractive in the historical period, and unattractive in the forecast period.

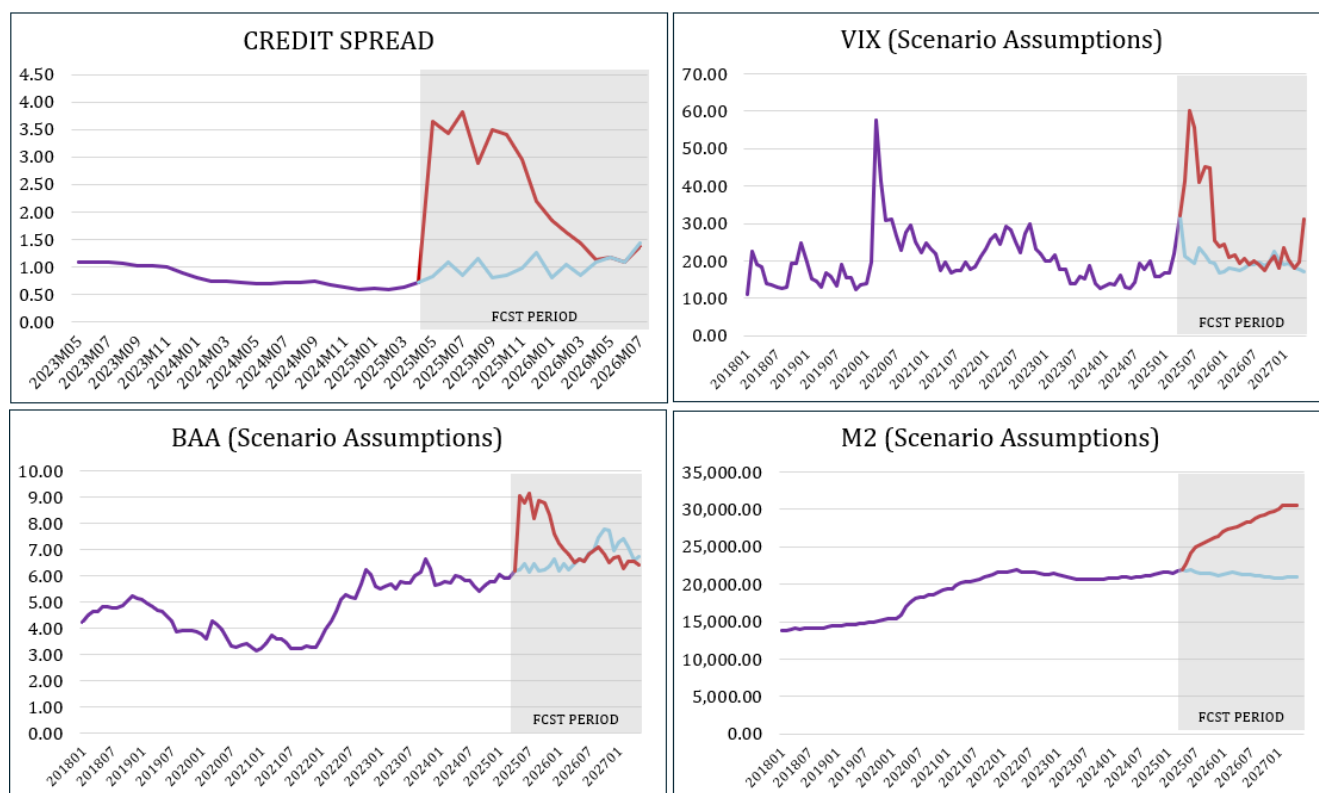
Multi-Equation Model Alternate Scenario Forecast

The *alternate scenario* forecasted is a "financial crisis." In these events, the impact on financial markets are often harsh and immediate. The infamous crisis of 2008 saw business failure in multiple sectors simultaneously (banking, industrial, and entertainment). The markets were not ignorant of this. On September 29th, 2008, the SPX and NASDAQ printed single day returns of -8.8% and -9.1% respectively. This portfolio experienced a 20% decline between September and October that year. And BAA corporate bonds spiked over 240 basis points. Broad based indices, individual stocks, and corporate credit all responded to the event in dramatic fashion. A financial crisis typically has consequences that impact *multiple* areas of the markets, but it would be an incomplete characterization if we only consider the market's reaction.

These events are also characterized by the policy response. Monetary policy rapidly shifted to measures intended to stimulate the economy. The FED moved to a near-zero interest rate level and kept that policy in place for nearly 10 years. They believed interest rate cuts were not enough to bring the economy back to its previous strength, and they began increasing the money supply through large-scale purchases of mortgage-backed securities and long-term US Treasuries. The Fiscal policy response included multiple stimulus packages designed to provide relief to those impacted. They implemented a "Troubled Asset Relief Program" (TARP) to bail out troubled banks, automakers, and insurance to the tune of over \$700 Billion and an "American Recovery and Reinvestment Act of 2009 (ARRA) of over \$787 Billion for infrastructure projects, health care system, and unemployment benefits. Crisis struck again in March 2020 when the government recognized the spread of the coronavirus as a "global pandemic" and mandated its citizens and businesses to stay indoors to abate further disease. BLS data on non-farm payrolls showed job losses of over 22 million between March and April 2020 (BLS, 2021). This crisis led to the bankruptcy of several retailers and it is estimated that 34% of all small businesses in the US closed (Ghosh, 2021). This level of economic displacement and extinction was matched by over \$5 Trillion in government expenditures (Dean, 2022),

and a 42% growth in the money supply. History has proven that financial markets and policy makers respond to crisis in major ways.

Charts 27-29: Baseline Scenario (Blue) vs Alternate Scenario (Red) Forecasts Scenario Forecast Assumptions



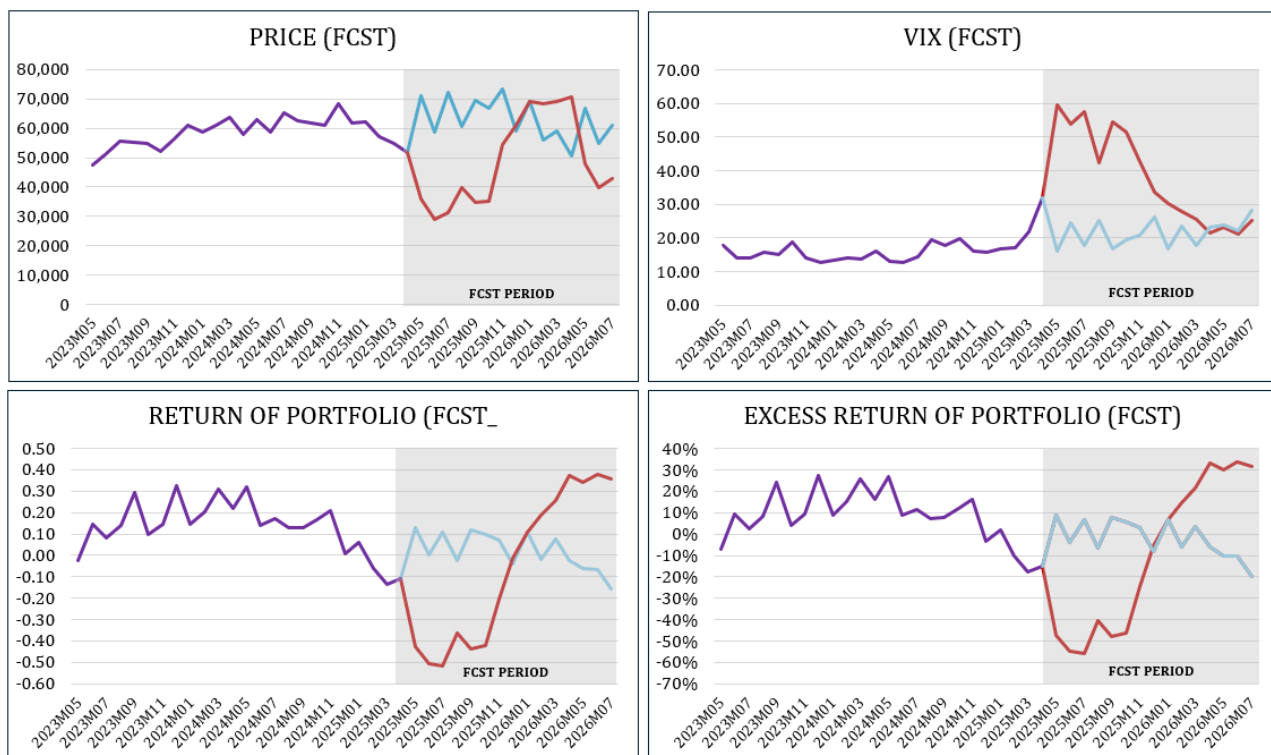
Credit Spread: In the financial crisis scenario, the spread between risky (BAA) and safe (AAA) bonds widens sharply. The baseline and scenario intersect and track closely to one another starting in 2026M05 then diverge before finishing the forecast period less than 150 basis points apart.

VIX: The Baseline scenario assumes volatility falls from over 31 and does not severely spike again over the forecast period. The OOS baseline average is 19.24. The Financial Crisis scenario assumes a level of volatility similar to the Covid-19 pandemic. Spiking to 60 in the second OOS period. Then falling back to historical levels. There is a smaller spike at the end of the OOS period due to “aftermath” events that have historically occurred in financial crisis scenarios. In 2011, S&P Global downgraded US sovereign credit from AAA to AA+. This is the most minor rating change possible, but it was the first time the US was below the AAA rating. This crisis assumes that a later, smaller, spike will be attributable to some government policy or credit re-rating event in reaction to the aftermath of a financial crisis.

BAA: In the crisis scenario, risky corporate bond yields immediately spike to historically high levels (over 9%). They do not stay there permanently. The corporate bond market eventually calms down, and yields fall back towards their previous levels.

M2: In the crisis scenario, Money supply takes a similar path to the COVI-19 pandemic. It immediately jumps 3% then 6% MoM, then gradually increases over the remaining forecast period. From the start of the forecast to the finish, M2 increases 38% overall. The baseline scenario does not assume extraordinary activity. April 2025 is the all-time-high value for M2. With no major economic impetus, the FED chooses not to make drastic changes in either direction. M2 stays around \$21 Trillion over the forecast, and MoM changes do not exceed 1% growth or 1.5% contraction.

Charts 30-33: Baseline Scenario (Blue) vs Alternate Scenario (Red) Forecasts Scenario Forecast Results



PRICE (FCST): The Portfolio dramatically falls in first couple OOS periods (30% then 20% MoM) under the financial crisis scenario. An interesting pattern appears where large spikes and large drawdowns occur in similar magnitude 12 periods apart. Spikes of over 50% occur in 2025M11 and 2026M11. Drawdowns of 32% and 37% occur in 2026M05 and 2027M05 respectively. The portfolio price OOS forecast in 2027M05 finishes .30% higher than the end of the historical period 2025M04. The ending value of the portfolio under the financial crisis is 11% lower than the baseline forecast (51.7K vs 57.4K respectively).

VIX (FCST): Under the Financial crisis scenario, the VIX is forecast to enter a prolonged period of high levels with an average forecasted value of 50 in the first 7 OOS periods. This slowly subsides over the next 7 periods but does not return to the post 2008 average value of 19. Historically, “post-crisis”, the average market volatility is about 20% higher than realized historical volatility post 2008 (VIX reading of 23 in scenario vs 19 historical). The Baseline forecast produces a VIX that is bouncing between 18 and 23 for the first 14 periods OOS, then spikes to 49 over 4 periods as the BAA bond rises in levels. After the spike, volatility remains at elevated levels between 38 and 23.

Competing Forecasts

Making decisions based on a single forecast result is an option, but with the technology and resources available to today’s analysts it is unadvisable. Competing the results from multiple forecasts is a smarter way to make decisions. Having multiple results allows analysts to validate their performance, gain insight into the amount of uncertainty in their model, and capture relationships in their data.

Vector Autoregression is another form of simultaneous estimation that we can use to estimate a set of endogenous variables. This provides an alternative framework for forecasting time series data. It is less restrictive than a structural model because it does not impose theoretical relationships between the endogenous variables. The structure of the VAR can take multiple forms. Decisions made about the lag length and estimation technique can impact the forecast result. Incorporating a long lag series in a VAR can capture persistent relationships in a time series and reduce autocorrelation in residuals, but it also adds estimated coefficients (estimation error). A highly parameterized model will also be more sensitive to individual data points and is prone to picking up the data’s “noise.”

An analyst might also choose to estimate Bayesian techniques over Ordinary Least Squares. Bayesian techniques do not focus on minimizing the squared residuals, rather it selects a prior probability distribution of the data and applies shrinkage to the coefficient estimates to pull them towards a value of zero. Analysts can tune these hyperparameters to adjust the influence that higher order lagged values and exogenous variables have on the endogenous variables.

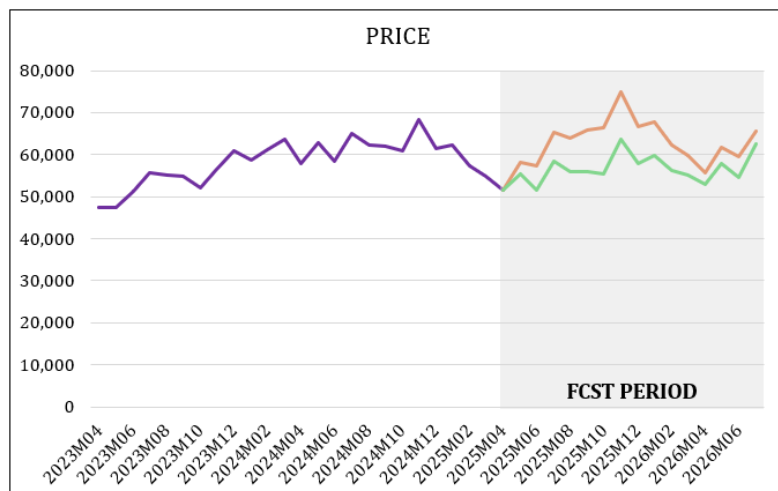
Table 10: Competing VAR #1

Endogenous variables: PRICE, VIX, AAA Exogenous variables: CS, CMA				
VAR_1	Stable? (Y/N)	Lag-Length Used	Variable Form	# of Estimated Parameters (% of Sample)
<u>Estimated VAR</u>	Y	14	Price (Diff) AAA (levels) VIX (levels) CS (levels) CMA (levels)	135 (32.9%)
<u>Estimation</u>	LS			
Criteria ->	<u>LR</u>	<u>FPE</u>	<u>AIC</u>	<u>HQ</u>
<u>Lag Length</u> <u>Criteria Results</u>	25	14	13	2

Table 11: Competing VAR #2

Endogenous variables: PRICE, VIX, AAA Exogenous variables: TP, IFD, M2				
VAR_2	Stable? (Y/N)	Lag-Length Used	Variable Form	# of Estimated Parameters (% of Sample)
<u>Estimated VAR</u>	Y	30	Price (Diff) AAA (levels) VIX (levels) CMA (levels) TP (Diff) M2 (Diff) IFD (levels)	282 (71.5%)
<u>Estimation</u>	BVAR			
Criteria ->	<u>LR</u>	<u>FPE</u>	<u>AIC</u>	<u>HQ</u>
<u>Lag Length</u> <u>Criteria Results</u>	30	30	30	13

Charts 34-35: VAR Forecast Results

**VAR 1 (LS)**

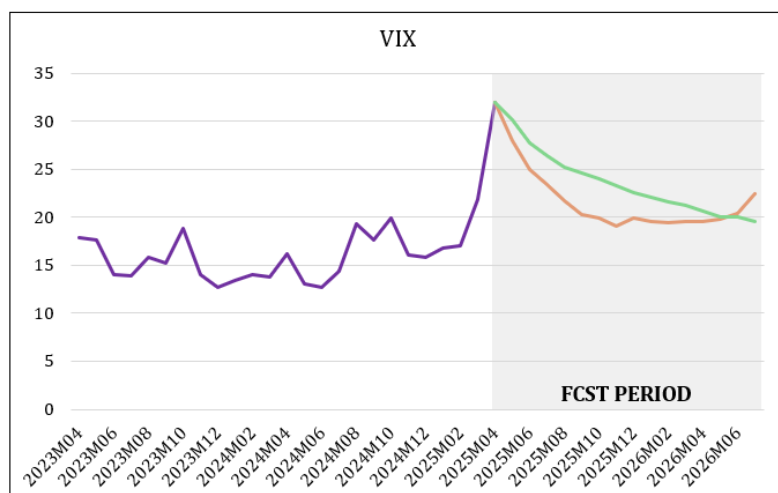
RMSE 13,297

MAPE 20.0

VAR 2 (BVAR)

RMSE 5,626

MAPE 8.3

**VAR 1 (LS)**

RMSE 9

MAPE 56.3

VAR 2 (BVAR)

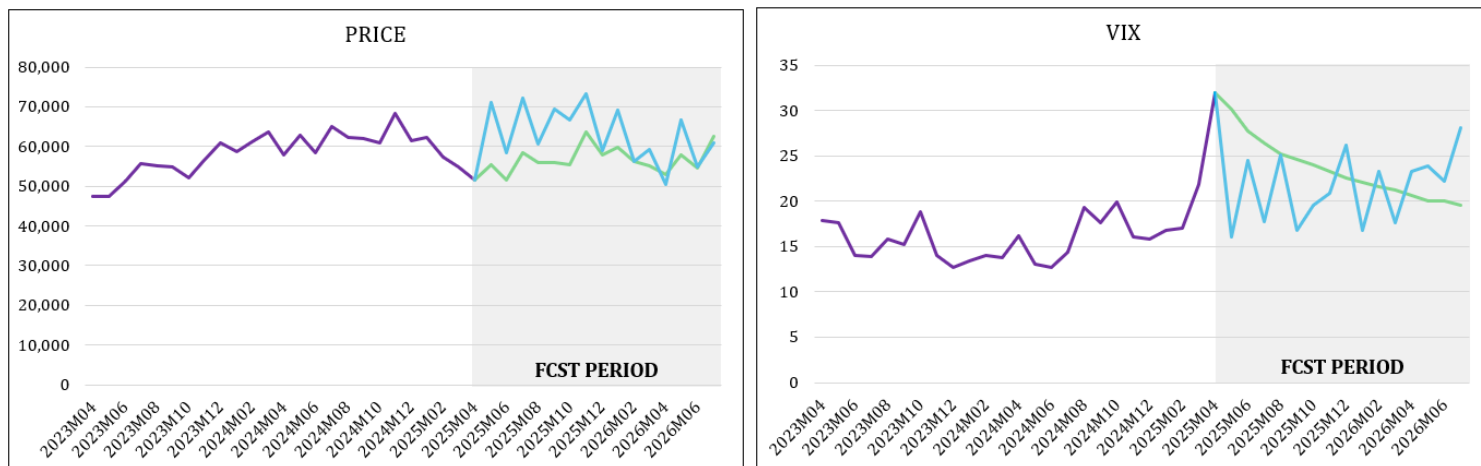
RMSE 4

MAPE 23.7

The VAR_1 (LS) model shows marginally better in-sample accuracy for PRICE, but its performance on VIX forecasts is worse. It is forecasting an atypical path that US stock market volatility prices do not follow. The model's tendency to overreact to price swings further suggests potential instability during volatile periods. In contrast, VAR_4 (BVAR) delivers substantially better volatility (VIX) forecasts – both in terms of magnitude and consistency – thanks to Bayesian regularization. While its PRICE accuracy is slightly lower, the smoother forecast path indicates less susceptibility to outlier shocks. Given the competitive accuracy for price and superior volatility forecasts, VAR_4 (BVAR) is a stronger candidate to challenge the baseline scenario, particularly when robust volatility modeling is important to decision-making.

VAR versus Baseline Forecast

Chart 36-37: Forecast Comparison Baseline Structural (Blue) vs VAR #2 (Green)



Competing ARIMA Forecast

A forecaster would want a competing ARIMA forecast for one of their dependent variables to gauge the variables' sensitivity to their historical values. Another desirable feature of univariate inertial modeling is that typically (not always) they are more parsimonious than a structural model. Resulting in a forecast build with less estimation error.

Chart 38: ARIMA Forecast of PRICE

ARIMA (2, 1, 2) + (0, 0, 0)



Methods Comparison

The analysis performed includes Structural, univariate inertial, and VARX. The exogenous variable set included in the structural come from economic theory. This predictor asserts a direction of causality that is supported by empirical research. Exogenous predictors asset the same causality as the structural model. The inertial model does not use economic theory for forecasting; it strictly relies on the time-series dynamics contained in the dependent variable.

From the structural model forecasts, we will only use the baseline forecast for analysis.

Charts 39: PRICE FCST Structural
MEM

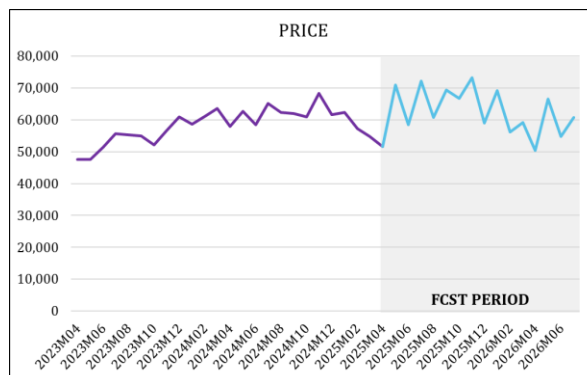


Chart 40: Price FCST Least Squares VAR

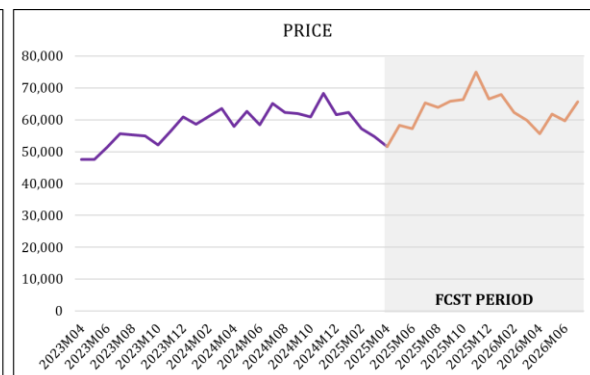


Chart 41: PRICE FCST Bayesian VAR

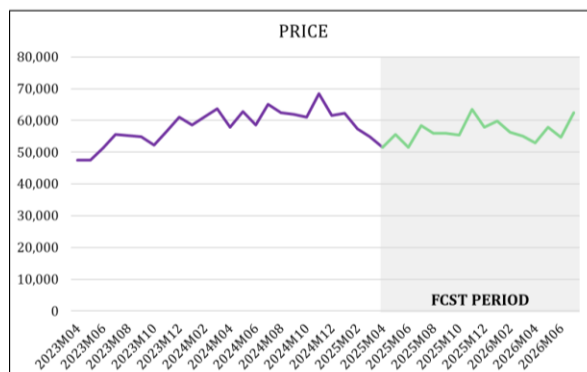
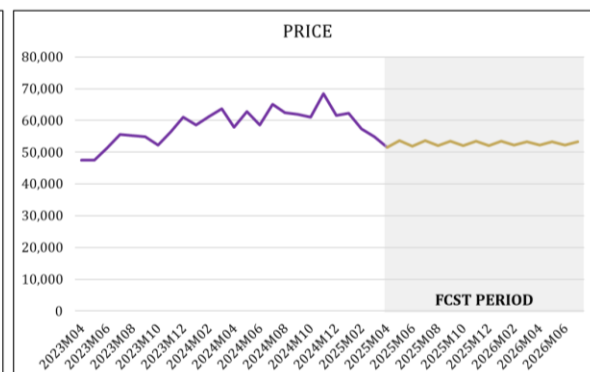


Chart 42: PRICE FCST ARIMA



Among the 4 techniques, the structural model produces the most volatile forecast, and the ARIMA produces the least volatile forecast. The Least Squares VAR produces the highest maximum price of 74,993 in November 2025. Each forecast utilizing exogenous variables forecasts new “all-time-high” high value in the portfolio in November 2025. The ARIMA’s peak forecast value is the first period out-of-sample (May 2025). The average month over month change in the out of sample range for the structural and

least squares VAR are both slightly negative (-.12% and -.41% respectively). In the Bayesian VAR and ARIMA, the month-over-month change is positive (2.55% and .17% respectively).

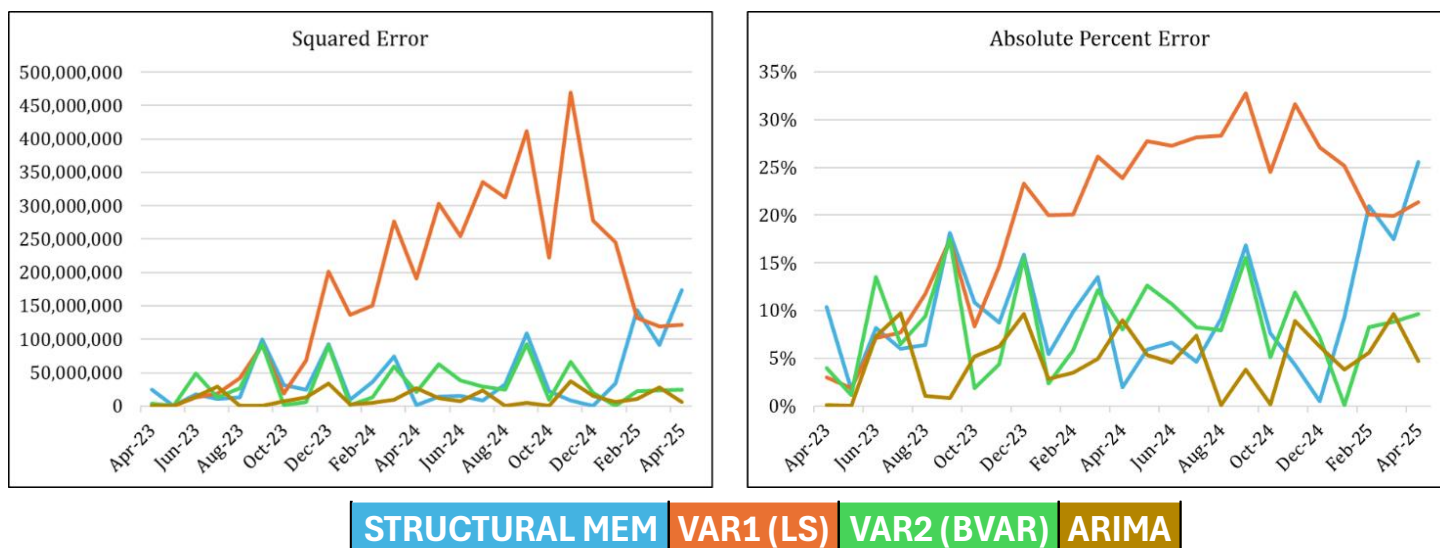
In Sample Error Analysis

An analyst would perform in-sample error analysis for multiple reasons. A major reason would be to gain a reading on the accuracy of their forecast over a certain period. Another would be to gain insight into how accuracy evolves/devolves over the sample period. Here, an analyst would aim to understand the prediction error and adjust their model if possible. If the analyst performed multiple forecasts, error analysis could be helpful in selecting a model for deployment or publishing their findings. Error analysis will be over the final 24 periods of my historical data set (April 2023 to April 2025). The error measurements this analysis is concerned with Root Mean Squared Error (RMSE) and Mean Averages Percent Error (MAPE). Mathematical computation of these error measurements are shown below.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (A_t - F_t)^2}$$

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

Chart 43-44: In Sample Error Measurement



Structural MEM

The squared error of the structural model is the 2nd lowest among the 4 forecasting methods. The overall RMSE over this forecast is 6,604 and MAPE is 9.9%. The error rate follows a pattern for most of the in-sample period. The error rate starts to spike in the 5th period out of sample, followed by two consecutive spikes 3 periods apart; then spikes again 5 periods later. In the first 16 periods in-sample the squared errors are lower than average, suggesting that this model is better at predicting in the near & mid-term. The final 7 periods in-sample have the highest error rate among any of the subsections (MAPE of 49%). If the final 7 in-sample periods were

excluded from the forecast the RMSE of the structural model would drop by 22% from 6,604 to 5,127 and fall by 90 basis points to 8.90%.

VAR 1 – Least Squares

The squared error of the structural model is the highest among the 4 forecasting methods. The overall RMSE over this forecast is 13,297 and MAPE is 20.0%. Like the structural model the first spike in the error rate comes in the 5th period in-sample. However, the pattern in the error is not discernable. After the initial spike, errors are consistently (and substantially) higher. This method exhibited a strong tendency to *underestimate* the actual price of the portfolio.

Additionally, spikes in the error were associated with a strong overreaction to a small drop in the price of the portfolio, or, not reacting to a large upward/downward movement in the portfolio.

VAR 4 – Bayesian VAR

The squared error of the structural model is the 2nd lowest among the 4 forecasting methods. The overall RMSE over this forecast is 5,626 and MAPE is 8.3% (still outperforming the structural model by 50 basis points if the final 7 high error periods were removed). The error pattern of the BVAR follows the structural model pattern but performing noticeably better in the final 7 periods in-sample. Like the least squares VAR, the BVAR also exhibited a strong tendency to *under* estimate the actual price of the portfolio. Additionally, spikes in the error were associated with a “spillover magnitude” or underreaction. When actual price of the portfolio fell for 2 consecutive periods, the BVAR would directionally follow it both times. In one of these instances the predicted % price change would match the actual % price change, but in the other period predicted % price change would be much greater than the actual % price drop. This occurred between July – August 2024 and January – February 2025 in sample.

ARIMA

The squared error of the ARIMA model is the lowest among the 4 forecasting methods. The overall RMSE over this forecast is 3,411 and MAPE is 4.83%. Spikes in this forecast consistently occurred between each 3-5 period subsection in-sample. Spikes in the error of this forecast were associated with a directional miss in the predicted price. ARIMA had a balanced count of predicted values under and over the actual value of the portfolio.

The error output from the individual techniques outputting results that are both higher and lower than the baseline forecast under the structural model. The other models also use different information sets to predict “price.” Often, we can combine techniques to create a forecast with reduced error than any one of the single methods.

Combination Forecasting

Combining the individual techniques should be done in some logically sound way to ensure that the combination forecast outputs more accurate results than any of the individual techniques. One way to go about this is to assign each technique an equal “weight” in its contribution to the combination. This method is humble, as it does not assert that one technique will perform “better” than another. An analyst can take the opposite stance and assign unequal weights to individual techniques: typically favoring the techniques with lower error rates in sample. The weights can be assigned in such a way that is “optimal” on a certain metric such as MSE, or MAPE.

Three different optimal weighting techniques were developed to produce optimal weight forecasts that we can compare to the equal weight technique.

Optimal Weight #1 – Measured each individual forecast methods contribution to the overall squared error of the forecast. Initially it assigned an equal (25%) weight to each technique. Any

individual technique that contributed > 25% of the overall share of the squared error was downweighted, and techniques with <25% of the error share were upweighted.

Resulting Forecast:

$$PRICE_t = .254(PRICE_{MEM_t}) + .064(PRICE_{VAR1_t}) + .264(PRICE_{BVAR_t}) + .388(PRICE_{ARIMA_t})$$

Optimal Weight #2 – Regressed the historical values of Price on each of the four forecasts of price (Structural, VAR 1, BVAR, ARIMA) and used their estimated slope coefficients as the forecast weight. Normalization was applied to the coefficient estimates so that the slope coefficients from the regression added up to 1 (i.e. 100% weight allocation). Unlike equal-weight or the Optimal Weight 1, this technique permits *negative* weights to be assigned to forecast techniques. The regression model for this optimal weight forecast took the following form:

$$PRICE_t = -1 + \beta_1 PRICE_{MEM_t} + \beta_2 PRICE_{VAR1_t} + \beta_3 PRICE_{BVAR_t} + \beta_4 PRICE_{ARIMA_t} + \varepsilon_t$$

The estimated Betas from this regression are the coefficients that minimize the sum of the squared errors between the actual value and the forecasted value.

Resulting Forecast:

$$PRICE_t = .208(PRICE_{MEM_t}) + .327(PRICE_{VAR1_t}) + (-.387)(PRICE_{BVAR_t}) + .851(PRICE_{ARIMA_t})$$

Optimal Weight #3 – Initially assigned equal-weights to the individual forecasts and applied a constrained optimization to minimize the combination forecast MAPE. The constraint mandated that no individual technique could receive less than a 1% weight allocation to the combination.

$$PRICE_t = .01(PRICE_{MEM_t}) + .094(PRICE_{VAR1_t}) + .01(PRICE_{BVAR_t}) + .886(PRICE_{ARIMA_t})$$

Optimal weighting methods #2 and #3 result in a high chance for one of the individual techniques to receive a dominant weighting allocation.

Methods and Combination Forecast Results

METHOD	RMSE	EW	OPT1	OPT2	OPT3
STRUCTURAL	6,604	25%	28.4%	20.8%	1.0%
VAR 1 (LS)	13,297	25%	6.4%	32.7%	9.4%
VAR 2 (BVAR)	5,626	25%	26.4%	-38.7%	1.0%
ARIMA	3,411	25%	38.8%	85.1%	88.6%
Combination (Equal Weight)	5,667				
Combination (Optimal # 1)	4,294				
Combination (Optimal # 2)	4,746				
Combination (Optimal # 3)	3,425				

METHOD	MAPE	EW	OPT1	OPT2	OPT3
STRUCTURAL	9.8%	25%	28.4%	20.8%	1.0%
VAR 1 (LS)	20.0%	25%	6.4%	32.7%	9.4%
VAR 2 (BVAR)	8.3%	25%	26.4%	-38.7%	1.0%
ARIMA	4.8%	25%	38.8%	85.1%	88.6%

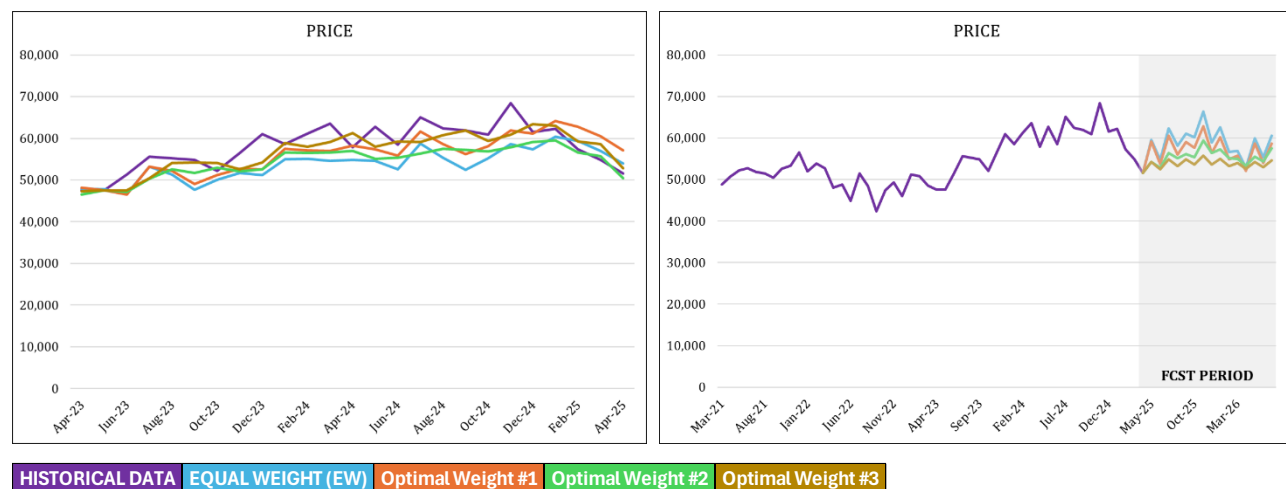
Combination (Equal Weight) 8.2%

Combination (Optimal # 1) 6.2%

Combination (Optimal # 2) 6.4%

Combination (Optimal # 3) 4.5%

Combination (Optimal #3) was able to outperform each of the individual forecasting techniques on the both accuracy metrics. This combination relied heavily on ARIMA. All four combination forecasts outperformed the other individual techniques. This combination forecast operated at the edge of the constraint observing that both the structural model and Bayesian VAR were shrunk to 1% allocation.



Each optimal weighting method performed better than the equal-weight combination by a meaningful margin on the error metrics. Judging the forecasts from a visually point of view, the equal-weight forecast looks like a more realistic forecast for this portfolio. As the Optimal weighting techniques converged towards an ARIMA dominated forecast, the volatility of price falls.

Each optimal weighting method performed better than the equal-weight combination by a meaningful margin on the error metrics. Judging the forecasts from a visually point of view, the equal-weight forecast looks like a more realistic forecast for this particular portfolio. As the Optimal weighting techniques converged towards an ARIMA dominated forecast, the volatility of price falls.

Out of Sample Forecast Conclusion

The best forecast is chosen after performing analysis from positive and normative perspective. Positively, the best forecast will be one that is highly correlated with history and competitively

accurate. Normatively, the best forecast will have a visual appearance that is logically consistent with what the analyst already knows about the concept in question.

The concept here is the price of a portfolio filled with capital assets. More specifically, capital assets that are small, modestly valued, and empirically found to be sensitive to multiple external risk factors. Michael Aked and Amie Ko of Researchaffiliates had a relevant comment to make regard volatility:

“The long-term volatility of asset returns is only known after the fact; over any finite time horizon, the true volatility remains an estimate with associated uncertainty.”

The Structural Model, while it ranks 3rd among the modeling techniques in terms of accuracy. Its volatility is closer significantly closer to this actual portfolio’s realized volatility than the other three forecast techniques. If we choose a forecast based on the volatility over a relatively short time horizon, then we may fail to capture the true characteristics of how small market cap

Chart 45: Best of Forecasts

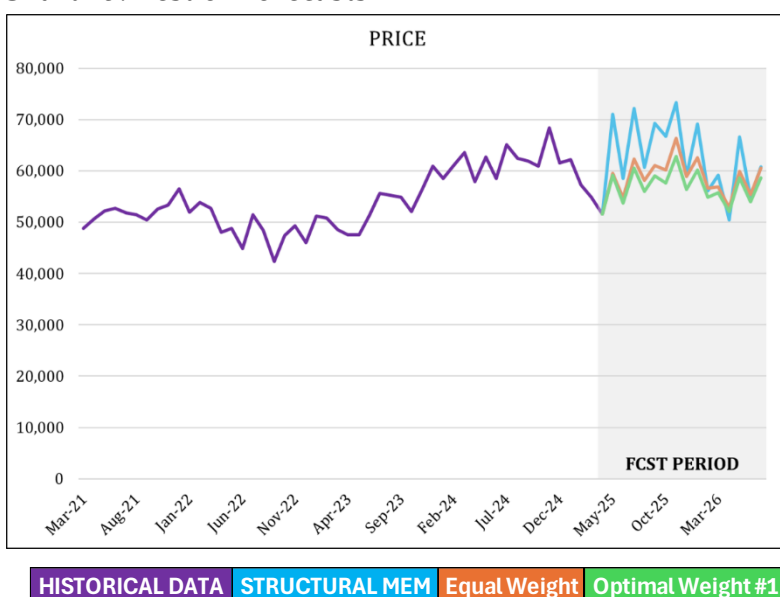


Table 12-13: SD Table

Standard Deviation of variable: PRICE	
In-Sample range (2023-2025)	5,221
2014 - 2025	13,652
2000 - 2025	16,654
1984-2025	16,385
Full History (1963-2025)	15,019

Standard Deviation of variable PRICE		Out of
Model-Based FCSTs	In Sample	Sample
Structural	7,080	7,037
VAR1 - Least Squares	2,395	4,939
VAR 2 - BVAR	5,140	3,256
ARIMA	5,462	699

stock portfolios behave. The Structural model provides a level of volatility over the in-sample and out of sample range that is *safer* to advertise to a client.

Takeaways

This analysis was produced after a thorough review of empirical literature, theoretical considerations, data gathering, programming, and consultations with faculty and classmates. Each area of the work process was necessary and required significant time and attention. Model based forecasting was allotted a majority of the content space; this area combined all steps of the work process I mentioned before. The structural model combined theory from macroeconomics, asset pricing, and financial markets before taking on an initial functional form. Not surprisingly, the initial functional form was incorrect. Re-specification and variable transformations on both the left- and right-hand sides were necessary before finding the empirically valid, final functional form of the model. The scenario forecasts that the structural model supported required special consideration and care given that there were multiple risk factors that played a relevant role in the baseline and alternate scenarios. Out of sample paths for M2, VIX, and BAA were forecasted at a monthly frequency to reflect theoretical behavior

under the scenarios. For both scenarios I evaluated current and historical patterns to choose paths that were realistic.

VAR based models required a similar, but less labor intensive, empirical rigor in finding the correct functional form. Some re-specification effort was required here. ARIMA modeling took far less effort to develop.

Error analysis of the model-based forecasts required programming in two software environments (R and Excel). Visual inspection of the error metrics created brief opportunity for model re-specification in the VAR based model. This process also offered insight into how the different models tend to forecast a particular variable.

The initial motivation for this research was to gain a better understanding of US equity markets, and how an individual might develop an effective strategy for securing their welfare over the long term. The US stock market is gigantic in terms of the amount of assets it holds, and the number of participants it attracts. And over the course of a century, it has transformed greatly. It has transformed in so many ways: how investment products are offered, how performance is tracked, and how its data is disseminated. The literature behind asset pricing revealed to me how the individual participants in the market have transformed as well. Initial research in this area (Markowitz & Ross) concerned themselves with a single asset being a single stock, but the research now frames the (Fama & French) “single asset” as a portfolio of stocks. Their findings were so impactful that they now influence the behavior of market participants both big and small. The philosophy of “Factor Investing” that Fama & French pioneered with their research was first put in practice on a large scale at their firm, Dimensional Fund Advisors. However, their philosophy has spread far across industry. Large investment banks and asset manager such as Blackrock, J.P. Morgan, and Fidelity now offer products that employ the factor investment strategy. What is particularly interesting is how their research may/may not shape the asset management & hedge fund industries in the future. The five-factor model is relatively new. There are many managers who do not adopt this strategy. The CFA institute published in 2021 that between 2008-2016 hedge funds failed to outperform the broad-based SPX index (Davis, 2021). It will be interesting to see if more firms adopt the factor investing philosophy in favor or passive indexing or stock picking.

If more time was available to work on this project I would try to find better data on the corporate credit market, and macroeconomic surprises. Researchers I had read in the literature review Han et al (2016) did some clever data engineering on credit spreads. A key strength of their paper was the data used, and their methodology. The team had access to Lehman Brothers and TRACE (Trade Reporting and Compliance Engine) databases which allowed them yield data on Credit Default Swaps (CDS), and Corporate Bonds over the short (1-2 years) to medium (4-9 years) term structure. The CDS market is also much more liquid than corporate bonds. A CDS will trade daily, whereas corporate bonds are usually held to maturity and sometimes not traded at all (Howard, 2007). This allowed them to construct detailed yield curves over the short to medium term (1 to 9 years), and regress the portfolio returns on the yield curve slopes. Having a credit market variable such as this would have added strength to my first structural regression.

My initial attempt at modeling the “macroeconomic surprise” effect in the first structural regression failed. The inflation forecast error did not capture a statistically significant effect in the value of the portfolio. Researchers Wachter & Zhu (2019) were

able to find evidence that macroeconomic announcements had a statistically significant impact on equity returns by constructing indicators variables for the *days* on which various data was shared (CPI, employment, and GDP). Finding a better way to model in positive or negative macroeconomic news that was not anticipated by markets likely would have strengthened by first structural regression.

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Appendix 2 – AI Search results

(Note: all AI searches were used with search engine named “Perplexity”

This search engine can be accessed at <https://www.perplexity.ai>)

Prompt:

what is the SMB and HML factors representative of in the Fama French model?

Response:

The Fama-French Three-Factor Model incorporates two additional factors beyond the market risk factor used in the Capital Asset Pricing Model (CAPM): SMB (Small Minus Big) and HML (High Minus Low). These factors aim to capture specific risk premiums associated with company size and value characteristics.

SMB (Small Minus Big)

The SMB factor represents the size premium in stock returns:

- It measures the excess return of small-cap stocks over large-cap stocks.
- SMB accounts for the historical observation that smaller companies tend to outperform larger ones over the long term.
- The factor is calculated by taking the average return of small-cap portfolios and subtracting the average return of large-cap portfolios.

A positive SMB value indicates that smaller companies are outperforming larger ones, while a negative value suggests the opposite.

HML (High Minus Low)

The HML factor represents the value premium in stock returns:

- It measures the excess return of stocks with high book-to-market ratios (value stocks) over those with low book-to-market ratios (growth stocks).
- HML captures the historical tendency of value stocks to outperform growth stocks in the long run.
- The factor is computed by taking the average return of high book-to-market portfolios and subtracting the average return of low book-to-market portfolios.

A positive HML value indicates that value stocks are outperforming growth stocks, while a negative value suggests the opposite.

Prompt:

What is the CMA factor supposed to represent?

Response:

The CMA (Conservative Minus Aggressive) factor in the Fama-French Five-Factor Model represents the investment premium in stock returns. Specifically:

Definition

CMA is the difference between the returns of firms that invest conservatively and firms that invest aggressively.

Interpretation

- Conservative firms are those that require little ongoing capital investment to maintain and grow their operations.
- Aggressive firms are those that direct more profit towards major growth projects.

Investment Premium

The CMA factor captures the observation that companies with lower investment rates (conservative) tend to outperform those with higher investment rates (aggressive) over time.

Calculation

CMA is computed as the average return on conservative investment portfolios minus the average return on aggressive investment portfolios.

Significance

- The inclusion of CMA in the Five-Factor Model aims to better explain stock returns by accounting for the relationship between a company's internal investment practices and its stock market performance.
- CMA, along with the profitability factor (RMW), was added to the original Three-Factor Model in 2015 to enhance its explanatory power.

By incorporating the CMA factor, the Fama-French model attempts to capture the tendency for companies that invest more conservatively to generate higher returns compared to those that invest more aggressively in the long run.

Prompt

Is there any US policy ongoing right now that might stress US corporations balance sheets in the future ? as soon as end of 2026?

Response:

Several US policies and regulatory changes currently underway could place stress on US corporations' balance sheets as soon as the end of 2026. Key areas of concern include tax policy shifts, monetary policy, tariffs, and regulatory requirements.

1. Expiration and Changes to Tax Cuts and Jobs Act (TCJA) Provisions

- Bonus Depreciation Phase-Out: The ability for businesses to fully expense most equipment purchases (bonus depreciation) is set to completely expire at the end of 2026. This will increase taxable income and reduce cash flow for corporations making large capital investments.

- **International Tax Provisions:** Starting in 2026, tax rates on provisions such as the global intangible low-taxed income (GILTI) tax, the base erosion and anti-abuse tax (BEAT), and the foreign-derived intangible income (FDII) tax will increase, raising the tax burden for multinational corporations.
- **Interest Deduction Limits:** Deductions for net interest expenses are now based on EBIT instead of EBITDA, making it harder for highly leveraged companies to deduct interest and potentially increasing their effective tax rate.
- **R&D Expensing:** Since 2022, deductions for R&D costs must be spread over several years, reducing immediate tax benefits for innovation investments.

2. Federal Reserve Monetary Policy

- **Quantitative Tightening (QT):** The Federal Reserve has slowed the pace of its balance sheet reduction but is expected to keep its balance sheet flat through at least mid-2026. This ongoing QT and potential for higher interest rates could maintain tighter financial conditions, increasing borrowing costs and pressuring corporate balance sheets.
- **Interest Rate Uncertainty:** While some rate cuts are projected, rates are expected to remain above pre-pandemic levels, which could increase debt servicing costs and stress companies with significant leverage.

3. Tariffs and Trade Policy

- **Tariff Hikes:** The current administration has signaled a willingness to impose steep, across-the-board tariff hikes. These tariffs could increase input costs, disrupt supply chains, and reduce profit margins for US corporations, especially those reliant on global trade.
- **Retaliatory Trade Actions:** The risk of trade wars and retaliatory tariffs from other countries could further stress corporate finances by reducing export opportunities and increasing costs.

4. ESG and Regulatory Compliance

- **ESG Disclosure Requirements:** New and pending regulations, particularly from states like California, require extensive climate and ESG disclosures. Compliance costs are expected to rise, especially for tracking scope 3 emissions throughout corporate value chains.
- **Federal and State Regulatory Shifts:** Ongoing regulatory changes at both the federal and state level may require additional reporting, operational changes, or investments, further pressuring corporate resources.

5. Small Business Contracting Rule Changes

- **SBA Recertification Rules:** Effective January 17, 2026, new Small Business Administration rules will change how small businesses maintain eligibility for federal contracts after mergers or acquisitions. This could reduce the attractiveness and value of certain businesses, impacting M&A activity and balance sheet valuations.

Prompt

which companies failed during the 08 financial crisis?

Response:

Company Name	Industry	Outcome/Notes
Lehman Brothers	Investment Banking	Largest bankruptcy in U.S. history; collapsed in Sept. 2008, sparking global panic[1][2][3][4].
Washington Mutual (WaMu)	Banking	Seized by federal regulators; assets acquired by JPMorgan Chase[1].
General Motors (GM)	Automotive	Filed for bankruptcy in June 2009; later restructured with government assistance[1][5].
Chrysler	Automotive	Filed for bankruptcy in April 2009; purchased by Fiat[5].
Bear Stearns	Investment Banking	Suffered a liquidity crisis and was acquired by JPMorgan Chase with Fed support[6].
AI&G (American International Group)	Insurance	Received a government bailout to prevent collapse; did not file bankruptcy[6][7].
Merrill Lynch	Investment Banking	Sold to Bank of America to avoid bankruptcy[8].
Fannie Mae & Freddie Mac	Mortgage/Finance	Placed into federal conservatorship[8].
CIT Group	Commercial Finance	Filed for bankruptcy in 2009[9].
LandAmerica Financial Group	Financial Services	Filed for bankruptcy in 2008[10].
Linens 'n Things	Retail	Filed for bankruptcy in 2008[10].
Six Flags	Entertainment	Filed for bankruptcy in 2009[5].

Prompt:

what are the current NYSE quartiles for companies Book Equity, Market Capitalization, and operating profitability

Response:

NYSE Quartiles for Book Equity, Market Capitalization, and Operating Profitability (2025)

Current detailed NYSE-wide quartiles for these financial metrics are not publicly released as a single tabulated list, but authoritative academic and industry sources provide standard methods and recent breakpoints for these variables based on wide-scale U.S. equity data.

Market Capitalization

- Quartile calculation: NYSE quartiles for market capitalization traditionally use the market equity of all companies listed at the end of June each year.
- Recent Benchmarks:
 - The median (50th percentile) NYSE market capitalization as of mid-2025 is near \$2.6b–\$2.7b; this defines the boundary between "small cap" and "big cap" stocks on NYSE.
 - Approximate 25th and 75th percentile cutoffs (based on academic data libraries and S&P methodology) are:
 - 25th percentile (Q1): Roughly \$0.9b–\$1.2b
 - 50th percentile (Median): \$2.6b–\$2.7b

Quartile	Approximate Market Cap (USD, 2025)
Q1	$\leq \$1.2\text{b}$
Q2	$\$1.2\text{b} - \2.7b
Q3	$\$2.7\text{b} - \10b
Q4	$> \$10\text{b}$

- 75th percentile (Q3): About \$10b

Book Equity

- Definition: Book equity is the value of company assets as reported in the balance sheet minus liabilities and preferred stock.
- Quartiles: For research purposes, book-to-market breakpoints (book equity / market equity) are routinely calculated at the 30th and 70th percentiles of NYSE stocks, serving as proxies for "growth" versus "value" categories.
- Explicit dollar-value quartiles for raw book equity are less commonly reported, but the book-to-market ratio cutoffs are well-used:
 - Low (Q1; 0-30th pct): "Growth" companies, low BE/ME (below $\sim 0.35-0.40$ for most recent years)
 - High (Q3/4; 70-100th pct): "Value" companies, high BE/ME (above $\sim 0.75-1.0$)
- The absolute book equity dollar value quartiles vary considerably by industry; sector-level ratios are available, e.g., total market PBV ratio ~ 2.6 .

Operating Profitability

- Definition: Usually measured as operating income before depreciation divided by book equity (Fama-French OP metric) or operating profit margin.
- Quartile Estimation: According to established research data sets, typical cutoff points (cross-sectional on NYSE stocks) for operating profitability in recent U.S. data are:
 - Q1 (bottom quartile): $\leq 2\%-3\%$ (lowest profitability)
 - Q2-Q3: $\sim 3\%-12\%$
 - Q4 (top quartile): $\geq 12\%-15\%$ (highest profitability)
 - Value-weighted averages will vary by sector and index, but these ranges reflect current breakpoints for operating profitability used in common portfolio sorts.

Quartile	Operating Profitability (% of Book Equity)
Q1	$\leq 3\%$
Q2	3%–8%
Q3	8%–15%
Q4	$>15\%$

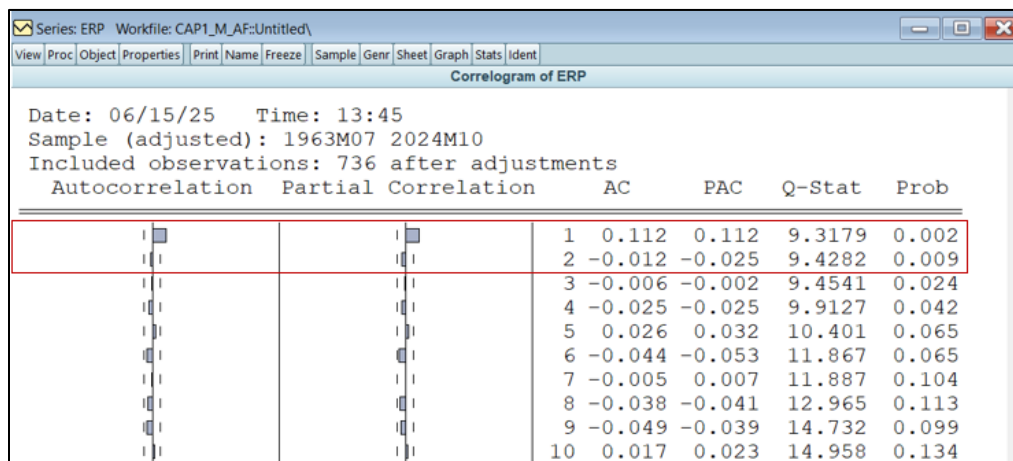
How Quartiles Are Used

- **Portfolio Construction:** Academic and professional managers use these quartiles to construct value, size, and profitability-sorted strategies, as in the Fama-French three-factor and five-factor models.
- **Annual Update:** Breakpoints (especially for market cap and book-to-market) are reset annually in June based on the cross-section of NYSE data.

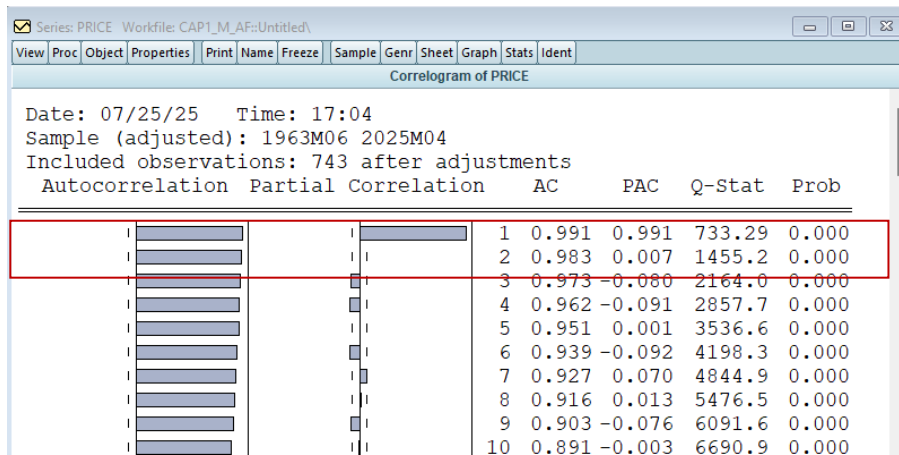
Note: For the precise, up-to-date quartile figures by company, refer to proprietary NYSE data services or major database providers. The above ranges are based on published academic protocols and recent data and provide accurate representations for most NYSE-listed stocks as of July 2025.

Appendix 3 – Eviews Machine Output

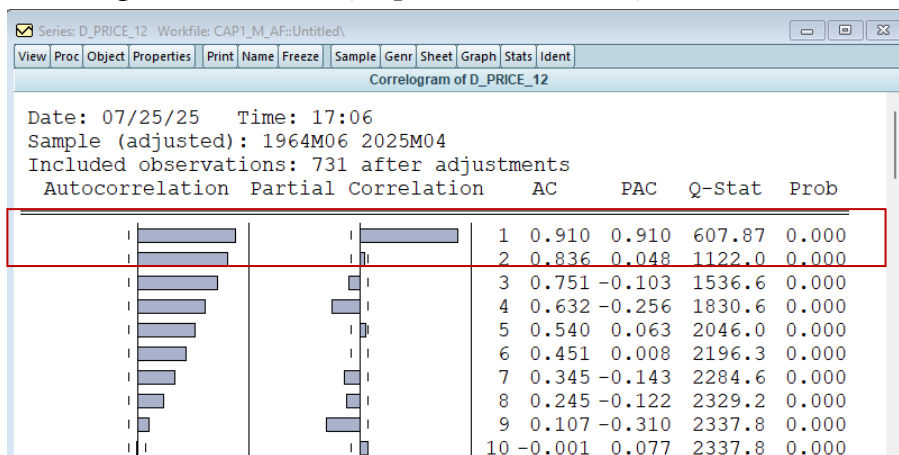
Correlogram on ERP



Correlogram on PRICE (non-differenced)



Correlogram on PRICE (12 period difference)



Unit Root Testing for Log PRICE change (KDV)

Augmented Dickey-Fuller Unit Root Test on LOG(PRICE)-LOG(PRICE(-12))				
A	B	C	D	E
Null Hypothesis: LOG(PRICE)-LOG(PRICE(-12)) has a unit root				
Exogenous: Constant				
Lag Length: 13 (Automatic - based on SIC, maxlag=19)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-5.775252	0.0000
Test critical values:				
1% level			-3.439243	
5% level			-2.865355	
10% level			-2.568858	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LOG(PRICE)-LOG(PRICE(-12)))				
Method: Least Squares				
Date: 07/25/25 Time: 16:38				
Sample (adjusted): 1965M08 2025M04				
Included observations: 717 after adjustments				
Variable	Coeffic...	Std. Error	t-Statistic	Prob.
LOG(PRICE(-1))-LOG(PRICE(-13))	-0.109832	0.019018	-5.775252	0.0000
D(LOG(PRICE(-1))-LOG(PRICE(-13)))	0.215106	0.038608	5.571472	0.0000
D(LOG(PRICE(-2))-LOG(PRICE(-14)))	0.080678	0.034302	2.352004	0.0189
D(LOG(PRICE(-3))-LOG(PRICE(-15)))	0.037264	0.034390	1.083573	0.2789
D(LOG(PRICE(-4))-LOG(PRICE(-16)))	0.028189	0.034445	0.818372	0.4134
D(LOG(PRICE(-5))-LOG(PRICE(-17)))	0.085945	0.034414	2.497370	0.0127
D(LOG(PRICE(-6))-LOG(PRICE(-18)))	0.053370	0.034476	1.548032	0.1221
D(LOG(PRICE(-7))-LOG(PRICE(-19)))	0.072404	0.034515	2.097766	0.0363
D(LOG(PRICE(-8))-LOG(PRICE(-20)))	0.047383	0.034547	1.371555	0.1706

Unit Root Testing for VIX (KDV 2)

Augmented Dickey-Fuller Unit Root Test on VIX				
A	B	C	D	E
Null Hypothesis: VIX has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=17)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-6.059094	0.0000
Test critical val...	1% level		-3.444724	
	5% level		-2.867772	
	10% level		-2.570153	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(VIX)				
Method: Least Squares				
Date: 07/25/25 Time: 16:50				
Sample (adjusted): 1990M02 2027M05				
Included observations: 448 after adjustments				
Variable	Coeffic...	Std. Error	t-Statistic	Prob.
VIX(-1)	-0.151893	0.025069	-6.059094	0.0000
C	2.946519	0.521459	5.650529	0.0000

STRUCTURAL REGRESSION RESULTS

SR-1

Equation: SR_1 Workfile: CAP1_M_AF:Untitled

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: LOG(PRICE)-LOG(PRICE(-12))
Method: Least Squares
Date: 08/17/25 Time: 13:27
Sample (adjusted): 1990M01 2025M04
Included observations: 424 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West
fixed bandwidth = 6.0000)

Variable	Coeffic...	Std. Error	t-Statistic	Prob.
C	0.347621	0.056837	6.116055	0.0000
CMA(-12)-CMA(-19)	-0.003784	0.002348	-1.611716	0.1078
CS	-0.159649	0.056720	-2.814670	0.0051
@PCY(M2(-6))	0.014485	0.005369	2.697744	0.0073
VIX	-0.009332	0.002654	-3.515853	0.0005
R-squared	0.349268	Mean dependent var	0.098311	
Adjusted R-squared	0.343056	S.D. dependent var	0.202586	
S.E. of regression	0.164200	Akaike info crit...	-0.763735	
Sum squared resid	11.29699	Schwarz criterion	-0.715979	
Log likelihood	166.9119	Hannan-Quinn cri...	-0.744867	
F-statistic	56.22253	Durbin-Watson stat	0.226859	
Prob(F-statistic)	0.000000	Wald F-statistic	7.674072	
Prob(Wald F-stati...	0.000006			

SR-2

Equation: SR_2 Workfile: CAP1_M_AF:Untitled

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: VIX
Method: Least Squares
Date: 08/17/25 Time: 13:27
Sample (adjusted): 1990M01 2025M04
Included observations: 424 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West
fixed bandwidth = 6.0000)

Variable	Coeffic...	Std. Error	t-Statistic	Prob.
C	8.084559	1.606693	5.031802	0.0000
CS	13.37927	1.401692	9.545086	0.0000
D(IFD)	1.409748	0.549822	2.564008	0.0107
TP(-9)	-1.276069	0.541263	-2.357577	0.0189
D(LOG(PRICE)-LOG(PRICE(-12)))	-13.10824	4.553051	-2.879000	0.0042
SPIKE_2020	28.52398	1.978189	14.41924	0.0000
R-squared	0.487105	Mean dependent var	19.49679	
Adjusted R-squared	0.480970	S.D. dependent var	7.477521	
S.E. of regression	5.387084	Akaike info crit...	6.219935	
Sum squared resid	12130.64	Schwarz criterion	6.277243	
Log likelihood	-1312.626	Hannan-Quinn cri...	6.242577	
F-statistic	79.39645	Durbin-Watson stat	0.408136	
Prob(F-statistic)	0.000000	Wald F-statistic	1454.364	
Prob(Wald F-statistic)	0.000000			

SR-3

Equation: SR_3 Workfile: CAP1_M_AF:Untitled\				
View	Proc	Object	Print	Name
Freeze	Estimate	Forecast	Stats	Resids
Dependent Variable: AAA Method: Fully Modified Least Squares (FMOLS) Date: 08/17/25 Time: 13:27 Sample (adjusted): 1962M02 2025M04 Included observations: 759 after adjustments Cointegrating equation deterministics: C Long-run covariance estimate (Quadratic-Spectral kernel, Andrews bandwidth = 94.7993)				
Variable	Coeffic...	Std. Error	t-Statistic	Prob.
TBTY	0.897870	0.030714	29.23354	0.0000
C	1.630932	0.201036	8.112652	0.0000
R-squared	0.969828	Mean dependent var	6.874572	
Adjusted R-squared	0.969789	S.D. dependent var	2.692983	
S.E. of regression	0.468080	Sum squared resid	165.8576	
Long-run variance	6.235755			

Unit Root Testing for Structural Regression Residuals

SR1

Augmented Dickey-Fuller Unit Root Test on SR_1_RESIDS				
A	B	C	D	E
Null Hypothesis: SR_1_RESIDS has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=17)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-5.061670	0.0000
Test critical val...	1% level	-3.445590		
	5% level	-2.868153		
	10% level	-2.570357		
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(SR_1_RESIDS)				
Method: Least Squares				
Date: 07/25/25 Time: 16:42				
Sample (adjusted): 1990M02 2025M04				
Included observations: 423 after adjustments				
Variable	Coeffic...	Std. Error	t-Statistic	Prob.

SR 2

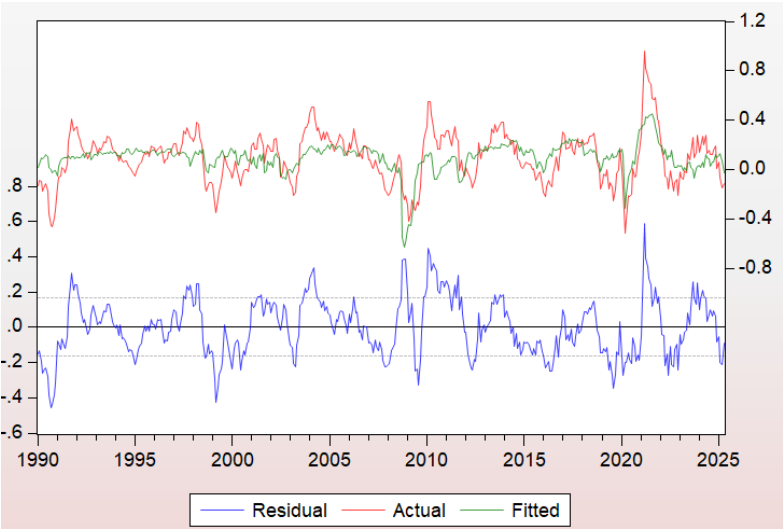
Augmented Dickey-Fuller Unit Root Test on SR_2_RESIDS				
A	B	C	D	E
Null Hypothesis: SR_2_RESIDS has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=12)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-6.312213	0.0000
Test critical val...			1% level	-3.445590
			5% level	-2.868153
			10% level	-2.570357
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(SR_2_RESIDS)				
Method: Least Squares				
Date: 07/25/25 Time: 16:37				
Sample (adjusted): 1990M02 2025M04				
Included observations: 423 after adjustments				

SR3

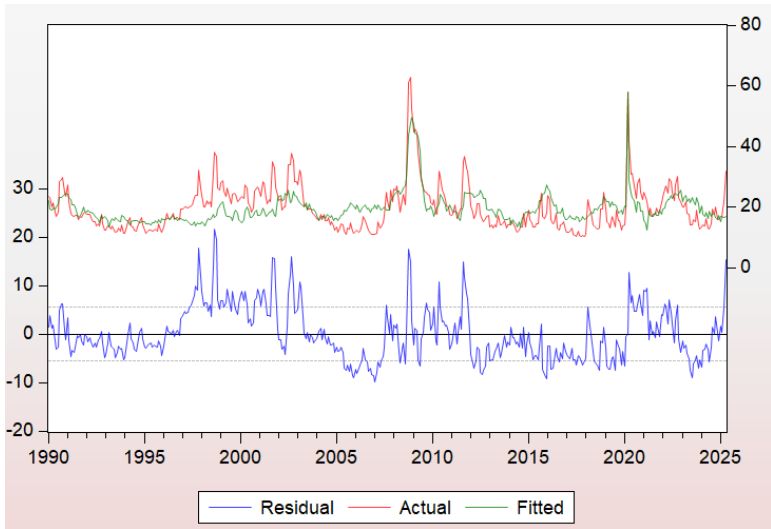
Augmented Dickey-Fuller Unit Root Test on SR_3_RESIDS				
A	B	C	D	E
Null Hypothesis: SR_3_RESIDS has a unit root				
Exogenous: Constant				
Lag Length: 2 (Automatic - based on SIC, maxlag=12)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.461433	0.0093
Test critical val... 1% level			-3.438773	
5% level			-2.865148	
10% level			-2.568746	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(SR_3_RESIDS)				
Method: Least Squares				
Date: 07/25/25 Time: 16:37				
Sample (adjusted): 1962M05 2025M04				
Included observations: 756 after adjustments				
Variable	Coeffic...	Std. Error	t-Statistic	Prob.

Actual Fitted Residuals for Structural Regressions

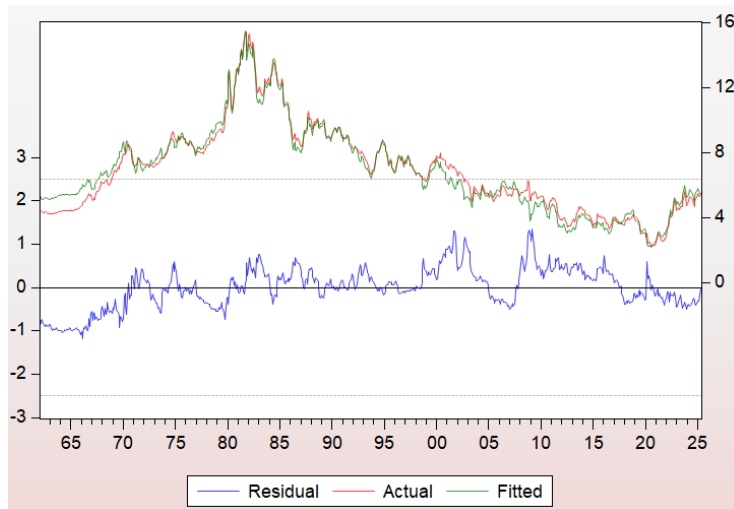
SR 1



SR 2



SR 3

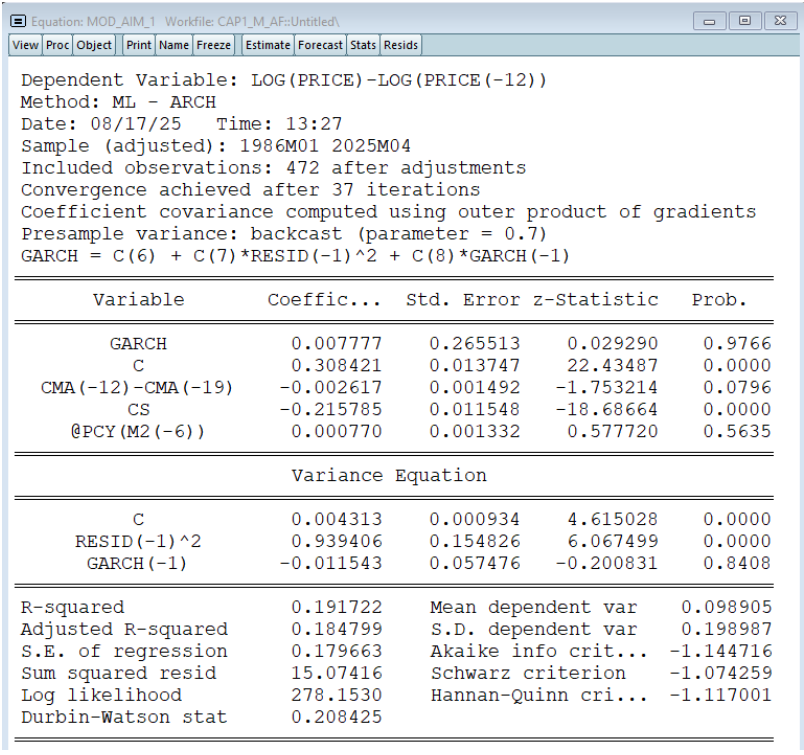


INERTIAL VOLATILITY REGRESSION RESULTS

GARCH MODEL

Equation: MOD_1.G1 Workfile: CAP1_M_AF:Untitled1				
View	Proc	Object	Print	Name
Freeze	Estimate	Forecast	Stats	Resids
Dependent Variable: LOG(PRICE)-LOG(PRICE(-12)) Method: ML - ARCH Date: 08/17/25 Time: 13:27 Sample (adjusted): 1986M01 2025M04 Included observations: 472 after adjustments Convergence achieved after 38 iterations Coefficient covariance computed using outer product of gradients Presample variance: backcast (parameter = 0.7) GARCH = C(5) + C(6)*RESID(-1)^2 + C(7)*GARCH(-1)				
Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	0.308552	0.013589	22.70637	0.0000
CMA(-12)-CMA(-19)	-0.002615	0.001491	-1.753443	0.0795
CS	-0.215780	0.011500	-18.76399	0.0000
@PCY(M2(-6))	0.000783	0.001327	0.590090	0.5551
Variance Equation				
C	0.004313	0.000930	4.638824	0.0000
RESID(-1)^2	0.939327	0.154528	6.078690	0.0000
GARCH(-1)	-0.011462	0.056777	-0.201872	0.8400
R-squared	0.191396	Mean dependent var	0.098905	
Adjusted R-squared	0.186213	S.D. dependent var	0.198987	
S.E. of regression	0.179507	Akaike info crit...	-1.148949	
Sum squared resid	15.08023	Schwarz criterion	-1.087299	
Log likelihood	278.1520	Hannan-Quinn cri...	-1.124699	
Durbin-Watson stat	0.208292			

ARCH IN MEAN MODEL

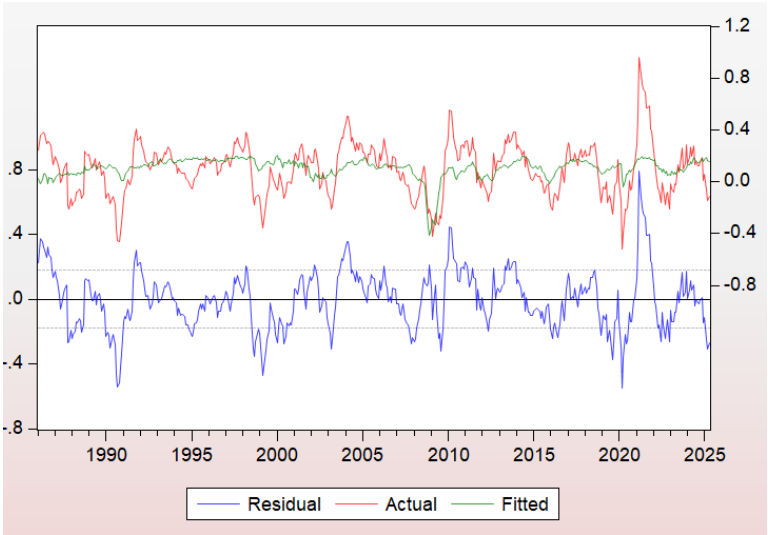


Actual Fitted Residuals for Inertial Volatility Regressions

GARCH Model



ARCH In-Mean Model



UNIT ROOT TEST OF RESIDUALS FOR INERTIAL VOLATILITY MODELS

GARCH

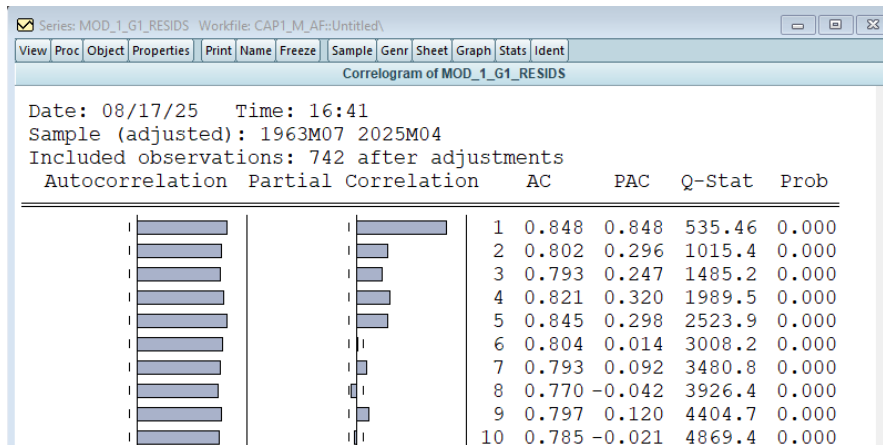
Table: GARCH_RESIDS_TEST Workfile: CAP1_M_AF:Untitled\					
View	Proc	Object	Print	Name	Edit+/- CellFmt Grid+/- Title Comments+/-
Augmented Dickey-Fuller Unit Root Test on MOD_1_G1_RESIDS					
	A	B	C	D	E
1	Null Hypothesis: MOD_1_G1_RESIDS has a unit root				
2	Exogenous: Constant				
3	Lag Length: 4 (Automatic - based on SIC, maxlag=12)				
4					
5	t-Statistic				Prob.*
6					
7	Augmented Dickey-Fuller test statistic			-2.263597	0.1843
8	Test critical values:			1% level	-3.438996
9				5% level	-2.865246
10				10% level	-2.568799
11					
12	*MacKinnon (1996) one-sided p-values.				
13					
14					
15	Augmented Dickey-Fuller Test Equation				
16	Dependent Variable: D(MOD_1_G1_RESIDS)				
17	Method: Least Squares				
18	Date: 08/17/25 Time: 13:27				
19	Sample (adjusted): 1963M12 2025M04				
20	Included observations: 737 after adjustments				
21					
22	Variable	Coeffic...	Std. Error	t-Statistic	Prob.
23					
24	MOD_1_G1_RESIDS(-1)	-0.039628	0.017507	-2.263597	0.0239
25	D(MOD_1_G1_RESIDS(-1))	-0.628334	0.037467	-16.77017	0.0000
26	D(MOD_1_G1_RESIDS(-2))	-0.554579	0.039319	-14.10471	0.0000
27	D(MOD_1_G1_RESIDS(-3))	-0.504215	0.038598	-13.06318	0.0000
28	D(MOD_1_G1_RESIDS(-4))	-0.312346	0.035094	-8.900246	0.0000
29	C	-0.645895	0.694696	-0.929751	0.3528
30					
31	R-squared	0.366734	Mean dependent var	0.093297	

ARCH IN-MEAN

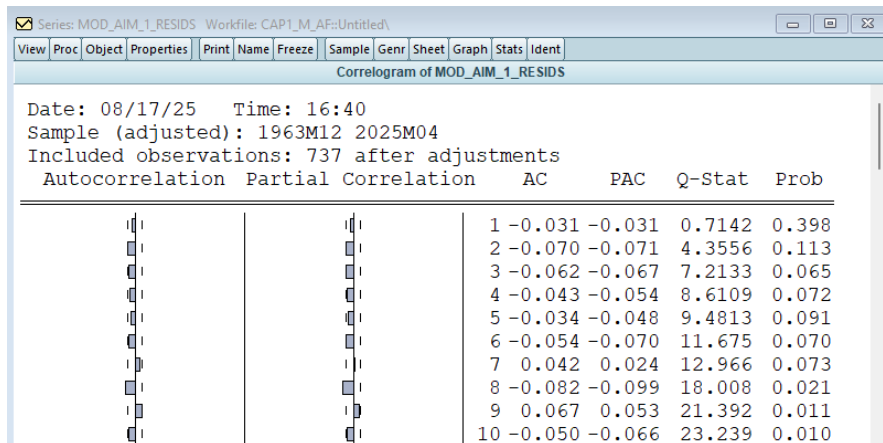
Table: AIM_RESIDS_TEST Workfile: CAP1_M_AF:Untitled\					
View	Proc	Object	Print	Name	Edit+/- CellFmt Grid+/- Title Comments+/-
Augmented Dickey-Fuller Unit Root Test on MOD_AIM_1_RESIDS					
	A	B	C	D	E
1	Null Hypothesis: MOD_AIM_1_RESIDS has a unit root				
2	Exogenous: Constant				
3	Lag Length: 0 (Automatic - based on SIC, maxlag=12)				
4					
5	t-Statistic				Prob.*
6					
7	Augmented Dickey-Fuller test statistic			-27.94974	0.0000
8	Test critical values:			1% level	-3.439008
9				5% level	-2.865251
10				10% level	-2.568802
11					
12	*MacKinnon (1996) one-sided p-values.				
13					
14					
15	Augmented Dickey-Fuller Test Equation				
16	Dependent Variable: D(MOD_AIM_1_RESIDS)				
17	Method: Least Squares				
18	Date: 08/17/25 Time: 13:27				
19	Sample (adjusted): 1964M01 2025M04				
20	Included observations: 736 after adjustments				
21					
22	Variable	Coeffic...	Std. Error	t-Statistic	Prob.
23					
24	MOD_AIM_1_RESIDS(-1)	-1.031067	0.036890	-27.94974	0.0000
25	C	-0.558032	0.550852	-1.013033	0.3114
26					
27	R-squared	0.515572	Mean dependent var	0.007164	
28	Adjusted R-squared	0.514912	S.D. dependent var	21.44228	
29	S.E. of regression	14.93418	Akaike info crit...	8.247896	

CORRELOGRAM OF RESIDUALS FOR INERTIAL VOLATILITY MODELS

GARCH

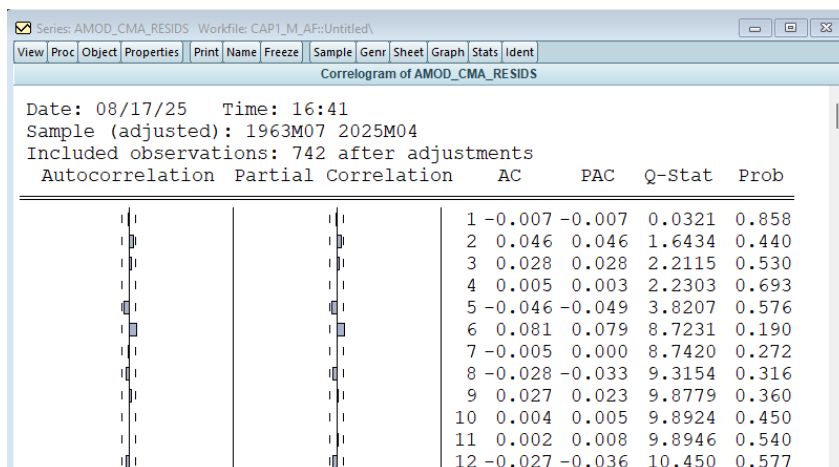


ARCH IN-MEAN

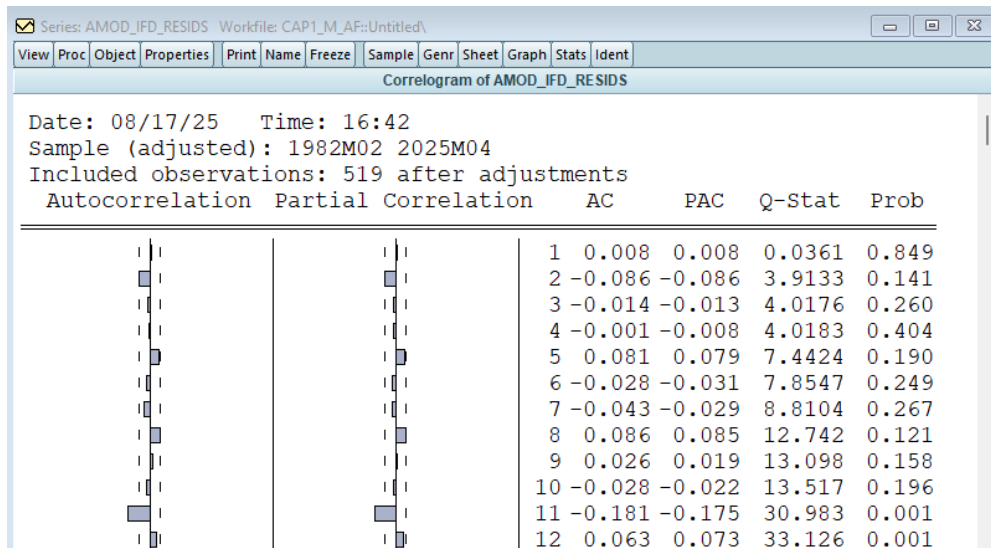


EXOGENOUS OOS ARIMA CORRELOGRAMS

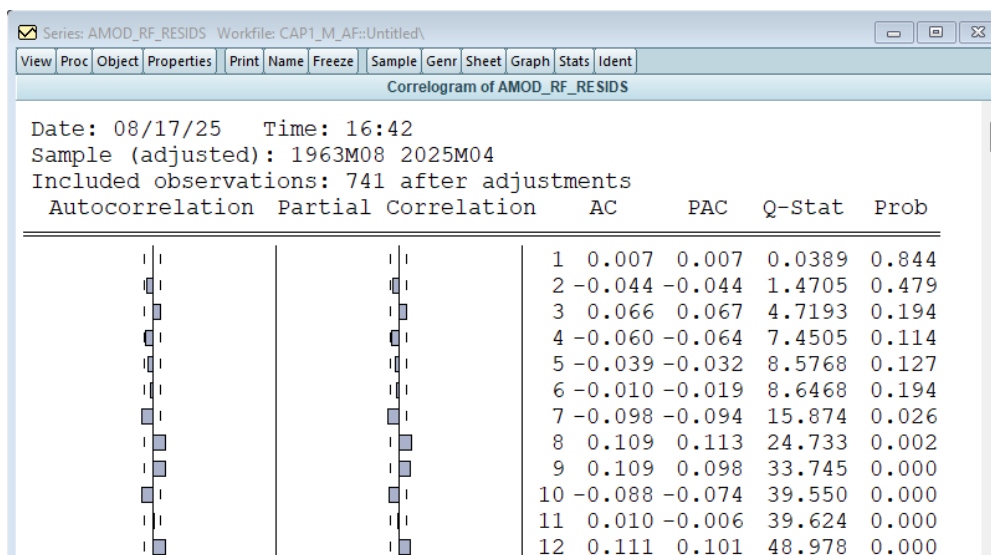
CMA



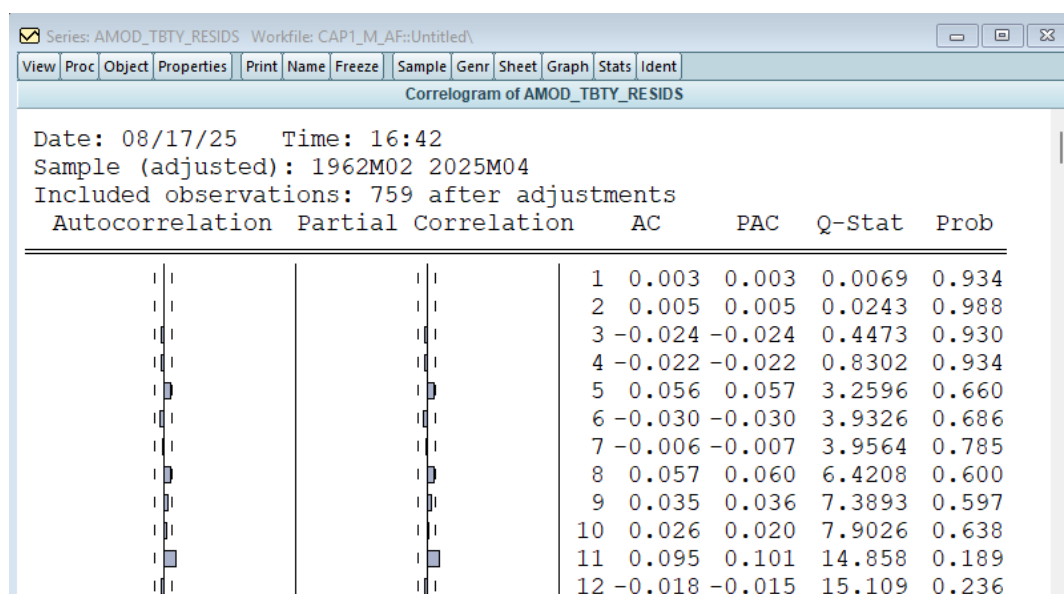
IFD



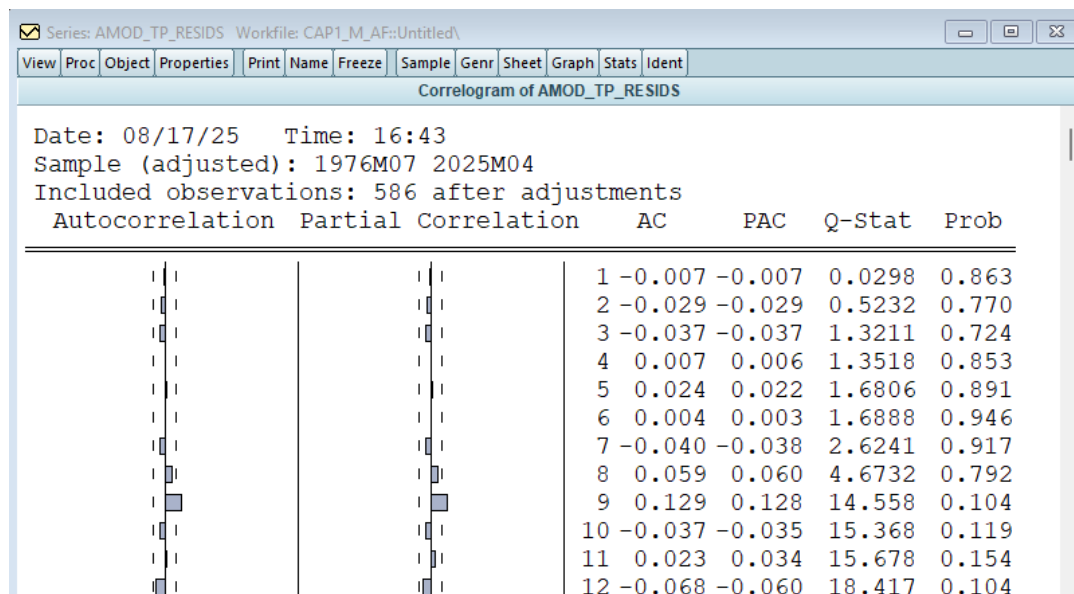
RF



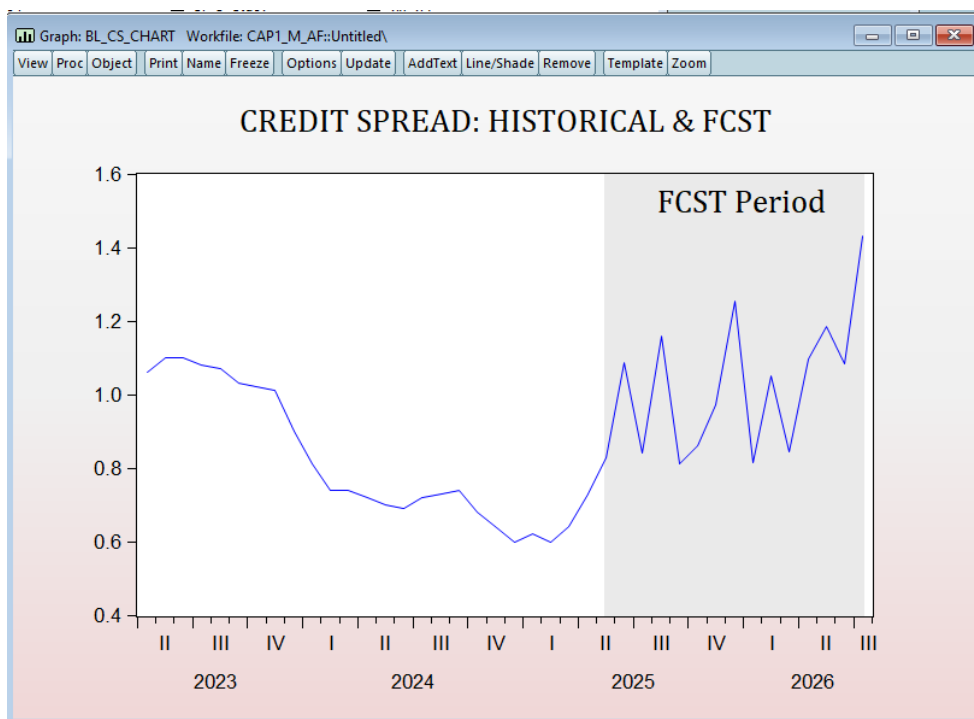
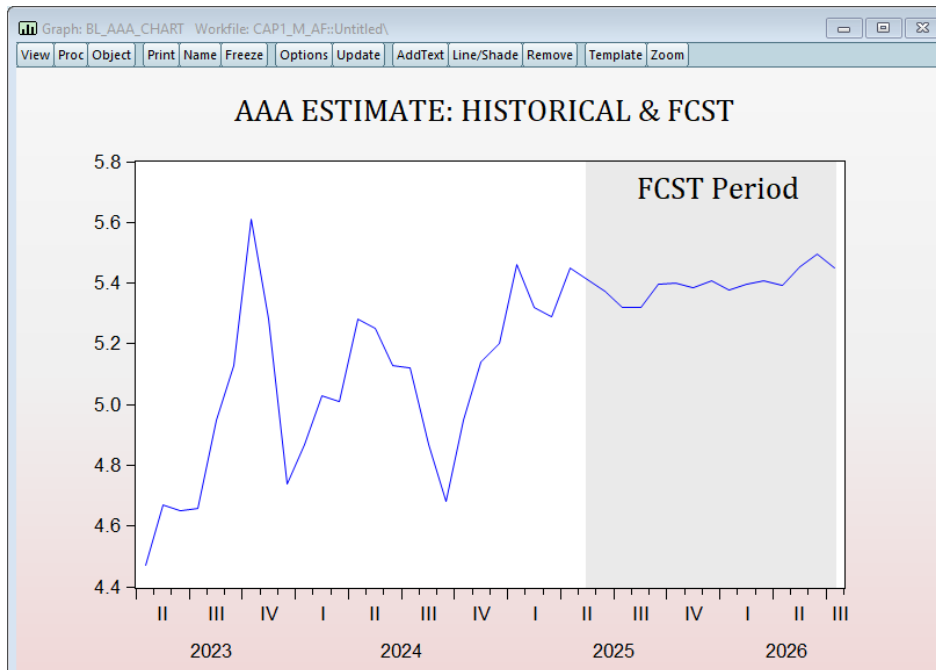
TBTY

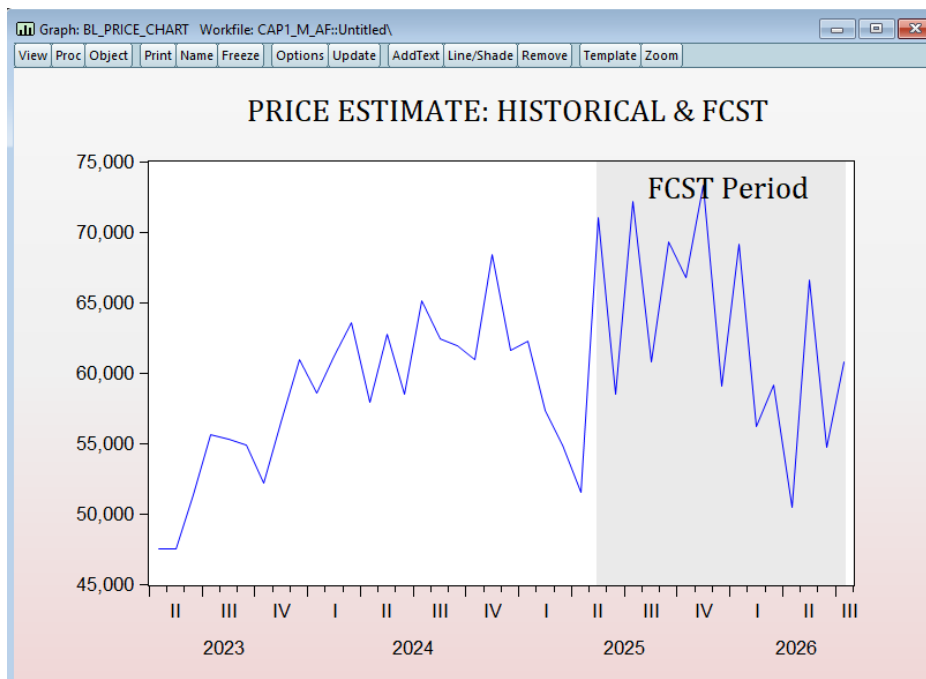
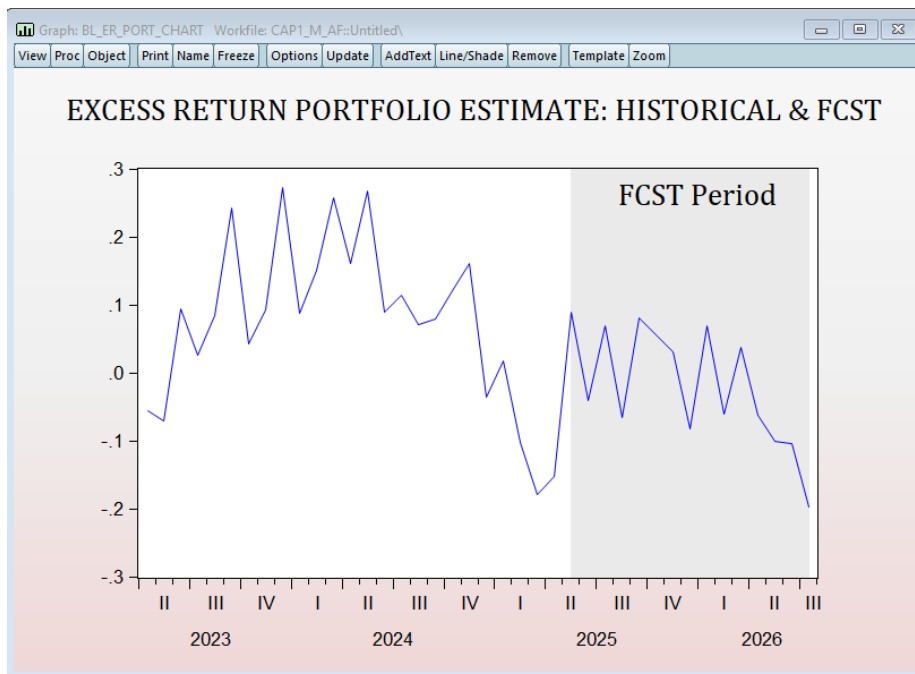


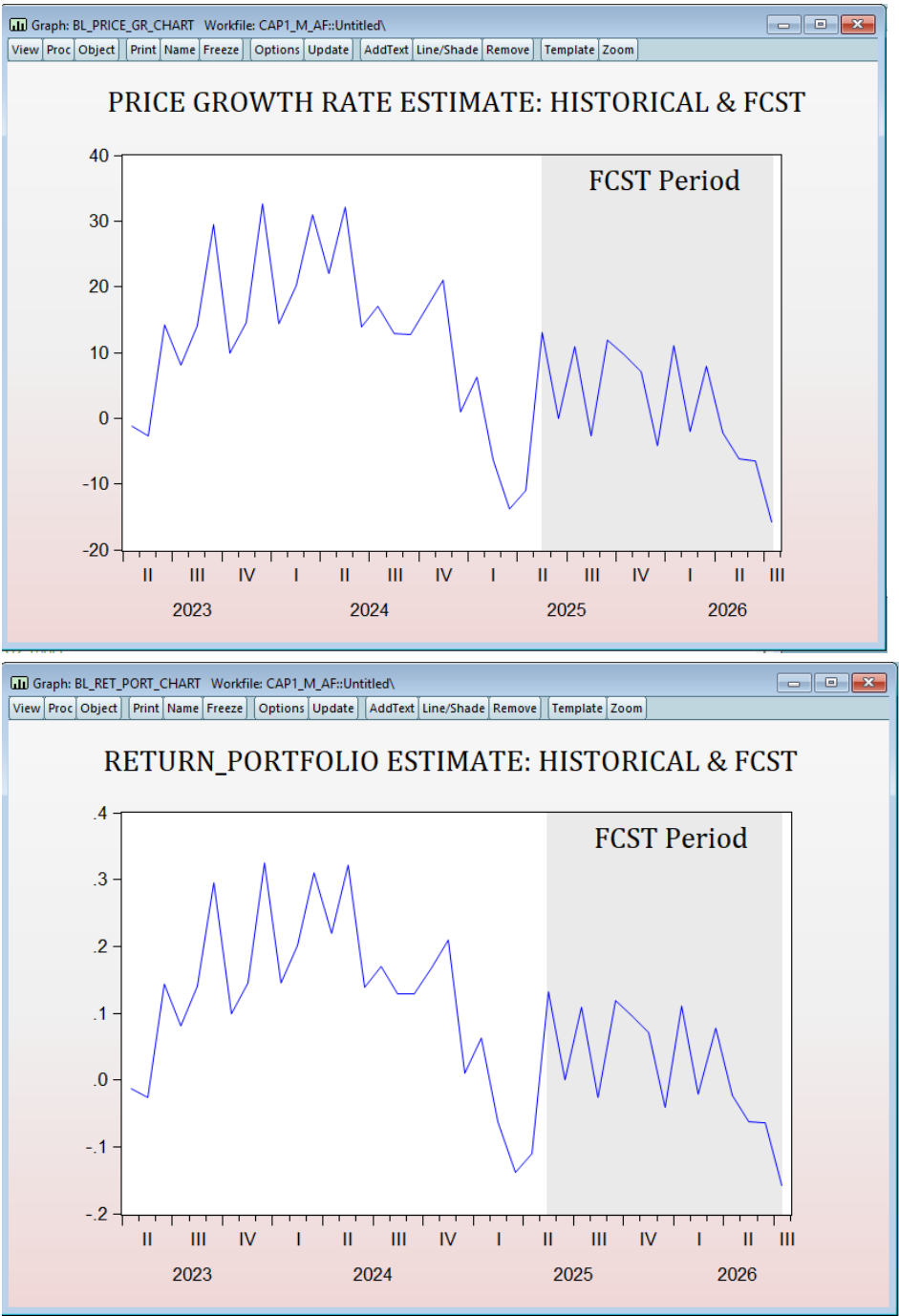
TP

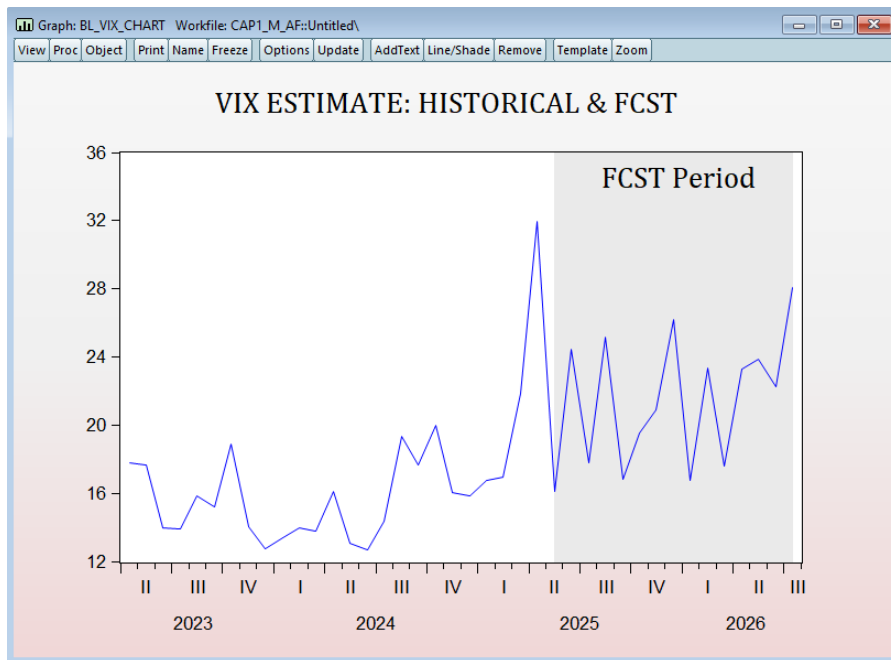


BASELINE FORECAST CHARTS

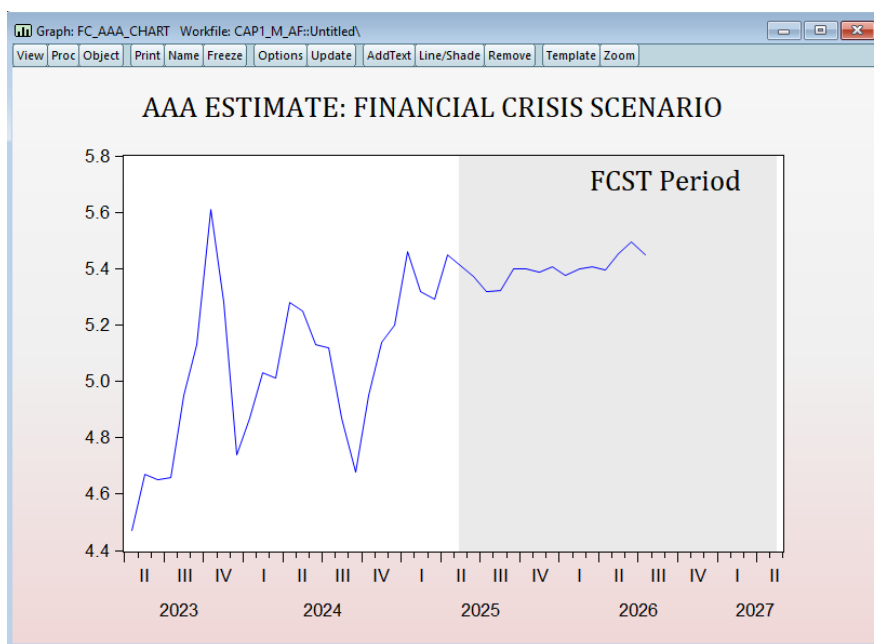


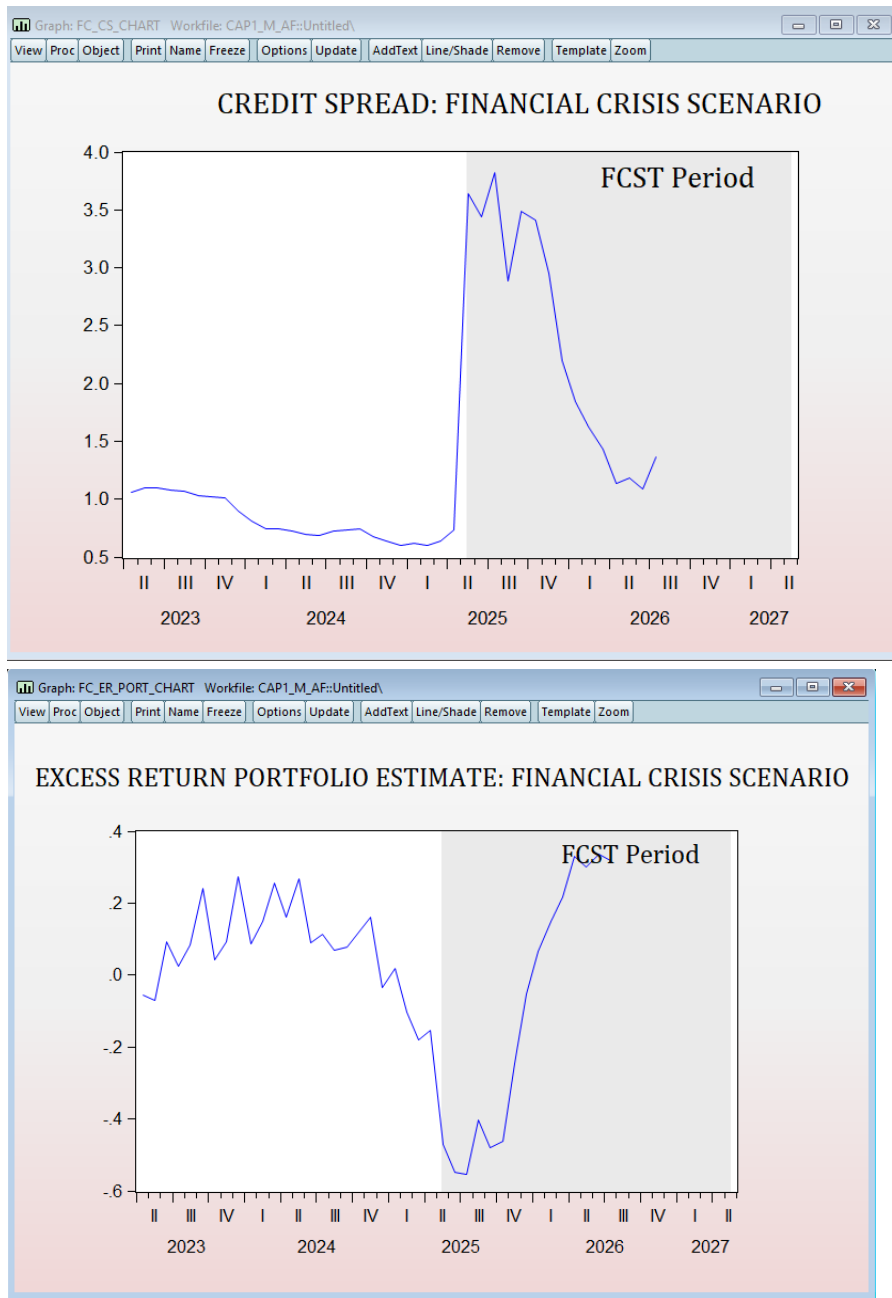


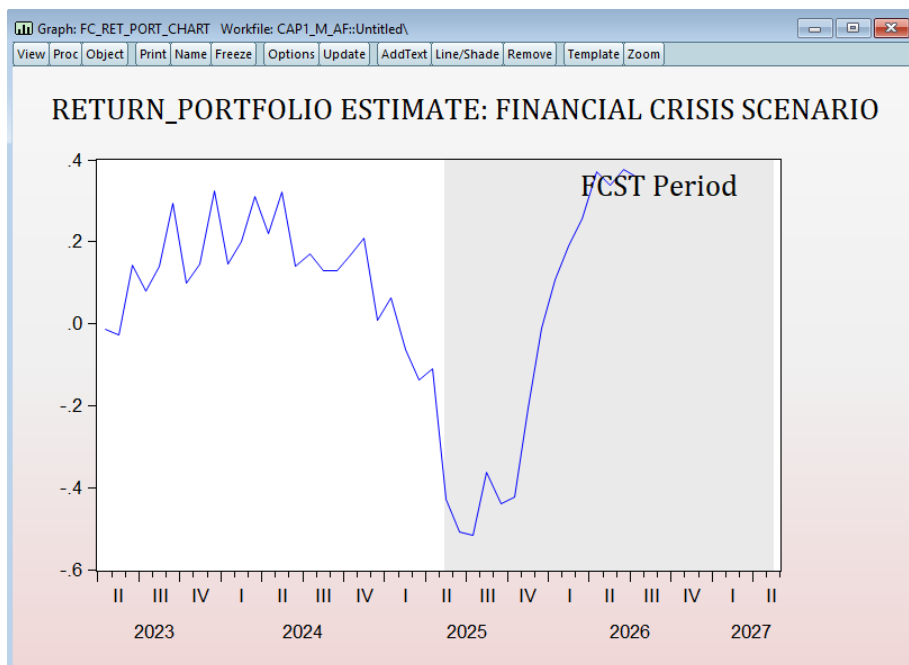
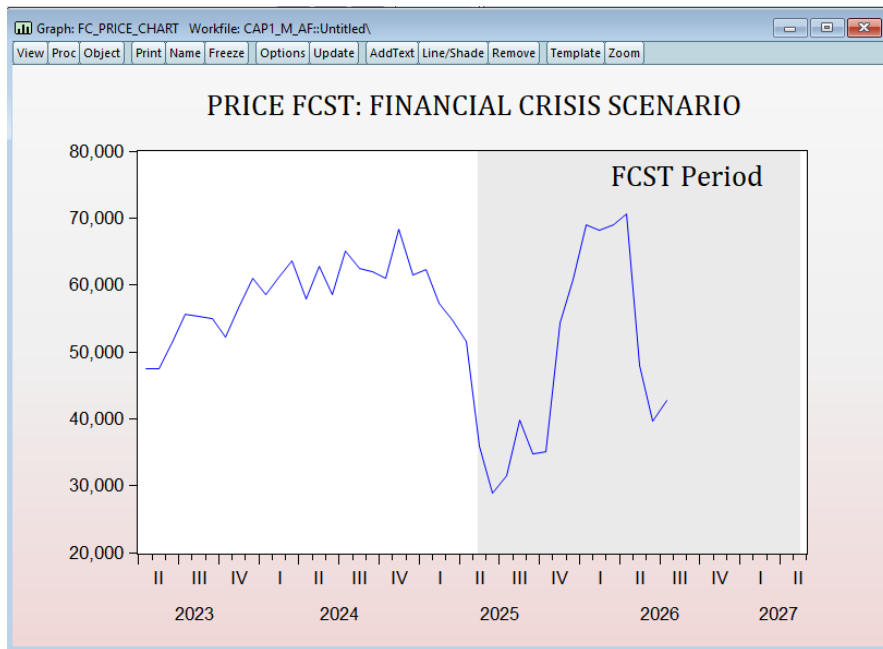


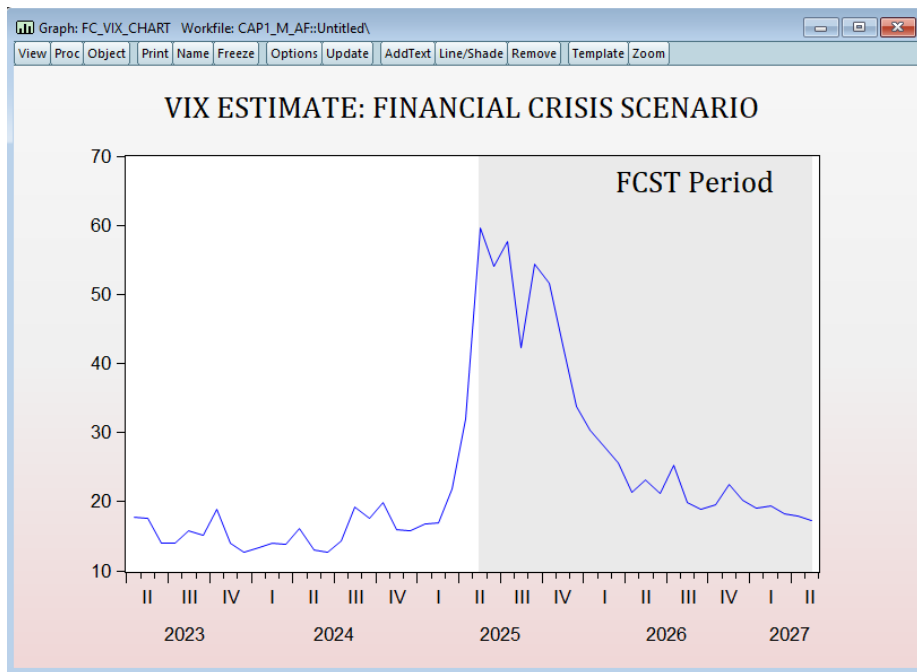


SCENARIO FORECAST CHARTS









VAR MODELS

VAR-1

Vector Autoregression Estimates			
Vector Autoregression Estimates			
Date: 08/17/25 Time: 13:27			
Sample (adjusted): 1991M03 2025M04			
Included observations: 410 after adjustments			
Standard errors in () & t-statistics in []			
	LOG (PRICE...	AAA	VIX
R-squared	0.903132	0.990182	0.769715
Adj. R-squared	0.891455	0.988999	0.741955
Sum sq. resids	1.512390	11.05382	5347.076
S.E. equation	0.064370	0.174024	3.827469
F-statistic	77.34132	836.6460	27.72717
Log likelihood	566.7408	158.9783	-1108.235
Akaike AIC	-2.545077	-0.555992	5.625537
Schwarz SC	-2.104279	-0.115194	6.066335
Mean dependent	0.109222	5.484293	19.36844
S.D. dependent	0.195380	1.659158	7.534668
Determinant resid covariance (do...		0.001323	
Determinant resid covariance		0.000933	
Log likelihood		-315.0380	
Akaike information criterion		2.195308	
Schwarz criterion		3.517701	
Number of coefficients		135	

VAR_1 STABILITY CONDITIONS

Var: VAR_1 Workfile: CAP1_M_AF::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Impulse Resids

VAR Stability Condition Check

Roots of Characteristic Polynomial
 Endogenous variables: LOG(PRICE)
 -LOG(PRICE(-12)) AAA VIX
 Exogenous variables: BAA-AAA CMA C
 Lag specification: 1 14
 Date: 08/17/25 Time: 16:56

Root	Modulus
------	---------

No root lies outside the unit circle.
 VAR satisfies the stability condition.

VAR - 2 RESULTS (NOTE: VAR_2 IS LABELED "VAR_4")

Var: VAR_4 Workfile: CAP1_M_AF::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Impulse Resids

Bayesian VAR Estimates

Bayesian VAR Estimates
 Date: 08/17/25 Time: 13:27
 Sample (adjusted): 1992M07 2025M04
 Included observations: 394 after adjustments
 Prior type: Litterman / Minnesota
 Initial residual covariance: Univariate AR
 Constant included in covariance calculation
 Hyper-parameters: Mul: 1, L1: 0.1, L2: 0.99, L3: 1, L4: inf
 Standard errors in ()

	D_PRICE_12	AAA	VIX
--	------------	-----	-----

R-squared	0.855003	0.990390	0.749666
Adj. R-squared	0.810054	0.987411	0.672063
Sum sq. resids	1.54E+09	9.289280	5784.758
S.E. equation	2266.271	0.175967	4.391187
F-statistic	19.02160	332.4469	9.660225
Mean dependent	1790.122	5.359213	19.46688
S.D. dependent	5199.921	1.568316	7.668082

Var: VAR_4 Workfile: CAP1_M_AF::Untitled\

View	Proc	Object	Print	Name	Freeze	Estimate	Forecast	Stats	Impulse	Resids
------	------	--------	-------	------	--------	----------	----------	-------	---------	--------

VAR Stability Condition Check

Roots of Characteristic Polynomial
Endogenous variables: D_PRICE_12 AAA
 VIX
Exogenous variables: @PCY(M2(-12))
 D(TP) IFD C
Lag specification: 1 30
Date: 08/17/25 Time: 16:57

Root	Modulus
No root lies outside the unit circle. VAR satisfies the stability condition.	

Appendix 4 - Eviews Code

```
' Andrew Frohner          ECON_512          CAP1 '
'-----'

CD C:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025\Data

WORKFILE CAP1_M_AF.wf1    M 1919:01 2027:05

READ(B2, S=Monthly) MSQE_DCAP_V10.xls ER ERP SMB HML RMW CMA RF MOM TP IORR
FFRT UE IFE PCEI PPCE PCED AAA BAA CS IFD EPI VIX IP REC QE1 QE2 QE3 VOL MP
TBTY RM CPI M2 PRICE BAA_FC VIX_FC M2_FC SPX SPX_ADJ PSPX

'----- DATA DERIVATIONS -----'
series P = ERP + RF
SERIES D_PRICE = PRICE - PRICE(-1)
SERIES RET_PORT = (PRICE - PRICE(-12)) / PRICE(-12)
SERIES ER_PORT = RET_PORT - RF

SERIES D_AAA = AAA - AAA(-1)
SERIES D_BAA = BAA - BAA(-1)
SERIES D_CS = CS - CS(-1)
SERIES D_IFD = IFD - IFD(-1)
SERIES D_M2 = M2 - M2(-1)
SERIES D_RET_PORT = RET_PORT - ret_port(-1)

SERIES D_RF = RF - rf(-1)
SERIES D_TBTY = TBTY - TBTY(-1)
SERIES D_TP = TP - TP(-1)
SERIES D_VIX = VIX - VIX(-1)
SERIES D_PRICE_X = LOG(PRICE)-LOG(PRICE(-12))
SERIES D_PRICE_12 = PRICE-PRICE(-12)

'----- KDV #1 HISTOGRAM -----'
ERP.HIST
D_PRICE_X.HIST

'----- KDV #1 CORRELOGRAM -----'
ERP.CORREL(36)
PRICE.CORREL(36)
'----- UNIT ROOT TESTING -----'

FREEZE(PRICE_ADF_RAW) PRICE.UROOT(ADF, CONST, LAGMETHOD=SIC, MAXLAG=12)
FREEZE(PRICE_DFGLS_RAW) PRICE.UROOT(DFGLS, CONST, LAGMETHOD=SIC, MAXLAG=12)
FREEZE(PRICE_KPSS_RAW) PRICE.UROOT(KPSS, TREND, LAGMETHOD=SIC, MAXLAG=12)
```

```
FREEZE (PRICE_KDV_ADF_RAW) D_PRICE_X.UROOT (ADF, CONST, LAGMETHOD=SIC,
MAXLAG=12)
```

```
FREEZE (VIX_ADF_RAW) VIX.UROOT (ADF, CONST, LAGMETHOD=SIC, MAXLAG=12)
```

```
FREEZE (AAA_ADF_RAW) AAA.UROOT (ADF, CONST, LAGMETHOD=SIC, MAXLAG=12)
```

```
'-----SHOCK ANALYSIS REGRESSIONS -----'
```

```
'----- KDV 1 REGRESSION ON CONSTANT-----'
```

```
EQUATION E1_ERP.LS ERP C
```

```
      E1_ERP.STATS
```

```
      E1_ERP.MAKERESIDS E1_ERP_RESIDS
```

```
EQUATION E1_PRICE.LS D(PRICE) C
```

```
      E1_PRICE.STATS
```

```
      E1_PRICE.MAKERESIDS E1_PRICE_RESIDS
```

```
'-----SPIKE ITV VARIABLES -----'
```

```
SMPL @ALL
```

```
SERIES SPIKE_1987 = 0.0
```

```
SMPL 1987:10 1987:10
```

```
SERIES SPIKE_1987 = 1.0
```

```
SMPL @ALL
```

```
SERIES SPIKE_1975 = 0.0
```

```
SMPL 1975:01 1975:01
```

```
SERIES SPIKE_1975 = 1.0
```

```
SMPL @ALL
```

```
SERIES SPIKE_2008= 0.0
```

```
SMPL 2008:10 2008:10
```

```
SERIES SPIKE_2008 = 1.0
```

```
SMPL @ALL
```

```
SERIES SPIKE_2020= 0.0
```

```
SMPL 2020:03 2020:03
```

```
SERIES SPIKE_2020 = 1.0
```

```
SMPL @ALL
```

```
SERIES SPIKE_2020E= 0.0
```

```
SMPL 2020:11 2020:11
```

```
SERIES SPIKE_2020E = 1.0
```

```
SMPL @ALL
```

```

'---REGRESSIONS WITH SHOCK VARIABLES-----'

EQUATION E_ADF_1.LS D(PRICE) C D(PRICE(-1)) D(PRICE(-2)) D(PRICE(-3))
          E_ADF_1.STATS
          E_adf_1.MAKERESIDS E_ADF_1.RESIDS

EQUATION E_ADF_2.LS PRICE C D(PRICE(-1)) D(PRICE(-2)) D(PRICE(-3)) SPIKE_1975
SPIKE_1987 SPIKE_2008 SPIKE_2020 SPIKE_2020E

E_ADF_2.STATS
E_adf_2.MAKERESIDS E_ADF_2.RESIDS

'---UROOT TESTING: KDVS-----'

'---- ADF TEST WITH TREND - PRICE---'
FREEZE(PRICE_RAW) PRICE.UROOT(ADF, TREND, LAGMETHOD=SIC, MAXLAG=12)
FREEZE(PRICE_2) LOG(PRICE)-LOG(PRICE(-12)).UROOT(ADF, TREND, LAGMETHOD=SIC,
MAXLAG=12)

'---- ADF TEST WITH TREND - VIX AND AAA ---'
FREEZE(VIX_RAW) VIX.UROOT(ADF, TREND, LAGMETHOD=SIC, MAXLAG=12)

FREEZE(AAA_RAW) AAA.UROOT(ADF, TREND, LAGMETHOD=SIC, MAXLAG=12)

'-----ARCHTESTS-----'

E1_ERP.ARCHTEST(24)
E_ADF_1.ARCHTEST(24)
E_ADF_2.ARCHTEST(20)

'----- SHOCK SIZE MEASUREMENT-----'

SCALAR PRICE_ST = @STDEV(D_PRICE)
SCALAR PRICE_MEAN = @MEAN(D_PRICE)
SERIES PRICE_DSSM = (D_PRICE- PRICE_MEAN) / PRICE_ST
PRICE_DSSM.LINE

'-----C_1 --- REGRESSION ON INTERCEPT -----'

EQUATION C1_KDV.LS D(PRICE) C
          C1_KDV.STATS
'-----C_1----- GARCH MODELING-----'
'-- GARCH TESTING: MOD_1 -----'

```

```

EQUATION MOD_1_G1.ARCH(1,1) log(price)-log(price(-12)) C CMA(-12)-CMA(-19)
CS @PCY(M2(-6))

MOD_1_G1.STATS
MOD_1_G1.CORRELSQ(36)
MOD_1_G1.IDENT(36)

'-- GARCH MODEL 1 -----'

MOD_1_G1.MAKEGARCH MOD_1_GARCH_1
SERIES MOD_1_G1_RESIDS = RESID

'--GARCH MODEL RESIDUALS UNIT ROOT TEST-----'
FREEZE(GARCH_RESIDS_TEST) MOD_1_G1_RESIDS.UROOT(ADF, CONST, LAGMETHOD=SIC,
MAXLAG=12)

'----- ARCH IN MEAN MODEL NO VIX -----'
'-----'
EQUATION MOD_AIM_1.ARCH(1,1,ARCHM=VAR) log(price)-log(price(-12)) C CMA(-
12)-CMA(-19) CS @PCY(M2(-6))
MOD_AIM_1.STATS
series MOD_AIM_1_RESIDS = RESID

'--ARCH IN MEAN MODEL RESIDUALS UNIT ROOT TEST-----'
FREEZE(AIM_RESIDS_TEST) MOD_AIM_1_RESIDS.UROOT(ADF, CONST, LAGMETHOD=SIC,
MAXLAG=12)

'----- ARCH IN MEAN MODEL WITH VIX -----'
'-----'

EQUATION MOD_AIM_2.ARCH(1,1,ARCHM=VAR) LOG(PRICE)-LOG(PRICE(-12)) C CMA(-12)-
CMA(-19) CS @PCY(M2(-6)) @ VIX
MOD_AIM_2.STATS

'----- ARIMA MODELING FOR EXOGENOUS VARIABLES-----'

EQUATION AMOD_CMA.LS CMA C AR(1) AR(4) AR(10) MA(11) MA(22) AR(33) MA(41)
AMOD_CMA.STATS
SERIES AMOD_CMA_RESIDS = RESID
AMOD_CMA_RESIDS.IDENT(36)

'
EQUATION AMOD_CS.LS D(CS) C MA(1) MA(2) MA(8) MA(36)
MA(16) AR(16) AR(22) AR(36) AR(55) MA(90)
'
AMOD_CS.STATS
'
SERIES AMOD_CS_RESIDS = RESID
'
AMOD_CS_RESIDS.IDENT(36)

```

```

EQUATION AMOD_VIX.LS D(VIX) C AR(2) AR(10) MA(12)
MA(24) MA(32) AR(34) AR(46) MA(42) AR(31) AR(77) MA(80) MA(120)
AMOD_VIX.STATS
SERIES          AMOD_VIX_RESIDS = RESID
AMOD_VIX_RESIDS.IDENT(36)

EQUATION AMOD_TP.LS D(TP) C AR(1) MA(1) AR(4) MA(14) MA(21) AR(32) MA(36)
AR(60) MA(70) AR(75)
AMOD_TP.STATS
SERIES          AMOD_TP_RESIDS = RESID
AMOD_TP_RESIDS.IDENT(36)

EQUATION AMOD_TBTY.LS D(TBTY) C MA(1) MA(2) AR(7) MA(30) AR(44) MA(55) AR(21)
AMOD_TBTY.STATS
SERIES          AMOD_TBTY_RESIDS = RESID
AMOD_TBTY_RESIDS.IDENT(36)

'
EQUATION AMOD_M2.LS D(M2) C AR(1) MA(1) AR(12) MA(12)
AR(26) MA(36)
'
AMOD_M2.STATS
'
SERIES          AMOD_M2_RESIDS = RESID
'
AMOD_M2_RESIDS.IDENT(36)

EQUATION AMOD_IFD.LS D(IFD) C MA(6) AR(6) MA(24) AR(27) AR(45) MA(50) AR(37)
AMOD_IFD.STATS
SERIES          AMOD_IFD_RESIDS = RESID
AMOD_IFD_RESIDS.IDENT(36)

EQUATION AMOD_AAA.LS D(AAA) AR(1) AR(5) MA(1) MA(5) AR(30) MA(34) AR(48)
SERIES          AMOD_AAA_RESIDS = RESID
AMOD_AAA_RESIDS.IDENT(36)

'
EQUATION AMOD_BAA.LS D(BAA) AR(1) AR(5) MA(1) MA(5)
AR(20) MA(20) AR(33) MA(33)
'
SERIES          AMOD_BAA_RESIDS = RESID
'
AMOD_BAA_RESIDS.IDENT(36)

EQUATION AMOD_RF.LS D(RF) AR(1) AR(5) MA(1) MA(5) MA(25) AR(25)
SERIES          AMOD_RF_RESIDS = RESID
AMOD_RF_RESIDS.IDENT(36)

'
EQUATION AMOD_RET_PORT.LS RET_PORT C AR(1) MA(1)
AR(12) MA(12)
'
SERIES          AMOD_RET_PORT_RESIDS = RESID
'
AMOD_RET_PORT_RESIDS.IDENT(36)

```

```
'----- ARIMA FORECASTING FOR EXOGENOUS VARIABLES-----'
```

```
EXPAND 1963M07 2027M05
```

```
SMPL 2025M05 2027M05
```

```
AMOD_CMA.FORECAST CMA_F
```

```
'      AMOD_CS.FORECAST CS_F
```

```
'      AMOD_VIX.FORECAST VIX_F
```

```
AMOD_TP.FORECAST TP_F
```

```
AMOD_TBTY.forecast TBTY_F
```

```
'      AMOD_M2.FORECAST m2_F
```

```
AMOD_IFD.FORECAST IFD_F
```

```
AMOD_AAA.FORECAST AAA_F
```

```
'      AMOD_BAA.FORECAST BAA_F
```

```
AMOD_RF.FORECAST RF_F
```

```
'      AMOD_RET_PORT.FORECAST RET_PORT_F
```

```
'----- REVERT TO STANDARD NMEONICS FOR OOS PERIOD -----'
```

```
SMPL 2025M05 2027M05
```

```
SERIES CMA = cma_f
```

```
' SERIES CS = CS_F
```

```
'SERIES VIX = VIX_F
```

```
SERIES TP  = TP_F
```

```
SERIES TBTY = TBTY_F
```

```
'SERIES M2 = M2_F
```

```
SERIES IFD = IFD_F
```

```
'SERIES AAA = AAA_F
```

```
'SERIES BAA = BAA_F
```

```
SERIES RF = RF_F
```

```
'SERIES RET_PORT = RET_PORT_F
```

```
SERIES SPIKE_2020 = 0
```

```
'----- EXOGENOUS OOS FORECAST CHARTS-----'
```

```
'
```

```
SMPL 2023M04 2027M05
```

```
GROUP G_CMA CMA CMA_F
```

```
G_CMA.LINE
```

```
'GROUP G_CS CS CS_F
```

```
'G_CS.LINE
```

```
'GROUP G_VIX VIX VIX_F
```

```

'G_VIX.LINE

GROUP G_TP TP TP_F
G_TP.LINE

GROUP G_TBTY TBTY TBTY_F
graph TBTY_CHART.line G_TBTY
freeze(CHART_TBTY) TBTY_CHART
CHART_TBTY.addtext(0.5, 0.95, "TBTY OOS FCST")
CHART_TBTY.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
CHART_TBTY.addtext(0.85, 0.05, "FCST Period")
show CHART_TBTY

'GROUP G_M2 M2 M2_F
'G_M2.LINE

GROUP G_IFD IFD IFD_F
G_IFD.LINE

GROUP G_AAA AAA AAA_F
G_AAA.LINE

'GROUP G_BAA BAA BAA_F
'G_BAA.LINE

GROUP G_RF RF RF_F
G_RF.LINE

'GROUP G_RET_PORT RET_PORT RET_PORT_F
'G_RET_PORT.LINE

'----- CORREL OF RAW EXOGENOUS DATA -----'
SMPL @ALL
AAA.IDENT(24)
BAA.IDENT(24)
CMA.IDENT(24)
CS.IDENT(24)
IFD.IDENT(24)
M2.IDENT(24)
RET_PORT.IDENT(24)
RF.IDENT(24)
TBTY.IDENT(24)
TP.IDENT(24)
VIX.IDENT(24)

```

```

'----- CORREL OF PW EXOGENOUS DATA -----'
SMPL @ALL
D_CS.IDENT(24)
D_VIX.IDENT(24)
D_TP.IDENT(24)
D_TBTY.IDENT(24)
D_M2.IDENT(24)
D_IFD.IDENT(24)
D_BAA.IDENT(24)
D_RF.IDENT(24)
D_RET_PORT.IDENT(24)

'-----C_1 MEM CONSTRUNCTION -----'
'---- C_1: DEFINING STOCHASTIC EQUATIONS FOR MEM-----'
SMPL 1919M01 2025M04

EQUATION SR_1.LS(N) LOG(PRICE)-LOG(PRICE(-12)) C CMA(-12)-CMA(-19) CS
@PCY(M2(-6)) VIX
                SR_1.STATS
SERIES SR_1_RESIDS = RESID
FREEZE(SR_1_ADF) SR_1_RESIDS.uroot(ADF,CONST,LAGMETHOD=SIC, MAXLAG=12)

EQUATION SR_2.LS(n) VIX C CS D(IFD) TP(-9) D(log(price)-log(price(-12)))
spike_2020
                SR_2.STATS
SERIES SR_2_RESIDS = RESID

FREEZE(SR_2_ADF) SR_2_RESIDS.uroot(ADF,CONST,LAGMETHOD=SIC, MAXLAG=12)

EQUATION SR_3.COINTREG(METHOD=FMOLS, KERN=QUADSPEC, BW=ANDREWS) AAA C TBTY
                SR_3.STATS
SERIES SR_3_RESIDS = RESID
FREEZE(SR_3_ADF) SR_3_RESIDS.UROOT(ADF,CONST,LAGMETHOD=SIC, MAXLAG=12)

'EQUATION SR_4.LS AAA C TBTY
'
                SR_4.STATS
'SERIES SR_4_RESIDS = RESID

'-----SHOCK SIZE MEASUREMENTS-----'

SMPL @ALL

SCALAR SR_1_STDEV = @STDEV(sr_1_resids)
Summer 2025

```

```

SERIES SR_1_RSSM = sr_1_RESIDS / SR_1_STDEV

SCALAR SR_2_STDEV = @STDEV(sr_2_resids)
SERIES SR_2_RSSM = sr_2_RESIDS / SR_2_STDEV

SCALAR SR_3_STDEV = @STDEV(sr_3_resids)
SERIES SR_3_RSSM = sr_3_RESIDS / SR_3_STDEV

'-----C_1: MODEL OBJECT CONSTRUCTION-----'

MODEL MEM_C1_AF

    MEM_C1_AF.MERGE SR_1
    MEM_C1_AF.MERGE SR_2
    MEM_C1_AF.MERGE SR_3
    MEM_C1_AF.APPEND RET_PORT = (PRICE - PRICE(-12)) / PRICE(-12)
    MEM_C1_AF.APPEND CS = BAA - AAA
    MEM_C1_AF.APPEND ER_PORT = RET_PORT - RF

'----- HISTORICAL PERIOD FORECAST SOLVE-----'
SMPL 2023m04 2025m04
MEM_C1_AF.APPEND ASSIGN @ALL _SH
MEM_C1_AF.SOLVE

'----- OOS PERIOD FORECAST SOLVE-----'
EXPAND 1963M07 2026M07
SMPL 2025M05 2027M05
MEM_C1_AF.APPEND ASSIGN @ALL _SF
MEM_C1_AF.SOLVE

'----- MEM FCST CHARTS -----'
SMPL 2023M04 2026M07

GROUP BL_PRICE PRICE_SF
GROUP BL_PRICE_GR @PCY(PRICE_SF)
GROUP BL_AAA AAA_SF
GROUP BL_VIX VIX_SF
GROUP BL_RET_PORT RET_PORT_SF
GROUP BL_CS CS_SF
GROUP BL_ER_PORT ER_PORT_SF

'GROUP BL_PRICE PRICE_SH PRICE_SF @PCA(PRICE_SH) @PCA(PRICE_SF)
'GROUP BL_AAA AAA_SH AAA_SF @PCA(AAA_SH) @PCA(AAA_SF)

```

```

'GROUP BL_VIX VIX_SH VIX_SF @PCA(VIX_SH) @PCA(VIX_SF)
'GROUP BL_RET_PORT RET_PORT_SH RET_PORT_SF @PCA(RET_PORT_SH)
@PCA(RET_PORT_SF)
'GROUP BL_CS CS_SH CS_SF @PCA(CS_SH) @PCA(CS_SF)
'GROUP BL_ER_PORT ER_PORT_SH ER_PORT_SF @PCA(ER_PORT_SH) @PCA(ER_PORT_SF)

'GROUP EXPORT BL_PRICE    BL_AAA    BL_VIX    BL_RET_PORT    BL_CS    BL_ER_PORT

GRAPH BL_PRICE_CHART.line BL_PRICE
BL_PRICE_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "PRICE ESTIMATE:
HISTORICAL & FCST"
BL_PRICE_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_PRICE_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
bl_price_chart.setelem(1) legend(PRICE - HISTORICAL FCST)
bl_price_chart.setelem(2) legend(PRICE - OOS FCST)
BL_PRICE_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025\Data\EVIEWS\
CAP_CHARTS

GRAPH BL_PRICE_GR_CHART.line BL_PRICE_GR
BL_PRICE_GR_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "PRICE GROWTH RATE
ESTIMATE: HISTORICAL & FCST"
BL_PRICE_GR_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_PRICE_GR_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
BL_PRICE_GR_CHART.setelem(1) legend(PRICE GR - HISTORICAL FCST)
BL_PRICE_GR_CHART.setelem(2) legend(PRICE GR - OOS FCST)
BL_PRICE_GR_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025\Data\EVIEWS\
CAP_CHARTS

GRAPH bl_AAA_chart.line BL_AAA
BL_AAA_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "AAA ESTIMATE: HISTORICAL &
FCST"
BL_AAA_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_AAA_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
bl_AAA_chart.setelem(1) legend(AAA - HISTORICAL FCST)
bl_AAA_chart.setelem(2) legend(AAA - OOS FCST)
BL_AAA_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025

GRAPH BL_VIX_CHART.line BL_VIX
BL_VIX_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "VIX ESTIMATE: HISTORICAL &
FCST"
BL_VIX_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05

```

```

BL_VIX_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
bl_vix_chart.setelem(1) legend(VIX - HISTORICAL FCST)
bl_vix_chart.setelem(2) legend(VIX - OOS FCST)
BL_VIX_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025

```

```

GRAPH BL_RET_PORT_CHART.line BL_RET_PORT
  BL_RET_PORT_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "RETURN_PORTFOLIO
ESTIMATE: HISTORICAL & FCST"
BL_RET_PORT_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_RET_PORT_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
bl_RET_PORT_chart.setelem(1) legend(RET_PORT - HISTORICAL FCST)
bl_RET_PORT_chart.setelem(2) legend(RET_PORT - OOS FCST)
BL_RET_PORT_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025

```

```

GRAPH BL_CS_CHART.line BL_CS
BL_CS_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "CREDIT SPREAD: HISTORICAL &
FCST"
BL_CS_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_CS_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
  bl_CS_chart.setelem(1) legend(CS - HISTORICAL FCST)
bl_CS_chart.setelem(2) legend(CS - OOS FCST)
BL_CS_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025

```

```

GRAPH BL_ER_PORT_CHART.line BL_ER_PORT
  BL_ER_PORT_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "EXCESS RETURN
PORTFOLIO ESTIMATE: HISTORICAL & FCST"
BL_ER_PORT_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
BL_ER_PORT_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
bl_ER_PORT_chart.setelem(1) legend(ER_PORT - HISTORICAL FCST)
bl_ER_PORT_chart.setelem(2) legend(ER_PORT - OOS FCST)
BL_ER_PORT_CHART.SAVE(T=PNG, U=IN, W=5, D=300)
F:\Users\andre\OneDrive\Desktop\School\MSQE\CAPSTONE\2_2024_2025

```

```

'---- C_2: DEFINING STOCHASTIC EQUATIONS FOR SCENARIO FCST MEM-----'
SMPL @ALL

```

```

EQUATION SR_FC_1.LS(N) LOG(PRICE)-LOG(PRICE(-12)) C CMA(-12)-CMA(-19) CS
@PCY(M2_FC(-6)) VIX_FC
          SR_FC_1.STATS
SERIES SR_FC_1_RESIDS = RESID

```

```
EQUATION SR_FC_2.LS VIX C CS IFD TP(-2) D(log(price)-log(price(-12)))
spike_2020
```

```
SR_FC_2.STATS
```

```
SERIES SR_FC_2_RESIDS = RESID
```

```
EQUATION SR_FC_3.COINTREG(METHOD=FMOLS, KERN=QUADSPEC, BW=ANDREWS) AAA C
TBTY
```

```
SR_FC_3.STATS
```

```
SERIES SR_FC_3_RESIDS = RESID
```

```
'-----C_2: MODEL OBJECT CONSTRUCTION -- SCENARIO: FIN CRISIS -----'
```

```
MODEL MEM_C2_AF
```

```
MEM_C2_AF.MERGE SR_FC_1
```

```
MEM_C2_AF.MERGE SR_FC_2
```

```
MEM_C2_AF.MERGE SR_FC_3
```

```
MEM_C2_AF.APPEND RET_PORT = (PRICE - PRICE(-12)) / PRICE(-12)
```

```
MEM_C2_AF.APPEND CS = BAA_FC - AAA
```

```
MEM_C2_AF.APPEND ER_PORT = RET_PORT - RF
```

```
'----- HISTORICAL PERIOD FORECAST SOLVE SCENARIO-----'
```

```
SMPL 2023M04 2025M04
```

```
MEM_C2_AF.APPEND ASSIGN @ALL _SHS
```

```
MEM_C2_AF.SOLVE
```

```
'----- OOS PERIOD FORECAST SOLVE SCENARIO-----'
```

```
SMPL 2025M05 2026M07
```

```
MEM_C2_AF.APPEND ASSIGN @ALL _SFS
```

```
MEM_C2_AF.SOLVE
```

```
'----- MEM 2 FCST CHARTS: FINANCIAL CRISIS -----'
```

```
SMPL 2023M04 2027M05
```

```
GROUP FC_PRICE PRICE_SFS
```

```
GROUP FC_PRICE_GR @PCY(PRICE_SFS)
```

```
GROUP FC_AAA AAA_SFS
```

```
GROUP FC_VIX VIX_SFS
```

```
GROUP FC_RET_PORT RET_PORT_SFS
```

```
GROUP FC_CS CS_SFS
```

```
GROUP FC_ER_PORT ER_PORT_SFS
```

```
GRAPH FC_PRICE_CHART.line FC_PRICE
```

```
FC_PRICE_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "PRICE FCST: FINANCIAL  
CRISIS SCENARIO"
```

```
FC_PRICE_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_PRICE_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_price_chart.setelem(1) legend(PRICE - OOS FCST)
```

```
GRAPH FC_PRICE_GR_CHART.line FC_PRICE_GR
```

```
FC_PRICE_GR_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "PRICE GROWTH RATE  
ESTIMATE: FINANCIAL CRISIS SCENARIO"
```

```
FC_PRICE_GR_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_PRICE_GR_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_PRICE_GR_CHART.setelem(1) legend(PRICE GR - OOS FCST)
```

```
GRAPH FC_AAA_chart.line FC_AAA
```

```
FC_AAA_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "AAA ESTIMATE: FINANCIAL  
CRISIS SCENARIO"
```

```
FC_AAA_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_AAA_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_AAA_chart.setelem(1) legend(AAA - OOS FCST)
```

```
GRAPH FC_VIX_CHART.line FC_VIX
```

```
FC_VIX_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "VIX ESTIMATE: FINANCIAL  
CRISIS SCENARIO"
```

```
FC_VIX_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_VIX_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_VIX_chart.setelem(1) legend(VIX - OOS FCST)
```

```
GRAPH FC_RET_PORT_CHART.line FC_RET_PORT
```

```
FC_RET_PORT_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "RETURN_PORTFOLIO  
ESTIMATE: FINANCIAL CRISIS SCENARIO"
```

```
FC_RET_PORT_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_RET_PORT_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_RET_PORT_chart.setelem(1) legend(RET_PORT - OOS FCST)
```

```
GRAPH FC_CS_CHART.line FC_CS
```

```
FC_CS_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "CREDIT SPREAD: FINANCIAL  
CRISIS SCENARIO"
```

```
FC_CS_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```
FC_CS_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
```

```
FC_CS_chart.setelem(1) legend(CS - HISTORICAL FCST)
```

```
GRAPH FC_ER_PORT_CHART.line FC_ER_PORT
```

```
FC_ER_PORT_CHART.addtext(.70, -.5 , FONT(CAMBRIA,16)) "EXCESS RETURN  
PORTFOLIO ESTIMATE: FINANCIAL CRISIS SCENARIO"
```

```
FC_ER_PORT_CHART.draw(shade, bottom, @rgb(234,234,234)) 2025M05 2027M05
```

```

FC_ER_PORT_CHART.addtext(3.54, 0.05, FONT(CAMBRIA,16)) "FCST Period"
FC_ER_PORT_chart.setelem(1) legend(ER_PORT - OOS FCST)

'-----VAR EQUATIONS-----'
SMPL @ALL

VAR VAR_1.LS 1 14 LOG(PRICE)-LOG(PRICE(-12)) AAA VIX @ BAA-AAA CMA
VAR_1.STATS
FREEZE(ROOTS_1) VAR_1.ARROOTS

VAR_1.MAKEMODEL(VAR_1_MOD)

VAR VAR_4.BVAR 1 30 D_PRICE_12 AAA VIX @ @PCY(M2(-12)) D(TP) IFD
VAR_4.STATS
FREEZE(ROOTS_4) VAR_4.ARROOTS

VAR_4.MAKEMODEL(VAR_4_MOD)

'VAR VAR_X.BVAR(prior=lit, l1=0.2, l2=0.95, l3=1.0, INITCOV=DIAG) 1 12
D_PRICE_12 AAA VIX @ @PCY(M2(-6)) D(TP) D(BAA) IFD CMA
'VAR_X.STATS
'FREEZE(ROOTS_X) VAR_X.ARROOTS
'VAR_X.makemodel(VAR_X_MOD)

'VAR VAR_2.BVAR 1 6 D_PRICE_X AAA VIX @ CMA @PCY(M2(-6)) D(TP) D(BAA)
'VAR_2.STATS
'FREEZE(ROOTS_2) VAR_2.ARROOTS

'VAR VAR_3.BVAR 1 6 PRICE AAA VIX @ CMA @PCY(M2(-6)) D(TP) D(BAA) IFD
'VAR_3.STATS
'FREEZE(ROOTS_3) VAR_3.ARROOTS
'VAR_3.MAKEMODEL(VAR_3_MOD)

'VAR VAR_5.BVAR(prior=lit, l1=0.2, l2=0.95, l3=1.0, INITCOV=DIAG) 1 12
D_PRICE_12 AAA VIX @ @PCY(M2(-6)) D(TP) D(BAA) IFD CMA
'VAR_5.STATS
'FREEZE(ROOTS_5) VAR_5.ARROOTS
'VAR_5.MAKEMODEL(VAR_5_MOD)

'VAR VAR_6.LS 1 14 LOG(PRICE)-LOG(PRICE(-12)) AAA VIX @ BAA-AAA CMA
'VAR_6.STATS
'FREEZE(ROOTS_6) VAR_6.ARROOTS
'VAR_6.MAKEMODEL(VAR_6_MOD)

```

```

'VAR VAR_1.LS 1 4 LOG (PRICE) -LOG (PRICE (-12)) AAA VIX @ BAA-AAA @PCY (M2 (-12))
CMA
'VAR_1.STATS
'FREEZE (ROOTS_1) VAR_1.ARROOTS
'VAR_1.MAKEMODEL (VAR_1_MOD)

'----- VAR FORECAST SOLVE-----'
'----- HISTORICAL FORECAST SOLVE-----'

VAR_1_MOD.APPEND ASSIGN @ALL _VH1
'VAR_3_MOD.APPEND ASSIGN @ALL _VH3
VAR_4_MOD.APPEND ASSIGN @ALL _VH4
'VAR_5_MOD.APPEND ASSIGN @ALL _VH5
'VAR_6_MOD.APPEND ASSIGN @ALL _VH6
'VAR_X_MOD.APPEND ASSIGN @ALL _VHX

SMPL 2023M04 2025M04

VAR_1_MOD.SOLVE
'VAR_3_MOD.SOLVE
VAR_4_MOD.SOLVE
'VAR_5_MOD.SOLVE
'VAR_6_MOD.SOLVE
'VAR_X_MOD.SOLVE

'----- OOS FORECAST SOLVE-----'

VAR_1_MOD.APPEND ASSIGN @ALL _VF1
'VAR_3_MOD.APPEND ASSIGN @ALL _VF3
VAR_4_MOD.APPEND ASSIGN @ALL _VF4
'VAR_5_MOD.APPEND ASSIGN @ALL _VF5
'VAR_6_MOD.APPEND ASSIGN @ALL _VF6
'VAR_X_MOD.APPEND ASSIGN @ALL _VFX

SMPL 2025M05 2026M07

VAR_1_MOD.SOLVE
'VAR_3_MOD.SOLVE
VAR_4_MOD.SOLVE
'VAR_5_MOD.SOLVE
'VAR_6_MOD.SOLVE
'VAR_X_MOD.SOLVE

```

```

'----- DATA EXPORTS TO EXCEL FOR CHARTING -----'
SMPL @ALL
GROUP EXOG_OOS TP TP_F TBTY TBTY_F IFD IFD_F RF RF_F BAA BAA VIX VIX M2 M2
      EXOG_OOS.TABLE
SMPL 2018M01 2026M07
WRITE(T=TXT, D=C, DATES) VAR_FCST.CSV EXOG_OOS

GROUP V_EXPORT PRICE PRICE_VF1 D_PRICE_12_VF4 PRICE_SF AAA AAA_VF1
AAA_VF4 AAA_SF VIX VIX_VF1 VIX_VF4 VIX_SF
      V_EXPORT.TABLE
SMPL 2025M05 2026M08
WRITE(T=TXT, D=C, DATES) VAR_FCST.CSV V_EXPORT

GROUP EXPORT2 PRICE PRICE_SF PRICE_SFS ER_PORT ER_PORT_SF ER_PORT_SFS
RET_PORT RET_PORT_SF RET_PORT_SFS AAA AAA_SF AAA_SFS VIX VIX_SF VIX_SFS CS
CS_SF CS_SFS

SMPL 2025M05 2026M07
WRITE(T=TXT, D=C, DATES) RETURNS_FCST.CSV EXPORT2

GROUP EXPORT3 @PCY(PRICE) @PCY(PRICE_SF) @PCY(PRICE_SFS)

SMPL 2025M05 2026M07
WRITE(T=TXT, D=C, DATES) GR_FCST.CSV EXPORT3

GROUP COMPETE_EXPORT PRICE PRICE_VH1 D_PRICE_12_VH4 VIX VIX_VH1 VIX_VH4
      COMPETE_EXPORT.TABLE
SMPL 2023M04 2026M07
WRITE(T=TXT, D=C, DATES) COMPETE_FCST.CSV COMPETE_EXPORT

'GROUP COMPETE_EXPORT2 PRICE PRICE_SH PRICE_SF PRICE_VH6 PRICE_VF6
D_PRICE_12_VH5 D_PRICE_12_VF5
      COMPETE_EXPORT2.TABLE
'SMPL 2023M04 2026M07
'WRITE(T=TXT, D=C, DATES) COMPETE_FCST_2.CSV COMPETE_EXPORT2

'----- EXOGENOUS ARIMA FORECAST DATA EXPORTS TO EXCEL FOR CHARTING -----'
SMPL 2023M04 2026M07

GROUP EXPORT_F CMA CMA_F TBTY TBTY_F RF RF_F IFD IFD_F

SMPL 2018M01 2026M07
WRITE(T=TXT, D=C, DATES) EXOG_FCST.CSV EXPORT_F

```

```

'----- PRICE DATA EXPORTS TO EXCEL FOR CHARTING in R -----
---'

SMPL 2018M01 2026M07

GROUP C4_KDV1_FCST PRICE PRICE_SH PRICE_SF PRICE_VH1 PRICE_VF1 D_PRICE_12_VH4
D_PRICE_12_VF4

SMPL 2018M01 2026M07
WRITE(T=TXT , D=C, DATES) C4_KDV1_FCST.CSV C4_KDV1_FCST

'----- DRAFT EQUATIONS-----'

SMPL @ALL

EQUATION SR_X.LS(N) LOG(PRICE)-LOG(PRICE(-12)) C CMA(-12)-CMA(-19) CS
@PCY(M2(-6)) VIX SPIKE_2020
                SR_X.STATS
                SERIES SR_X_RESIDS = RESID

EQUATION SR_X2.LS vix c cs ifd D(TP) log(price(-1))-log(price(-13))
spike_2020
                SR_X2.STATS
                SERIES SR_X2_RESIDS = RESID

EQUATION SR_X3.LS vix c cs ifd tp(-2) log(price)-log(price(-12)) spike_2020
                SR_X3.STATS
                SERIES SR_X3_RESIDS = RESID

EQUATION SR_X4.LS vix c cs ifd D(TP) log(price)-log(price(-12)) spike_2020
                SR_X4.STATS
                SERIES SR_X4_RESIDS = RESID

EQUATION SR_X7.LS LOG(PRICE)-LOG(PRICE(-12)) C SMB HML CMA RMW MOM
SR_X7.STATS

EQUATION SR_X8.COINTREG(METHOD=FMOLS, KERN=QUADSPEC, BW=ANDREWS) LOG(PRICE)-
LOG(PRICE(-12)) C CMA CS @PCY(M2(-6)) VIX
SR_X8.STATS

EQUATION SR_X9.LS ERP C SMB HML CMA RMW MOM
SR_X9.STATS

```

Appendix 5 – R code

```

---
title: "ECON_513_C4"
author: "Andrew Frohner"
date: "2025-07-31"
output: html_document
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

#-----#
# ECON_513_C4_PROGRAM.R      A.FROHNER AUG 2025  #
#-----#

```{r cars}
#import the C4_Rfile dataset
C4_Rfile <- read.csv("C4_RFILE.csv")

data <- C4_Rfile
```

```{r}
head(data)

```

```{r}
#load the forecast , xts, and stats packages
library(forecast)
library(xts)
library(stats)

```

```{r}
Convert Year_Per to yearmon
data$Year_Per_ym <- as.yearmon(data$Year_Per, "%YM%m")

Make xts object. Replace "Value" with your actual value column name
PRICE_SF <- xts(data$PRICE_SF, order.by = data$Year_Per_ym)
PRICE_SH <- xts(data$PRICE_SH, order.by = data$Year_Per_ym)
PRICE_VH1 <- xts(data$PRICE_VH1, order.by = data$Year_Per_ym)

```

```
PRICE_VF1 <- xts(data$PRICE_VF1, order.by = data$Year_Per_ym)
PRICE_VH4 <- xts(data$PRICE_VH4, order.by = data$Year_Per_ym)
PRICE_VF4 <- xts(data$PRICE_VF4, order.by = data$Year_Per_ym)
PRICE <- xts(data$PRICE, order.by = data$Year_Per_ym)
```

```{r}
#SAMPLE WINDOWS
PRICE_R_IS_SH <- window(PRICE_SH, start = "2023-04", end = "2025-04",
frequency=12)
PRICE_R_IS_VH1 <- window(PRICE_VH1, start = "2023-04", end = "2025-04",
frequency=12)
PRICE_R_IS_VH4 <- window(PRICE_VH4, start = "2023-04", end = "2025-04",
frequency=12)
PRICE_R_IS <- window(PRICE, start = "2023-04", end = "2025-04", frequency=12)
PRICE_R_OOS <- window(PRICE, start = "2025-05", end = "2026-07",
frequency=12)
PRICE_R_OOS_VF1 <- window(PRICE_VF1, start = "2025-05", end = "2026-07",
frequency=12)
PRICE_R_OOS_VF4 <- window(PRICE_VF4, start = "2025-05", end = "2026-07",
frequency=12)
PRICE_R_OOS_SF <- window(PRICE_SF, start = "2025-05", end = "2026-07",
frequency=12)
```

```{r}
#create an ARIMA model for PRICE using PRICE and ITV_1
PRICE_R_ARIMA <- Arima(PRICE_R_IS, order = c(2, 1, 2))

```

```{r}
#summary of the ARIMA model
summary(PRICE_R_ARIMA)
```

```{r}
#create an ARIMA model for PRICE using PRICE and ITV_1
PRICE_R_ARIMA2 <- Arima(PRICE_R_IS, order = c(2, 1, 0))

#summary of the ARIMA model
summary(PRICE_R_ARIMA2)

```

```{r}
#Create an object for the Fitted values
PRICE_R_AH <- fitted(PRICE_R_ARIMA)

#export the fitted values to a dataframe
```

```
PRICE_R_AH_df <- data.frame(
 Year_Per = time(PRICE_R_IS),
 Fitted = PRICE_R_AH
)

#create an object for the residuals
PRICE_R_AH_res <- residuals(PRICE_R_ARIMA)
````  
````{r}  
#export the dataframe to a csv
write.csv(PRICE_R_AH_df, "PRICE_R_AH_df.csv", row.names = FALSE)
````  
````{r}  

library(forecast)
Predict the next 15 months

#Future ITV
future_ITV_1 <- rep(0, 15) # Assuming ITV_1 is constant at 0 for the forecast
period

PRICE_R_OOS_AF <- forecast(PRICE_R_ARIMA, h = 15, xreg= future_ITV_1)

PRICE_R_AF <- as.data.frame(PRICE_R_OOS_AF)

````  
````{r}  
#export the forecast results to a csv
write.csv(PRICE_R_OOS_AF, "PRICE_R_OOS_AF.csv", row.names = FALSE)

Create a data frame with all 15 forecast period values
PRICE_R_OOS_df <- data.frame(
 Year_Per = seq(as.Date("2025-05-01"), by = "month", length.out = 15),
 Forecast = as.numeric(PRICE_R_OOS_AF$mean)
)
write.csv(PRICE_R_OOS_df, "PRICE_R_OOS_df.csv", row.names = FALSE)

````  
````{r}  
#plot the forecast
library(ggplot2)
ggplot(data = PRICE_R_OOS_df, aes(x = Year_Per, y = Forecast)) +
 geom_line(color = "blue") +
 labs(title = "Forecast of PRICE", x = "Year_Per", y = "Forecast") +
```

```

 theme_minimal() +
 scale_x_date(date_labels = "%Y-%m", date_breaks = "1 month") +
 theme(axis.text.x = element_text(angle = 45, hjust = 1))
#-----#
```

```{r}

#Create a data frame of PRICE_R_IS , PRICE_R_IS_SH, PRICE_R_IS_VH1,
PRICE_R_IS_VH4, PRICE_R_IS_AH
PRICE_IS_FRAME <- data.frame(
 Year_Per = time(PRICE_R_IS),
 PRICE_R_IS = as.numeric(PRICE_R_IS),
 PRICE_R_IS_SH = as.numeric(PRICE_R_IS_SH),
 PRICE_R_IS_VH1 = as.numeric(PRICE_R_IS_VH1),
 PRICE_R_IS_VH4 = as.numeric(PRICE_R_IS_VH4),
 PRICE_R_AH = as.numeric(PRICE_R_AH)
)

```

```{r}
#Create the squared deviations

PRICE_IS_FRAME$SE_SH <- (PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_SH)^2
PRICE_IS_FRAME$SE_VH1 <- (PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH1)^2
PRICE_IS_FRAME$SE_VH4 <- (PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH4)^2
PRICE_IS_FRAME$SE_AH <- (PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_AH)^2

#create a data frame of the squared deviations
PRICE_IS_SE_FRAME <- data.frame(
 Year_Per = PRICE_IS_FRAME$Year_Per,
 SE_SH = PRICE_IS_FRAME$SE_SH,
 SE_VH1 = PRICE_IS_FRAME$SE_VH1,
 SE_VH4 = PRICE_IS_FRAME$SE_VH4,
 SE_AH = PRICE_IS_FRAME$SE_AH
)

#write the squared deviations to a csv
write.csv(PRICE_IS_SE_FRAME, "PRICE_IS_SE_FRAME.csv", row.names = FALSE)

```

```{r}
#Compute the Root Mean Squared Error

```

```

RMSE_SH <- (mean(PRICE_IS_FRAME$SE_SH, na.rm = TRUE))^0.5
RMSE_VH1 <- (mean(PRICE_IS_FRAME$SE_VH1, na.rm = TRUE))^0.5
RMSE_VH4 <- (mean(PRICE_IS_FRAME$SE_VH4, na.rm = TRUE))^0.5
RMSE_AH <- (mean(PRICE_IS_FRAME$SE_AH, na.rm = TRUE))^0.5

```
```{r}
#sum the RMSEs
Total_RMSE <- sum(RMSE_SH, RMSE_VH1, RMSE_VH4, RMSE_AH)

RMSE_share_SH <- RMSE_SH / Total_RMSE
RMSE_share_VH1 <- RMSE_VH1 / Total_RMSE
RMSE_share_VH4 <- RMSE_VH4 / Total_RMSE
RMSE_share_AH <- RMSE_AH / Total_RMSE

```
```{r}
#plot the SEs
library(ggplot2)

ggplot(PRICE_IS_FRAME, aes(x = Year_Per)) +
 geom_line(aes(y = SE_SH, color = "SE_SH")) +
 geom_line(aes(y = SE_VH1, color = "SE_VH1")) +
 geom_line(aes(y = SE_VH4, color = "SE_VH4")) +
 geom_line(aes(y = SE_AH, color = "SE_AH")) +
 labs(title = "Squared Deviations of PRICE", x = "Year_Per", y = "Squared
Deviation") +
 scale_color_manual(name = "Fcst Technique", values = c("SE_SH" = "red",
"SE_VH1" = "green", "SE_VH4" = "blue", "SE_AH" = "purple"), labels =
c("SE_SH" = "Structural MEM", "SE_VH1" = "VAR (LS)", "SE_VH4" = "BVAR",
"SE_AH" = "ARIMA")) +
 theme_minimal()

```
```{r}
#compute the normalized Absolute Deviations
PRICE_IS_FRAME$MA_SH <- abs(PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_SH) /
PRICE_IS_FRAME$PRICE_R_IS
PRICE_IS_FRAME$MA_VH1 <- abs(PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH1) /
PRICE_IS_FRAME$PRICE_R_IS
PRICE_IS_FRAME$MA_VH4 <- abs(PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH4) /
PRICE_IS_FRAME$PRICE_R_IS
PRICE_IS_FRAME$MA_AH <- abs(PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_AH) /
PRICE_IS_FRAME$PRICE_R_IS

#Compute the MAPE

```

```

MAPE_SH <- mean(PRICE_IS_FRAME$MA_SH, na.rm = TRUE) * 100
MAPE_VH1 <- mean(PRICE_IS_FRAME$MA_VH1, na.rm = TRUE) * 100
MAPE_VH4 <- mean(PRICE_IS_FRAME$MA_VH4, na.rm = TRUE) * 100
MAPE_AH <- mean(PRICE_IS_FRAME$MA_AH, na.rm = TRUE) * 100

#Compute the APE
APE_SH <- abs((PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_SH) /
PRICE_IS_FRAME$PRICE_R_IS) * 100
APE_VH1 <- abs((PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH1) /
PRICE_IS_FRAME$PRICE_R_IS) * 100
APE_VH4 <- abs((PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_VH4) /
PRICE_IS_FRAME$PRICE_R_IS) * 100
APE_AH <- abs((PRICE_IS_FRAME$PRICE_R_IS - PRICE_R_IS_AH) /
PRICE_IS_FRAME$PRICE_R_IS) * 100

Create a data frame with the MAPE and APE values
PRICE_IS_MA_FRAME <- data.frame(
 Year_Per = PRICE_IS_FRAME$Year_Per,
 MA_SH = PRICE_IS_FRAME$MA_SH,
 MA_VH1 = PRICE_IS_FRAME$MA_VH1,
 MA_VH4 = PRICE_IS_FRAME$MA_VH4,
 MA_AH = PRICE_IS_FRAME$MA_AH
)

PRICE_IS_APE_FRAME <- data.frame(
 Year_Per = PRICE_IS_FRAME$Year_Per,
 APE_SH = APE_SH,
 APE_VH1 = APE_VH1,
 APE_VH4 = APE_VH4,
 APE_AH = APE_AH
)

#EXPORT TO CSV
write.csv(PRICE_IS_MA_FRAME, "PRICE_IS_MA_FRAME.csv", row.names = FALSE)
write.csv(PRICE_IS_APE_FRAME, "PRICE_IS_APE_FRAME.csv", row.names = FALSE)

```
```{r}
#plot the MADs
ggplot(PRICE_IS_FRAME, aes(x = Year_Per)) +
 geom_line(aes(y = MA_SH, color = "MA_SH")) +
 geom_line(aes(y = MA_VH1, color = "MA_VH1")) +
 geom_line(aes(y = MA_VH4, color = "MA_VH4")) +
 geom_line(aes(y = MA_AH, color = "MA_AH")) +

```

```

labs(title = "Normalized Absolute Deviations of PRICE", x = "Year_Per", y =
"Normalized Absolute Deviation") +
 scale_color_manual(name = " Fcst Technique" , values = c("MA_SH" = "red",
"MA_VH1" = "green", "MA_VH4" = "blue", "MA_AH" = "purple"), labels =
c("MA_SH" = "Structural MEM", "MA_VH1" = "VAR (LS)", "MA_VH4" = "BVAR",
"MA_AH" = "ARIMA")) +
 theme_minimal()

```

```{r}
#Plot the MAPE
ggplot(PRICE_IS_FRAME, aes(x = Year_Per)) +
 geom_line(aes(y = MA_SH, color = "MA_SH")) +
 geom_line(aes(y = MA_VH1, color = "MA_VH1")) +
 geom_line(aes(y = MA_VH4, color = "MA_VH4")) +
 geom_line(aes(y = MA_AH, color = "MA_AH")) +
 labs(title = "MAPE of PRICE", x = "Year_Per", y = "MAPE") +
 scale_color_manual(name = " Fcst Technique" , values = c("MA_SH" = "red",
"MA_VH1" = "green", "MA_VH4" = "blue", "MA_AH" = "purple"), labels =
c("MA_SH" = "Structural MEM", "MA_VH1" = "VAR (LS)", "MA_VH4" = "BVAR",
"MA_AH" = "ARIMA")) +
 theme_minimal()

```

```{r}
#export the IS FRAME to a csv
write.csv(PRICE_IS_FRAME, "PRICE_IS_FRAME.csv", row.names = FALSE)

```

```{r}
#Equal Weight combination Forecast
library(xts)
PRICE_R_IS_EWT <- (as.ts(PRICE_R_IS_SH) +
 as.ts(PRICE_R_IS_VH1) +
 as.ts(PRICE_R_IS_VH4) +
 as.ts(PRICE_R_IS_AH)) / 4

```

```{r}
#Optimal weight forecast #1 -- IS
#create a combination forecast using RMSE adjusted weightRMSE shares

OPT1_ADJ_SH <- .25 - RMSE_share_SH
OPT1_ADJ_VH1 <- .25 - RMSE_share_VH1
OPT1_ADJ_VH4 <- .25 - RMSE_share_VH4
OPT1_ADJ_AH <- .25 - RMSE_share_AH

PRICE_R_IS_C_OP1 <- (.25 + OPT1_ADJ_SH) * as.ts(PRICE_R_IS_SH) +

```

```

 (.25 + OPT1_ADJ_VH1) * as.ts(PRICE_R_IS_VH1) +
 (.25 + OPT1_ADJ_VH4) * as.ts(PRICE_R_IS_VH4) +
 (.25 + OPT1_ADJ_AH) * as.ts(PRICE_R_AH)

 ...

  ```{r}
#run a linear model with PRICE_R_IS as the dependent variable
#make the coefficients sum to 1
#remove the intercept
LM_OLS_W <- lm(PRICE_R_IS ~ PRICE_R_IS_SH + PRICE_R_IS_VH1 + PRICE_R_IS_VH4 +
PRICE_R_AH - 1)

# Normalize the coefficients to sum to 1
LM_OLS_W$coefficients <- LM_OLS_W$coefficients / sum(LM_OLS_W$coefficients)
# Display the summary of the model
summary(LM_OLS_W)

...

  ```{r}
#SAVE THE COEFFICIENTS TO AN OBJECT
coefficients <- coef(LM_OLS_W)

#sum the coefficients
sum_coefficients <- sum(coefficients)
print(sum_coefficients)

Create a data frame with the coefficients
coefficients_df <- data.frame(
 Variable = names(coefficients),
 Coefficient = coefficients
)

#export the coefficients to a csv
write.csv(coefficients_df, "coefficients.csv", row.names = FALSE)

...

  ```{r}
#Optimal weight forecast #2 -- IS
#Create a a forecast using the corresponding coefficients for each forecast
PRICE_R_IS_C_OP2 <- coefficients[1] * as.ts(PRICE_R_IS_SH) +
      coefficients[2] * as.ts(PRICE_R_IS_VH1) +
      coefficients[3] * as.ts(PRICE_R_IS_VH4) +
      coefficients[4] * as.ts(PRICE_R_AH)

```

```

```
```{r}
#Optimal weight forecast #3 -- IS
#Adjust Weights to Minimize the Forecasts APE
PRICE_R_IS_C_OP3 <- .0100 * as.ts(PRICE_R_IS_SH) +
                    .09241 * as.ts(PRICE_R_IS_VH1) +
                    .01000 * as.ts(PRICE_R_IS_VH4) +
                    .88759 * as.ts(PRICE_R_AH)
```
```{r}
#create a data frame with the equal-weight, OLS weight, adjusted weight, and
minimized SE forecasts
PRICE_R_IS_C_FRAME <- data.frame(
  Year_Per = time(PRICE_R_IS),
  PRICE_R_IS_EWT = as.numeric(PRICE_R_IS_EWT),
  PRICE_R_IS_C_OP1 = as.numeric(PRICE_R_IS_C_OP1),
  PRICE_R_IS_C_OP2 = as.numeric(PRICE_R_IS_C_OP2),
  PRICE_R_IS_C_OP3 = as.numeric(PRICE_R_IS_C_OP3)
)

#export the dataframe to a csv
write.csv(PRICE_R_IS_C_FRAME, "PRICE_IS_COMBO.csv", row.names = FALSE)
```
```{r}

# First, create the time index for your forecast period
forecast_dates <- seq(from = as.yearmon("2025-05"),
                      by = 1/12,
                      length.out = 15)

# Convert forecast to xts (using Point Forecast)
PRICE_R_AF_xts <- xts(PRICE_R_AF$`Point Forecast`,
                     order.by = forecast_dates)

#Create a data frame of PRICE_R_IS , PRICE_R_IS_SF, PRICE_R_IS_VF1,
PRICE_R_IS_VF4, PRICE_R_IS_AF
PRICE_OOS_FRAME <- data.frame(
  Year_Per = time(PRICE_R_OOS),
  PRICE_R_OOS = as.numeric(PRICE_R_OOS),
  PRICE_R_OOS_SF = as.numeric(PRICE_R_OOS_SF),
  PRICE_R_OOS_VF1 = as.numeric(PRICE_R_OOS_VF1),
  PRICE_R_OOS_VF4 = as.numeric(PRICE_R_OOS_VF4),
  PRICE_R_OOS_AFxts = as.numeric(PRICE_R_AF_xts)
)

```

```

#export the dataframe to a csv
write.csv(PRICE_OOS_FRAME, "PRICE_OOS_FRAME.csv", row.names = FALSE)
```
```{r}
numberSH <- .25 + OPT1_ADJ_SH
numberVH1 <- .25 + OPT1_ADJ_VH1
numberVH4 <- .25 + OPT1_ADJ_VH4
numberAH <- .25 + OPT1_ADJ_AH
# Print the numbers
print(paste("Adjusted SH:", numberSH))
print(paste("Adjusted VH1:", numberVH1))
print(paste("Adjusted VH4:", numberVH4))
print(paste("Adjusted AH:", numberAH))

```
```{r}
#Optimal Weight combination #1 -- OOS
#create a combination forecast using the ADJ values for OOS
PRICE_R_OOS_C_OP1 <- ( .25 + OPT1_ADJ_SH) * as.ts(PRICE_R_OOS_SF) +
  (.25 + OPT1_ADJ_VH1) * as.ts(PRICE_R_OOS_VF1) +
  (.25 + OPT1_ADJ_VH4) * as.ts(PRICE_R_OOS_VF4) +
  (.25 + OPT1_ADJ_AH) * as.ts(PRICE_R_AF_xts)

#Optimal Weight combination #2 -- OOS
#create a combination forecast using the coefficients
PRICE_R_OOS_C_OP2 <- coefficients[1] * as.ts(PRICE_R_OOS_SF) +
  coefficients[2] * as.ts(PRICE_R_OOS_VF1) +
  coefficients[3] * as.ts(PRICE_R_OOS_VF4) +
  coefficients[4] * as.ts(PRICE_R_AF_xts)

#optimal Weight combination #3 - OOS
#create a combination forecast using the minimized APE values
PRICE_R_OOS_C_OP3 <- .01000 * as.ts(PRICE_R_OOS_SF) +
  .09241 * as.ts(PRICE_R_OOS_VF1) +
  .01000 * as.ts(PRICE_R_OOS_VF4) +
  .88759 * as.ts(PRICE_R_AF_xts)

#Equal Weight combination forecast -- OOS
#create a combination forecast using the Equal Weight
PRICE_R_OOS_EWT <- (as.ts(PRICE_R_OOS_SF) +
  as.ts(PRICE_R_OOS_VF1) +
  as.ts(PRICE_R_OOS_VF4) +
  as.ts(PRICE_R_AF_xts)) / 4

```

```
#Create a data frame with the equal-weight, OLS weight, adjusted weight, and  
minimized SE forecasts
```

```
PRICE_R_OOS_C_FRAME <- data.frame(  
  Year_Per = time(PRICE_R_OOS),  
  PRICE_R_OOS_EWT = as.numeric(PRICE_R_OOS_EWT),  
  PRICE_R_OOS_C_OP1 = as.numeric(PRICE_R_OOS_C_OP1),  
  PRICE_R_OOS_C_OP2 = as.numeric(PRICE_R_OOS_C_OP2),  
  PRICE_R_OOS_C_OP3 = as.numeric(PRICE_R_OOS_C_OP3)  
)  
#export the dataframe to a csv  
write.csv(PRICE_R_OOS_C_FRAME, "PRICE_OOS_COMBO.csv", row.names = FALSE)  
````
```